ALGEBRA I · COMMON CORE HIGH SCHOOL MATH

Answer Key

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

84 question sets • 2100 questions

Page numbers match the Question Collection.

Every page shows its question ID (for example S40-F2-Q3).

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SET 1 • HSN-RN.A.1 • NUMBER & QUANTITY

Explain the meaning of rational exponents from the properties of integer exponents

MAIN	Algebra I main question	**HSN-RN.A.1**
BACKWARD 1	Apply the properties of integer exponents	**8.EE.A.1**
BACKWARD 2	Evaluate square roots and cube roots	**8.EE.A.2**
FORWARD 1	Express the solution of an exponential equation as a logarithm	**HSF-LE.A.4**
FORWARD 2	Use the inverse relationship between exponents and logarithms	**HSF-BF.B.5**

SET 1 • HSN-RN.A.1

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1

For the power-of-a-power property to hold, $(7^{1/4})^4$ must equal $7^{(1/4)\cdot 4} = 7$.

What number must $7^{1/4}$ be? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q1/5

9

QUESTION ID

S1-M-Q1

TYPE

Short answer

CORRECT ANSWER

$\sqrt[4]{7}$, the fourth root of 7 (the positive number whose fourth power is 7).

GRADING NOTE Any equivalent form, such as the fourth root of 7. The explanation must use the fact that $(7^{1/4})^4 = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q1/5

Use the properties of exponents to explain why $8^{2/3} = 4$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q2/5

10

QUESTION ID

S1-M-Q2

TYPE

Short answer

CORRECT ANSWER

$8^{2/3} = (8^{1/3})^2 = 2^2 = 4.$

GRADING NOTE Any equivalent reasoning, such as $8^{2/3} = (8^2)^{1/3} = \sqrt[3]{64} = 4$. The explanation must use a property of exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q2/5

For $x > 0$, which expression is equal to $x^{3/4}$?

A. $x^3 \div 4$

B. the fourth root of x, cubed

C. the cube root of x, to the fourth power

D. $3x \div 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q3/5

11

QUESTION ID

S1-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. the fourth root of x, cubed

GRADING NOTE $x^{3/4} = (x^{1/4})^3$: the denominator 4 is the root and the numerator 3 is the power.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q3/5

True or false?

$16^{1/2} = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q4/5

12

QUESTION ID

S1-M-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $16^{1/2}$ is $\sqrt{16} = 4$. An exponent of 1/2 is a square root, not a division by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q4/5

True or false?

For $x > 0$, $x^{1/2} \cdot x^{1/2} = x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q5/5

13

QUESTION ID

S1-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The product property gives $x^{1/2+1/2} = x^1 = x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSN-RN.A.1

BACKWARD 1

Apply the properties of integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power.

$2^3 \cdot 2^5$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q1/5

15

QUESTION ID

S1-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2^8

GRADING NOTE Form required: a single power. Add exponents when multiplying powers with the same base: 3 + 5 = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q1/5

Write as a single power.

$(3^2)^4$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q2/5

16

QUESTION ID

S1-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3^8

GRADING NOTE Form required: a single power. Multiply exponents for a power of a power: 2 · 4 = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q2/5

Write as a single power.

$5^6 \div 5^2$

A. 1^4

B. 5^4

C. 5^3

D. 5^8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q3/5

17

QUESTION ID

S1-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5^4

GRADING NOTE Subtract exponents when dividing powers with the same base: $6 - 2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q3/5

True or false?

$(7^2)^3 = 7^6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q4/5

18

QUESTION ID

S1-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Multiply the exponents: $2 \cdot 3 = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q4/5

True or false?

$4^0 = 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q5/5

19

QUESTION ID

S1-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Any nonzero number to the power 0 is 1, so $4^0 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • HSN-RN.A.1

BACKWARD 2

Evaluate square roots and cube roots • Grade 8 • Expressions & Equations • 8.EE.A.2

Find the value.

$\sqrt{81}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q1/5

21

QUESTION ID

S1-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form. $9^2 = 81$, and $\sqrt{}$ means the nonnegative root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q1/5

Find the value.

$\sqrt[3]{27}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q2/5

22

QUESTION ID

S1-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. $3^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

$\sqrt[3]{64}$

A. 32

B. 4

C. 8

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q3/5

23

QUESTION ID

S1-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4

GRADING NOTE $4^3 = 64$. (8 is $\sqrt{64}$, the square root.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q3/5

True or false?

$\sqrt[3]{(-8)} = -2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q4/5

24

QUESTION ID

S1-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(-2)^3 = -8$, and a cube root of a negative number is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q4/5

True or false?

$\sqrt{25} = \pm 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q5/5

25

QUESTION ID

S1-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The symbol √ means the nonnegative square root, so √25 = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q5/5

SET 1 • HSN-RN.A.1

FORWARD 1

Express the solution of an exponential equation as a logarithm • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $10^x = 500$. Write the solution as a logarithm, then use a calculator to round it to the nearest hundredth.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q1/5

27

QUESTION ID

S1-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$x = \log 500 \approx 2.70$

GRADING NOTE Form required: the logarithm and the rounded value. log means log base 10; log 500 = 2.69897...

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q1/5

A colony of 400 bacteria grows by a factor of 10 every hour: $N = 400 \cdot 10^t$. Solve $400 \cdot 10^t = 3{,}000$. Give the exact logarithm and round to the nearest hundredth. (log means base 10.)

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q2/5

28

QUESTION ID

S1-F1-Q2

TYPE

Short answer

CORRECT ANSWER

log 7.5 ≈ 0.88 hour

GRADING NOTE Form required: the exact logarithm and the rounded value. $10^t = 7.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q2/5

Which expression is the solution of $3 \cdot 10^x = 60$? (log means base 10.)

A. x = log 60

B. x = 20

C. x = log 57

D. x = log 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q3/5

29

QUESTION ID

S1-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = log 20

GRADING NOTE Divide both sides by 3 first: $10^x = 20$, so x = log 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q3/5

True or false?

The solution of $2 \cdot 10^t = 1{,}000$ is $t = \log 500 \approx 2.70$. (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q4/5

30

QUESTION ID

S1-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $10^t = 500$, and $\log 500 \approx 2.699$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q4/5

True or false?

The solution of $10^x = -10$ is $x = -1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q5/5

31

QUESTION ID

S1-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 10^x is positive for every real x, so $10^x = -10$ has no real solution. ($10^{-1} = 0.1$.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSN-RN.A.1

FORWARD 2

Use the inverse relationship between exponents and logarithms • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

$f(x) = 10^x$ and $g(x) = \log x$ (log means base 10).

Find f(g(100)) and g(f(2)), and say what they show.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q1/5

33

QUESTION ID

S1-F2-Q1

TYPE

Short answer

CORRECT ANSWER

f(g(100)) = 100 and g(f(2)) = 2: f and g undo each other, so they are inverse functions.

GRADING NOTE Form required: both values and the conclusion. g(100) = 2 and f(2) = 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q1/5

A sound level is $L = 10 \log(I \div I_0)$ decibels, where log means base 10.

Solve for the intensity I in terms of L.

I = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q2/5

34

QUESTION ID

S1-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$I_0 \cdot 10^{L/10}$

GRADING NOTE Any equivalent form. Divide by 10, then use 10 to the power of each side: $I \div I_0 = 10^{L/10}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q2/5

Which function is the inverse of $f(x) = 5^x$?

A. $f^{-1}(x) = x \div 5$

B. $f^{-1}(x) = x^5$

C. $f^{-1}(x) = \log_5 x$

D. $f^{-1}(x) = 5^{-x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q3/5

35

QUESTION ID

S1-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $f^{-1}(x) = \log_5 x$

GRADING NOTE $\log_5 x$ undoes 5^x: $\log_5(5^x) = x$ and $5^{\log_5 x} = x$ for $x > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q3/5

True or false?

$10^{\log_{10} 7} = 7$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q4/5

36

QUESTION ID

S1-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 10^x and log x are inverse functions, so each undoes the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q4**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q4/5

True or false?

$\log(10^x) = 10x$ (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q5/5

37

QUESTION ID

S1-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\log(10^x) = x$, because log undoes 10^x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q5/5

SET 2 • HSN-RN.A.2 • NUMBER & QUANTITY

Rewrite expressions with radicals and rational exponents

MAIN	Algebra I main question	**HSN-RN.A.2**
BACKWARD 1	Solve equations of the form $x^2 = p$ and $x^3 = p$	**8.EE.A.2**
BACKWARD 2	Use the meaning of rational exponents	**HSN-RN.A.1**
FORWARD 1	Solve radical equations and identify extraneous solutions	**HSA-REI.A.2**
FORWARD 2	Add, subtract, and multiply complex numbers	**HSN-CN.A.2**

SET 2 • HSN-RN.A.2

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2

Assume $x \geq 0$. Write $x^{5/2}$ using a radical.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q1/5

40

QUESTION ID

S2-M-Q1

TYPE

Short answer

CORRECT ANSWER

$\sqrt{(x^5)}$

GRADING NOTE Form required: radical form. $(\sqrt{x})^5$ is also correct. $x^{5/2} = (x^5)^{1/2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-M-Q1**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write $\sqrt[3]{(y^2)}$ using a rational exponent.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q2/5

41

QUESTION ID

S2-M-Q2

TYPE

Short answer

CORRECT ANSWER

$y^{2/3}$

GRADING NOTE Form required: a rational exponent. The cube root gives the denominator 3, and the power 2 gives the numerator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q2/5

Simplify.

$\sqrt{50} \cdot \sqrt{2}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q3/5

42

QUESTION ID

S2-M-Q3

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. $\sqrt{50} \cdot \sqrt{2} = \sqrt{100} = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-M-Q3**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q3/5

Which expression is equal to $(27x^6)^{1/3}$ for every real number x?

A. $27x^2$

B. $9x^2$

C. $3x^3$

D. $3x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q4/5

43

QUESTION ID

S2-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $3x^2$

GRADING NOTE $27^{1/3} = 3$ and $(x^6)^{1/3} = x^2$ (a cube root is defined for every real number).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q4/5

True or false?

$\sqrt{(x^2)} = x$ for every real number x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q5/5

44

QUESTION ID

S2-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{(x^2)} = |x|$. For $x = -3$, $\sqrt{9} = 3$, not -3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q5/5

SET 2 • HSN-RN.A.2

BACKWARD 1

Solve equations of the form $x^2 = p$ and $x^3 = p$ • Grade 8 • Expressions & Equations • 8.EE.A.2

Solve for x.

$x^3 = 125$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q1/5

46

QUESTION ID

S2-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. $5^3 = 125$, and $x^3 = 125$ has one real solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x. Give all solutions.

$x^2 = 36$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q2/5

47

QUESTION ID

S2-B1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 6 and x = −6

GRADING NOTE Any equivalent form (x = ±6). Both solutions are required: $6^2 = (-6)^2 = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q2/5

Solve for x.

$x^3 = -8$

A. ±2

B. −2

C. −4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q3/5

48

QUESTION ID

S2-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −2

GRADING NOTE $(-2)^3 = -8$. A cube equation has one real solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q3/5

True or false?

$x^2 = 7$ has the solutions $\sqrt{7}$ and $-\sqrt{7}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q4/5

49

QUESTION ID

S2-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both $(\sqrt{7})^2$ and $(-\sqrt{7})^2$ equal 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q4/5

True or false?

$x^3 = 27$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q5/5

50

QUESTION ID

S2-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only x = 3 works; $(-3)^3 = -27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • HSN-RN.A.2

BACKWARD 2

Use the meaning of rational exponents • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1

Find the value.

$25^{1/2}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q1/5

52

QUESTION ID

S2-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. $25^{1/2} = \sqrt{25} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q1/5

Find the value.

$27^{2/3}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q2/5

53

QUESTION ID

S2-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form. $27^{2/3} = (\sqrt[3]{27})^2 = 3^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q2/5

Find the value.

$32^{1/5}$

A. 160

B. 2

C. 6.4

D. $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q3/5

54

QUESTION ID

S2-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2

GRADING NOTE $2^5 = 32$, so the fifth root of 32 is 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q3/5

True or false?

$9^{3/2} = 27$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q4/5

55

QUESTION ID

S2-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $9^{3/2} = (\sqrt{9})^3 = 3^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$8^{1/3} = \frac{8}{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q5/5

56

QUESTION ID

S2-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $8^{1/3}$ is the cube root of 8, which is 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • HSN-RN.A.2

FORWARD 1

Solve radical equations and identify extraneous solutions • High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve √(x + 3) = 5. Check your solution.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q1/5

58

QUESTION ID

S2-F1-Q1

TYPE

Short answer

CORRECT ANSWER

22

GRADING NOTE Any equivalent form. Square both sides: x + 3 = 25, so x = 22. Check: √25 = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q1**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q1/5

Solve √(2x + 1) = x − 1. Check for extraneous solutions.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q2/5

59

QUESTION ID

S2-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 4 (x = 0 is extraneous)

GRADING NOTE Form required: the solution, with x = 0 identified as extraneous. Squaring gives $x^2 - 4x = 0$, so x = 0 or 4. x = 0 gives √1 = 1 but 0 − 1 = −1, so it fails; x = 4 gives 3 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q2/5

What are the solutions of $\sqrt{x} = -3$?

A. $x = 3$

B. $x = 9$

C. $x = -9$

D. There is no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q3/5

60

QUESTION ID

S2-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. There is no solution.

GRADING NOTE $\sqrt{x}$ is never negative, so no x works. Squaring would give $x = 9$, but $\sqrt{9} = 3$, not -3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q3/5

True or false?

Squaring both sides of an equation can introduce extraneous solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q4/5

61

QUESTION ID

S2-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Squaring can make a false equation true, for example $-3 = 3$ becomes $9 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q4**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q4/5

True or false?

$x = -4$ is a solution of $\sqrt{(x + 8)} = x + 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q5/5

62

QUESTION ID

S2-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{4} = 2$, but $-4 + 2 = -2$, so the equation is false at $x = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q5/5

SET 2 • HSN-RN.A.2

FORWARD 2

Add, subtract, and multiply complex numbers • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2

Add. Write the answer in the form a + bi.

(3 + 2i) + (1 − 5i)

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q1/5

64

QUESTION ID

S2-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4 − 3i

GRADING NOTE Form required: a + bi. Add real parts (3 + 1) and imaginary parts (2 − 5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q1**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q1/5

Multiply. Write the answer in the form a + bi.

$(2 + i)(3 - i)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q2/5

65

QUESTION ID

S2-F2-Q2

TYPE

Short answer

CORRECT ANSWER

7 + i

GRADING NOTE Form required: a + bi. $6 - 2i + 3i - i^2 = 6 + i + 1 = 7 + i$, because $i^2 = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q2**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q2/5

Simplify.

$i^2 \cdot i$

A. 1

B. $-i$

C. i

D. -1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q3/5

66

QUESTION ID

S2-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $-i$

GRADING NOTE $i^2 = -1$, so $i^2 \cdot i = -i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q3**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q3/5

True or false?

$(1 + i)^2 = 2i$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q4/5

67

QUESTION ID

S2-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(1 + i)^2 = 1 + 2i + i^2 = 1 + 2i - 1 = 2i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q4**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q4/5

True or false?

$i \cdot i = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q5/5

68

QUESTION ID

S2-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $i \cdot i = i^2 = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q5/5

SET 3 • HSN-RN.B.3 • NUMBER & QUANTITY

Prove that the sum of a rational number and an irrational number is irrational

MAIN	Algebra I main question	**HSN-RN.B.3**
BACKWARD 1	Classify numbers as rational or irrational	**8.NS.A.1**
BACKWARD 2	Write rational numbers as quotients of integers	**7.NS.A.2.b**
FORWARD 1	Find exact lengths and perimeters on the coordinate plane	**HSG-GPE.B.7**
FORWARD 2	Write numbers in the complex number system as a + bi	**HSN-CN.A.1**

SET 3 • HSN-RN.B.3

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3

Explain why 5 + √2 is irrational. (Suppose 5 + √2 = r for some rational number r.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q1/5

71

QUESTION ID

S3-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

5 + √2 is irrational.

GRADING NOTE

REQUIRED WORK: An argument like this: if 5 + √2 = r with r rational, then √2 = r − 5, a difference of rational numbers, which is rational. But √2 is irrational, so 5 + √2 cannot be rational.

Any equivalent form of the argument. The key step is that rational numbers are closed under subtraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q1/5

Explain why √3 − 4 is irrational. (Suppose √3 − 4 = r for some rational number r.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q2/5

72

QUESTION ID

S3-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

√3 − 4 is irrational.

GRADING NOTE

REQUIRED WORK: An argument like this: if √3 − 4 = r with r rational, then √3 = r + 4, a sum of rational numbers, which is rational. But √3 is irrational, so √3 − 4 cannot be rational.

Form required: the argument by contradiction. It must use the fact that a sum of rational numbers is rational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q2/5

Prove that if a is rational and x is irrational, then a + x is irrational.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q3/5

73

QUESTION ID

S3-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

a + x is irrational.

GRADING NOTE REQUIRED WORK: Suppose a + x = r with r rational. Then x = r − a, a difference of rational numbers, which is rational. That contradicts x being irrational, so a + x is irrational.

Form required: the general argument by contradiction, using closure of the rational numbers under subtraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q3/5

Which sum is irrational?

A. $\sqrt{2} + (-\sqrt{2})$

B. $0.25 + \frac{3}{4}$

C. $\sqrt{16} + 1.5$

D. $\sqrt{5} + 1.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q4/5

74

QUESTION ID

S3-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $\sqrt{5} + 1.5$

GRADING NOTE $\sqrt{5}$ is irrational and 1.5 is rational, so their sum is irrational. $\sqrt{16} + 1.5 = 5.5$, $0.25 + 0.75 = 1$, and $\sqrt{2} + (-\sqrt{2}) = 0$ are rational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q4/5

True or false?

The sum of two irrational numbers is always irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q5/5

75

QUESTION ID
S3-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $\sqrt{2} + (-\sqrt{2}) = 0$, which is rational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q5/5

SET 3 • HSN-RN.B.3

BACKWARD 1

Classify numbers as rational or irrational • Grade 8 • The Number System • 8.NS.A.1

Is √49 rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q1/5

77

QUESTION ID

S3-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Rational;

√49 = 7 = 7/1.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q1/5

The number 0.1010010001... continues with one more 0 before each 1. Is it rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q2/5

78

QUESTION ID

S3-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Irrational; its decimal never ends and never repeats.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q2/5

Which number is rational?

A. √2

B. √10

C. π

D. 0.2727... (the digits 27 repeat)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q3/5

79

QUESTION ID

S3-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0.2727... (the digits 27 repeat)

GRADING NOTE A repeating decimal is rational: 0.2727... = 27/99 = 3/11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

Every integer is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q4/5

80

QUESTION ID

S3-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Any integer n can be written as n/1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q4/5

True or false?

√20 is rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q5/5

81

QUESTION ID

S3-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 20 is not a perfect square, so √20 is irrational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • HSN-RN.B.3

BACKWARD 2

Write rational numbers as quotients of integers • Grade 7 • The Number System • 7.NS.A.2.b

Write 1.25 as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q1/5

83

QUESTION ID

S3-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5/4

GRADING NOTE Any equivalent form, such as 125/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q1/5

Write $-3\frac{1}{3}$ as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q2/5

84

QUESTION ID

S3-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−10/3

GRADING NOTE Any equivalent form, such as 10/(−3). $-3\frac{1}{3} = -(9/3 + 1/3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q2/5

Which quotient is equal to 0.6?

A. $\frac{6}{100}$

B. $\frac{5}{3}$

C. $\frac{3}{5}$

D. $\frac{1}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q3/5

85

QUESTION ID

S3-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3/5

GRADING NOTE $3 \div 5 = 0.6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q3/5

True or false?

$-7 = \frac{-7}{1}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q4/5

86

QUESTION ID

S3-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every integer is a quotient of itself and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q4/5

True or false?

$\frac{2}{0}$ is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q5/5

87

QUESTION ID

S3-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Division by 0 is undefined, so 2/0 is not a number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q5/5

SET 3 • HSN-RN.B.3

FORWARD 1

Find exact lengths and perimeters on the coordinate plane • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

Find the exact perimeter of the triangle with vertices (0, 0), (3, 0), and (0, 3).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q1/5

89

QUESTION ID
S3-F1-Q1

TYPE
Short answer

CORRECT ANSWER
6 + 3√2 units

GRADING NOTE Form required: exact form. The legs are 3 and 3, and the hypotenuse is √18 = 3√2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q1/5

A square has vertices (0, 0), (1, 1), (0, 2), and (−1, 1). Find its exact perimeter, and say whether it is rational or irrational.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q2/5

90

QUESTION ID
S3-F1-Q2

TYPE
Short answer

CORRECT ANSWER
4√2 units; irrational

GRADING NOTE Form required: exact form. Each side is $\sqrt{(1^2 + 1^2)} = \sqrt{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q2/5

What is the area of the triangle with vertices (0, 0), (4, 0), and (4, 3)?

A. 6

B. 12

C. 7

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q3/5

91

QUESTION ID
S3-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. 6

GRADING NOTE It is a right triangle with legs 4 and 3: $\frac{1}{2} \cdot 4 \cdot 3 = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q3**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q3/5

True or false?

The triangle with vertices (0, 0), (1, 0), and (0, 1) has an irrational perimeter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q4/5

92

QUESTION ID
S3-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The perimeter is 2 + √2, a rational number plus an irrational number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q4/5

True or false?

The triangle with vertices (0, 0), (6, 0), and (0, 8) has an irrational perimeter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q5/5

93

QUESTION ID
S3-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The sides are 6, 8, and 10, so the perimeter is 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSN-RN.B.3

FORWARD 2

Write numbers in the complex number system as a + bi • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1

Write $\sqrt{-49}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q1/5

95

QUESTION ID

S3-F2-Q1

TYPE

Short answer

CORRECT ANSWER

7i

GRADING NOTE Form required: bi. $\sqrt{-49} = \sqrt{49} \cdot \sqrt{-1} = 7i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q1**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q1/5

Name the real part and the imaginary part of $4 - 3i$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q2/5

96

QUESTION ID

S3-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Real part 4; imaginary part −3

GRADING NOTE Any equivalent wording. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q2**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q2/5

Which of these is a real number?

A. i

B. i^2

C. 2i

D. 1 + i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q3/5

97

QUESTION ID

S3-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. i^2

GRADING NOTE $i^2 = -1$, which is real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q3**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q3/5

True or false?

Every real number is a complex number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q4/5

98

QUESTION ID

S3-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A real number a is a + 0i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q4/5

True or false?

$i^2 = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q5/5

99

QUESTION ID

S3-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE i is defined so that $i^2 = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q5**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q5/5

SET 69 • HSN-RN.B.3 • NUMBER & QUANTITY

Prove that the product of a nonzero rational number and an irrational number is irrational

MAIN	Algebra I main question	**HSN-RN.B.3**
BACKWARD 1	Classify numbers as rational or irrational	**8.NS.A.1**
BACKWARD 2	Multiply and divide rational numbers	**7.NS.A.2.c**
FORWARD 1	Find exact lengths, perimeters, and areas with irrational side lengths	**HSG-GPE.B.7**
FORWARD 2	Factor sums of squares over the complex numbers	**HSN-CN.C.8**

SET 69 • HSN-RN.B.3

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3

Explain why $3\sqrt{7}$ is irrational. (Suppose $3\sqrt{7} = r$ for some rational number r.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q1/5

102

QUESTION ID

S69-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$3\sqrt{7}$ is irrational.

GRADING NOTE REQUIRED WORK: An argument like this: if $3\sqrt{7} = r$, then $\sqrt{7} = r \div 3$, a quotient of a rational number by a nonzero rational number, which is rational. But $\sqrt{7}$ is irrational, so $3\sqrt{7}$ is irrational.

Form required: the argument by contradiction. It must divide by the nonzero rational number 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q1/5

Prove that if a is a nonzero rational number and x is irrational, then ax is irrational.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q2/5

103

QUESTION ID

S69-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

ax is irrational.

GRADING NOTE REQUIRED WORK: Suppose ax = r with r rational. Since a ≠ 0, x = r ÷ a, a quotient of rational numbers with a nonzero divisor, which is rational. That contradicts x being irrational.

Form required: the general argument by contradiction, including where a ≠ 0 is used.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q2/5

Why does the statement "a rational number times an irrational number is irrational" need the rational number to be nonzero?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q3/5

104

QUESTION ID

S69-M-Q3

TYPE

Short answer

CORRECT ANSWER

If a = 0, then ax = 0, which is rational. The proof also divides by a, which needs a ≠ 0.

GRADING NOTE Any equivalent explanation that gives the counterexample 0 · x = 0 or the division by a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q3/5

Which product is irrational?

A. $\sqrt{5} \cdot \sqrt{5}$

B. $\sqrt{2} \cdot \sqrt{8}$

C. $0 \cdot \sqrt{5}$

D. $\frac{2}{3}\sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q4/5

105

QUESTION ID

S69-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 2/3√5

GRADING NOTE $\frac{2}{3}$ is a nonzero rational number and $\sqrt{5}$ is irrational. $0 \cdot \sqrt{5} = 0$, $\sqrt{5} \cdot \sqrt{5} = 5$, and $\sqrt{2} \cdot \sqrt{8} = 4$ are rational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q4/5

True or false?

The product of two irrational numbers is always irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q5/5

106

QUESTION ID
S69-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $\sqrt{2} \cdot \sqrt{2} = 2$, which is rational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q5/5

SET 69 • HSN-RN.B.3

BACKWARD 1

Classify numbers as rational or irrational • Grade 8 • The Number System • 8.NS.A.1

Is √64 rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q1/5

108

QUESTION ID

S69-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Rational: √64 = 8.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q1/5

The decimal 0.1010010001... has one more 0 between the 1s each time. Is it rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q2/5

109

QUESTION ID

S69-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Irrational: it never ends and never repeats.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q2/5

Which of these numbers is irrational?

A. $\sqrt{10}$

B. $-\frac{4}{7}$

C. $\sqrt{100}$

D. 0.333...

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q3/5

110

QUESTION ID

S69-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $\sqrt{10}$

GRADING NOTE 10 is not a perfect square. $\sqrt{100} = 10$, $0.333... = \frac{1}{3}$, and $-\frac{4}{7}$ are rational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q3/5

True or false?

Every terminating decimal is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q4/5

111

QUESTION ID

S69-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A terminating decimal is a fraction with a power of 10 in the denominator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q4/5

True or false?

$\pi = \frac{22}{7}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q5/5

112

QUESTION ID

S69-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE π is irrational; $\frac{22}{7} \approx 3.142857$ is only an approximation of $\pi \approx 3.141593$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q5/5

SET 69 • HSN-RN.B.3

BACKWARD 2

Multiply and divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find $(-\frac{2}{3}) \cdot \frac{9}{4}$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q1/5

114

QUESTION ID

S69-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−3/2

GRADING NOTE Any equivalent form, such as −1.5. −18/12 = −3/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q1/5

Find $\frac{5}{6} \div \left(-\frac{10}{3}\right)$.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q2/5

115

QUESTION ID

S69-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−1/4

GRADING NOTE Any equivalent form. $\frac{5}{6} \cdot \left(-\frac{3}{10}\right) = -\frac{15}{60}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q2/5

Which product is positive?

A. $(0.5)(-8)$

B. $0 \cdot 8$

C. $(-0.5)(-8)$

D. $(-0.5)(8)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q3/5

116

QUESTION ID

S69-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $(-0.5)(-8)$

GRADING NOTE A negative times a negative is positive: 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q3/5

True or false?

The quotient of two nonzero rational numbers is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q4/5

117

QUESTION ID

S69-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (a/b) ÷ (c/d) = ad/(bc), a quotient of integers with bc ≠ 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q4/5

True or false?

$\frac{3}{4} \div 0 = 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q5/5

118

QUESTION ID

S69-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Division by 0 is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q5/5

SET 69 • HSN-RN.B.3

FORWARD 1

Find exact lengths, perimeters, and areas with irrational side lengths • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

A square has vertices (0, 0), (2, 1), (1, 3), and (−1, 2). Find its exact side length and its area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q1/5

120

QUESTION ID
S69-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Side √5; area 5 square units

GRADING NOTE Form required: the exact side and the area. Each side is $\sqrt{(2^2 + 1^2)} = \sqrt{5}$, and $(\sqrt{5})^2 = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle has vertices (0, 0), (3, 3), (1, 5), and (−2, 2). Find its exact perimeter.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q2/5

121

QUESTION ID
S69-F1-Q2

TYPE
Short answer

CORRECT ANSWER
10√2 units

GRADING NOTE Form required: exact. The sides are √18 = 3√2 and √8 = 2√2, so the perimeter is 2(3√2 + 2√2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q2/5

What is the exact perimeter of the triangle with vertices (0, 0), (4, 0), and (0, 4)?

A. 4√2

B. 16

C. 8 + 2√2

D. 8 + 4√2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q3/5

122

QUESTION ID
S69-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. 8 + 4√2

GRADING NOTE The legs are 4 and 4, and the hypotenuse is √32 = 4√2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F1-Q3**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q3/5

True or false?

A square with side $\sqrt{5}$ has a rational area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q4/5

123

QUESTION ID
S69-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Its area is $(\sqrt{5})^2 = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q4/5

True or false?

A square with side √5 has a rational perimeter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q5/5

124

QUESTION ID
S69-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its perimeter is 4√5, a nonzero rational number times an irrational number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q5/5

SET 69 • HSN-RN.B.3

FORWARD 2

Factor sums of squares over the complex numbers • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 + 9$ over the complex numbers.

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q1/5

126

QUESTION ID
S69-F2-Q1

TYPE
Short answer

CORRECT ANSWER
(x + 3i)(x − 3i)

GRADING NOTE Form required: factored with complex numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand (x + 2i)(x − 2i).

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q2/5

127

QUESTION ID

S69-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 + 4$

GRADING NOTE Form required: expanded. $-(2i)^2 = -4i^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q2/5

Which is a factorization of $x^2 + 25$?

A. $(x + 5i)(x - 5i)$

B. $(x + 5)(x - 5)$

C. $(x + 5i)^2$

D. $(x - 5i)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q3/5

128

QUESTION ID

S69-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $(x + 5i)(x - 5i)$

GRADING NOTE $(x + 5i)(x - 5i) = x^2 - 25i^2 = x^2 + 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q3/5

True or false?

$x^2 + 16 = (x + 4i)(x - 4i)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q4/5

129

QUESTION ID

S69-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $-(4i)^2 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q4/5

True or false?

$x^2 + 4$ cannot be factored, even with complex numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q5/5

130

QUESTION ID

S69-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 + 4 = (x + 2i)(x - 2i)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q5/5

SET 70 • HSN-RN.B.3 • NUMBER & QUANTITY

Prove that sums and products of two rational numbers are rational

MAIN	Algebra I main question	**HSN-RN.B.3**
BACKWARD 1	Write quotients of integers as rational numbers	**7.NS.A.2.b**
BACKWARD 2	Divide fractions	**6.NS.A.1**
FORWARD 1	Show that sums and products of complex numbers are complex n…	**HSN-CN.A.2**
FORWARD 2	Show that rational expressions are closed under addition and mult…	**HSA-APR.D.7**

SET 70 • HSN-RN.B.3

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3

Explain why the sum of two rational numbers a/b and c/d (with integers a, b, c, d, and b ≠ 0, d ≠ 0) is rational.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q1
High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q1/5

QUESTION ID
S70-M-Q1

TYPE
Short answer

CORRECT ANSWER
a/b + c/d = (ad + bc)/(bd). ad + bc and bd are integers and bd ≠ 0, so the sum is a quotient of integers.

GRADING NOTE Any equivalent explanation that writes the sum as a quotient of integers with a nonzero denominator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q1
High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q1/5

Prove that the product of two rational numbers a/b and c/d (with integers a, b, c, d, and b ≠ 0, d ≠ 0) is rational.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q2/5

134

QUESTION ID

S70-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

The product is rational.

GRADING NOTE

REQUIRED WORK: (a/b)(c/d) = (ac)/(bd); ac and bd are integers, and bd ≠ 0 because b ≠ 0 and d ≠ 0, so the product is a quotient of integers.

Form required: the product written as a quotient of integers, with the reason the denominator is not 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q2/5

In the proof that (a/b)(c/d) is rational, why is bd ≠ 0?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q3/5

135

QUESTION ID

S70-M-Q3

TYPE

Short answer

CORRECT ANSWER

b ≠ 0 and d ≠ 0, and a product of two nonzero integers is not 0.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q3/5

Which statement shows that $(\frac{2}{5})(-\frac{15}{4})$ is rational?

A. It can be written as a decimal with a calculator.

B. It equals −30/20 = −3/2, a quotient of integers with a nonzero denominator.

C. It is negative.

D. Its first factor has an even numerator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q4/5

136

QUESTION ID

S70-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. It equals −30/20 = −3/2, a quotient of integers with a nonzero denominator.

GRADING NOTE Rational means a quotient of integers with a nonzero denominator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q4/5

True or false?

The difference of two rational numbers is always rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q5/5

137

QUESTION ID

S70-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE a/b − c/d = (ad − bc)/(bd), a quotient of integers with bd ≠ 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q5/5

SET 70 • HSN-RN.B.3

BACKWARD 1

Write quotients of integers as rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Write $-\frac{3}{4}$ as a quotient of integers in two different ways.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q1/5

139

QUESTION ID

S70-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−3/4 and 3/(−4)

GRADING NOTE Any two correct quotients, such as −6/8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is −12 ÷ 5 a rational number? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q2/5

140

QUESTION ID

S70-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. It is a quotient of integers with a nonzero divisor.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q2/5

Which quotient is undefined?

A. $-7 \div 2$

B. $0 \div 7$

C. $7 \div 0$

D. $7 \div (-1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q3/5

141

QUESTION ID

S70-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $7 \div 0$

GRADING NOTE Division by 0 is undefined; $0 \div 7 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q3/5

True or false?

For integers p and q with q ≠ 0,
−(p/q) = (−p)/q = p/(−q).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q4/5

142

QUESTION ID

S70-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each is the same negative number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q4/5

True or false?

Some integers are not rational numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q5/5

143

QUESTION ID

S70-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Every integer n equals n/1, so every integer is rational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q5/5

SET 70 • HSN-RN.B.3

BACKWARD 2

Divide fractions • Grade 6 • The Number System • 6.NS.A.1

Find $\frac{3}{4} \div \frac{1}{8}$.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q1/5

145

QUESTION ID

S70-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. $\frac{3}{4} \cdot 8 = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q1/5

Find $\frac{2}{3} \div \frac{5}{6}$.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q2/5

146

QUESTION ID

S70-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4/5

GRADING NOTE Any equivalent form. $\frac{2}{3} \cdot \frac{6}{5} = \frac{12}{15}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q2/5

Which number equals $\frac{5}{2} \div \frac{1}{4}$?

A. $\frac{5}{8}$

B. $\frac{1}{10}$

C. 10

D. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q3/5

147

QUESTION ID

S70-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 10

GRADING NOTE $\frac{5}{2} \cdot 4 = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q3/5

True or false?

$\frac{1}{2} \div \frac{1}{4} = 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q4/5

148

QUESTION ID

S70-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE There are two quarters in a half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q4/5

True or false?

$\frac{3}{5} \div 3 = \frac{9}{5}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q5/5

149

QUESTION ID

S70-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\frac{3}{5} \div 3 = \frac{1}{5}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q5/5

SET 70 • HSN-RN.B.3

FORWARD 1

Show that sums and products of complex numbers are complex numbers • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2

Add. Write the answer in the form a + bi.

(3 − 2i) + (−1 + 6i)

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q1/5

151

QUESTION ID

S70-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2 + 4i

GRADING NOTE Form required: a + bi. Add the real parts and the imaginary parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F1-Q1**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q1/5

Multiply. Write the answer in the form a + bi.

(2 + i)(3 + 2i)

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q2/5

152

QUESTION ID

S70-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4 + 7i

GRADING NOTE Form required: a + bi. 6 + 4i + 3i + 2i² = 6 + 7i − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F1-Q2**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q2/5

Which shows that the product of two complex numbers is a complex number?

A. $i^2 = 1$

B. $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$, and $ac - bd$ and $ad + bc$ are real numbers

C. $(a + bi)(c + di) = ac + bdi$

D. The product of two complex numbers is always real

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q3/5

153

QUESTION ID
S70-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B.
$(a + bi)(c + di) = (ac - bd) + (ad + bc)i$, and $ac - bd$ and $ad + bc$ are real numbers

GRADING NOTE Expanding and using $i^2 = -1$ gives a real part and an imaginary part.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q3/5

True or false?

The sum of two complex numbers is a complex number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q4/5

154

QUESTION ID

S70-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (a + bi) + (c + di) = (a + c) + (b + d)i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F1-Q4**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q4/5

True or false?

(1 + i) + (1 − i) is not a complex number because it has no i.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q5/5

155

QUESTION ID

S70-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The sum is 2 = 2 + 0i, a complex number (a real number).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q5/5

SET 70 • HSN-RN.B.3

FORWARD 2

Show that rational expressions are closed under addition and multiplication • High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Assume $x \neq 0$ and $x \neq -1$. Write $1 \div x + 2 \div (x + 1)$ as one fraction.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q1
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q1/5

157

QUESTION ID
S70-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$(3x + 1) \div (x(x + 1))$

GRADING NOTE Any equivalent form, such as $(3x + 1)/(x^2 + x)$. The common denominator is $x(x + 1)$: $(x + 1 + 2x) \div (x(x + 1))$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q1
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q1/5

Assume $x \neq 2$ and $x \neq -3$. Multiply and simplify:
$(x \div (x - 2)) \cdot ((x - 2) \div (x + 3))$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q2
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q2/5

158

QUESTION ID
S70-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$x \div (x + 3)$

GRADING NOTE Form required: simplified. The factor $x - 2$ cancels.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q2
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q2/5

Why is the sum of two rational expressions a rational expression?

A. Over a common denominator, the sum is a polynomial divided by a nonzero polynomial.

B. The denominators always cancel.

C. Rational expressions are always integers.

D. Any sum of fractions equals 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q3
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q3/5

159

QUESTION ID
S70-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. Over a common denominator, the sum is a polynomial divided by a nonzero polynomial.

GRADING NOTE This is the same argument as for rational numbers, with polynomials in place of integers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q3
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q3/5

True or false?

The quotient of two rational expressions, when the divisor is not 0, is a rational expression.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q4
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q4/5

160

QUESTION ID
S70-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Dividing multiplies by the reciprocal, which is also a rational expression.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q4
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q4/5

True or false?

$1 \div x + 1 \div y = 2 \div (x + y)$ for all nonzero x and y with $x + y \neq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q5
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q5/5

161

QUESTION ID
S70-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE For x = y = 1, the left side is 2 and the right side is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q5
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q5/5

SET 4 • HSN-Q.A.1 • NUMBER & QUANTITY

Use units to guide multistep solutions and to choose scales for graphs

MAIN	Algebra I main question	**HSN-Q.A.1**
BACKWARD 1	Convert measurement units with ratio reasoning	**6.RP.A.3.d**
BACKWARD 2	Compute unit rates with fractions	**7.RP.A.1**
FORWARD 1	Use density in a model	**HSG-MG.A.2**
FORWARD 2	Solve problems with velocity vectors — nearest related	**HSN-VM.A.3**

SET 4 • HSN-Q.A.1

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1

A car goes 32 miles on each gallon of gas, and gas costs $3.60 per gallon.

Use units to find the cost of gas for a 400-mile trip. Show how the units cancel.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q1/5

164

QUESTION ID

S4-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$45

GRADING NOTE REQUIRED WORK: A product of rates with units, such as 400 miles × (1 gallon ÷ 32 miles) × ($3.60 ÷ 1 gallon), with miles and gallons canceling to leave dollars.

Form required: the cost with the unit work shown. 400 ÷ 32 = 12.5 gallons, and 12.5 × $3.60 = $45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q1/5

Convert 72 kilometers per hour to meters per second.

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q2/5

165

QUESTION ID

S4-M-Q2

TYPE

Short answer

CORRECT ANSWER

20 meters per second

GRADING NOTE Any equivalent form. 72 km/h × 1,000 m/km ÷ 3,600 s/h = 20 m/s.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q2/5

The formula d = rt gives distance. A rate r is in meters per second and a time t is in minutes.

What must be done before multiplying?

A. Divide r by t.

B. Convert t to seconds (or r to meters per minute).

C. Nothing; the units already match.

D. Convert d to minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q3/5

166

QUESTION ID

S4-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Convert t to seconds (or r to meters per minute).

GRADING NOTE The time units must match: (m/s) × s = m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q3/5

A graph will show a town's population each year from 1990 to 2020. The populations range from 12,000 to 31,000.

Choose a scale for the vertical axis, and explain your choice.

Scale: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q4/5

167

QUESTION ID

S4-M-Q4

TYPE

Short answer

CORRECT ANSWER

For example, 0 to 35,000 in steps of 5,000, labeled "Population"; every value fits and the steps are equal.

GRADING NOTE Any equivalent form: any labeled scale with equal steps that includes 12,000 to 31,000 (starting at 10,000 instead of 0 is also correct if the break is shown or stated).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q4/5

True or false?

A speed of 5 miles per hour is about 7.3 feet per second.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q5/5

168

QUESTION ID

S4-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 5 × 5,280 ÷ 3,600 = 7.33... feet per second.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q5/5

SET 4 • HSN-Q.A.1

BACKWARD 1

Convert measurement units with ratio reasoning • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

How many inches are in 3.5 feet?

Inches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q1/5

170

QUESTION ID

S4-B1-Q1

TYPE

Short answer

CORRECT ANSWER

42

GRADING NOTE Any equivalent form. 3.5 × 12 = 42.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q1/5

How many kilograms are in 2,500 grams?

Kilograms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q2/5

171

QUESTION ID

S4-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE Any equivalent form. 2,500 ÷ 1,000 = 2.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q2/5

How many feet are in 4 yards?

A. 48

B. 36

C. $\frac{4}{3}$

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q3/5

172

QUESTION ID

S4-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 12

GRADING NOTE 1 yard = 3 feet, so 4 yards = 12 feet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q3/5

True or false?

1 hour = 3,600 seconds

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q4/5

173

QUESTION ID

S4-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 60 minutes × 60 seconds = 3,600 seconds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q4/5

True or false?

5 km = 500 m

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q5/5

174

QUESTION ID

S4-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1 km = 1,000 m, so 5 km = 5,000 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q5/5

SET 4 • HSN-Q.A.1

BACKWARD 2

Compute unit rates with fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A walker goes $\frac{3}{4}$ mile in $\frac{1}{4}$ hour.

How many miles per hour is this?

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q1/5

176

QUESTION ID

S4-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. (3/4) ÷ (1/4) = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q1/5

Half a pound of cheese costs $3.

What is the price per pound?

Price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q2/5

177

QUESTION ID

S4-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$6

GRADING NOTE Any equivalent form. $3 ÷ (1/2) = $6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q2/5

$\frac{2}{3}$ gallon of paint covers $\frac{1}{3}$ of a fence.

How many gallons cover the whole fence?

A. $\frac{1}{2}$

B. $\frac{2}{9}$

C. 2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q3/5

178

QUESTION ID

S4-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2

GRADING NOTE (2/3) ÷ (1/3) = 2 gallons per fence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q3/5

True or false?

Walking $\frac{1}{2}$ mile in $\frac{1}{6}$ hour is a speed of 3 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q4/5

179

QUESTION ID

S4-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (1/2) ÷ (1/6) = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q4/5

True or false?

Painting $\frac{1}{4}$ of a wall in $\frac{1}{2}$ hour is a rate of $\frac{1}{8}$ wall per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q5/5

180

QUESTION ID

S4-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (1/4) ÷ (1/2) = 1/2 wall per hour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • HSN-Q.A.1

FORWARD 1

Use density in a model • High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2

Copper has a density of 8.96 grams per cubic centimeter.

What is the mass of a 5 cubic-centimeter piece of copper?

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q1/5

182

QUESTION ID
S4-F1-Q1

TYPE
Short answer

CORRECT ANSWER
44.8 grams

GRADING NOTE Any equivalent form. Mass = density × volume = 8.96 × 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q1/5

A county has 240,000 people living in 600 square miles.

What is its population density?

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q2/5

183

QUESTION ID

S4-F1-Q2

TYPE

Short answer

CORRECT ANSWER

400 people per square mile

GRADING NOTE Any equivalent form, with units. 240,000 ÷ 600 = 400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q2/5

A 2 cm by 3 cm by 4 cm block has a mass of 60 grams.

What is its density?

A. 0.4 gram per cubic centimeter

B. 60 grams

C. 2.5 grams per cubic centimeter

D. 24 grams per cubic centimeter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q3/5

184

QUESTION ID

S4-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2.5 grams per cubic centimeter

GRADING NOTE The volume is 24 cubic centimeters, and 60 ÷ 24 = 2.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q3/5

True or false?

Population density is the number of people divided by the area they live on.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q4/5

185

QUESTION ID

S4-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Density is a quantity per unit of area (or volume).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q4/5

True or false?

An object with a volume of 10 cubic centimeters and a density of 3 grams per cubic centimeter has a mass of 3.3 grams.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q5/5

186

QUESTION ID

S4-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Mass = 3 × 10 = 30 grams.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSN-Q.A.1

FORWARD 2

Solve problems with velocity vectors • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Precalculus standard. Connection: Velocity vectors are rate quantities with units, like the unit rates of the MAIN set, with a direction added.

A boat heads east at 12 km/h across a river whose current flows north at 5 km/h.

What is the boat's resulting speed?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1
High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q1/5 • NEAREST RELATED

188

QUESTION ID
S4-F2-Q1

TYPE
Short answer

CORRECT ANSWER
13 km/h

GRADING NOTE Any equivalent form. The velocities are perpendicular, so the speed is $\sqrt{(12^2 + 5^2)} = 13$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1
High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q1/5 • NEAREST RELATED

A plane points north at 300 km/h, and the wind blows west at 40 km/h.

Write its velocity as (east part, north part) in km/h.

Velocity: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2
High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q2/5 • NEAREST RELATED

189

QUESTION ID
S4-F2-Q2

TYPE
Short answer

CORRECT ANSWER
(−40, 300)

GRADING NOTE Form required: (east part, north part). West is the negative east direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2
High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q2/5 • NEAREST RELATED

Two velocities, (3, 4) and (5, −1), are added.

What is the result?

A. (8, 5)

B. (8, 3)

C. (15, −4)

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3
High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q3/5 • NEAREST RELATED

190

QUESTION ID
S4-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. (8, 3)

GRADING NOTE Add the parts: (3 + 5, 4 + (−1)) = (8, 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3
High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

An object with velocity (6, 8) meters per second has a speed of 10 meters per second.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S4-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q4/5 • NEAREST RELATED

191

QUESTION ID

S4-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\sqrt{(6^2 + 8^2)} = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S4-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

Adding the velocities (3, 0) and (0, 4) gives a speed of 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5
High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q5/5 • NEAREST RELATED

192

QUESTION ID
S4-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The sum is (3, 4), whose speed is √(9 + 16) = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5
High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 5 • HSN-Q.A.2 • NUMBER & QUANTITY

Define appropriate quantities for descriptive modeling

MAIN	Algebra I main question	**HSN-Q.A.2**
BACKWARD 1	Use variables to represent quantities in a situation	**6.EE.B.6**
BACKWARD 2	Write and solve px + q = r from a word problem	**7.EE.B.4.a**
FORWARD 1	Use geometric quantities to make design decisions	**HSG-MG.A.3**
FORWARD 2	Recognize surveys, experiments, and observational studies and th…	**HSS-IC.B.3**

SET 5 • HSN-Q.A.2

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2

A school wants to model the total cost of a field trip by bus.

Name the quantities the model needs, with a unit for each.

Quantities: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q1/5

195

QUESTION ID

S5-M-Q1

TYPE

Short answer

CORRECT ANSWER

The number of students (students), the students each bus holds (students per bus), the cost of each bus (dollars per bus), and the admission price (dollars per student).

GRADING NOTE Any equivalent form: any complete, sensible set of quantities, each with a unit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q1/5

A club rents a room for $300 and pays $40 for each member who attends. Its cost model is C = 40n + 300.

Define C and n, with units.

Definitions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q2/5

196

QUESTION ID

S5-M-Q2

TYPE

Short answer

CORRECT ANSWER

C is the total cost in dollars; n is the number of members who attend.

GRADING NOTE Any equivalent wording, with both units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q2/5

Which quantity best describes how fast a phone battery runs down?

A. Number of apps on the phone

B. Percent of charge used per hour

C. Width of the screen in inches

D. Mass of the phone in grams

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q3/5

197

QUESTION ID

S5-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Percent of charge used per hour

GRADING NOTE A rate of drain needs an amount of charge per unit of time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q3/5

Two towns have different populations and different areas.

Define one quantity, with units, that compares how crowded they are.

Quantity: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q4/5

198

QUESTION ID

S5-M-Q4

TYPE

Short answer

CORRECT ANSWER

Population density: people per square mile (or per square kilometer).

GRADING NOTE Any equivalent quantity that is people per unit of area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q4/5

True or false?

Kilowatt-hours per month is a sensible quantity for a model of a household's electricity use.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q5/5

199

QUESTION ID

S5-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It measures energy used per unit of time, which is what an electricity bill is based on.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q5/5

SET 5 • HSN-Q.A.2

BACKWARD 1

Use variables to represent quantities in a situation • Grade 6 • Expressions & Equations • 6.EE.B.6

Tickets cost $8 each.

Write an expression for the cost of n tickets.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q1/5

201

QUESTION ID

S5-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8n

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A worker earns $15 per hour plus a $20 bonus.

Write an expression for the pay for h hours.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q2/5

202

QUESTION ID

S5-B1-Q2

TYPE

Short answer

CORRECT ANSWER

15h + 20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q2/5

A typist types 30 words per minute.

Which expression gives the number of words typed in m minutes?

A. $\frac{m}{30}$

B. $m + 30$

C. $30m$

D. $\frac{30}{m}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q3/5

203

QUESTION ID

S5-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 30m

GRADING NOTE Words = rate × time = 30m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q3/5

True or false?

If w is a number of weeks, 7w is the number of days.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q4/5

204

QUESTION ID

S5-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each week has 7 days.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q4/5

True or false?

If d is a number of dollars, d + 5 is 5 times as many dollars.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q5/5

205

QUESTION ID

S5-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 5 times as many is 5d; d + 5 is 5 more.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • HSN-Q.A.2

BACKWARD 2

Write and solve px + q = r from a word problem • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A gym charges a $25 fee plus $10 per class. Tom paid $95.

How many classes did he take?

Classes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q1/5

207

QUESTION ID

S5-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form. 25 + 10c = 95 gives c = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q1/5

A phone costs $50 plus $15 per month. The total so far is $140.

For how many months has it been paid?

Months: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q2/5

208

QUESTION ID

S5-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. 50 + 15m = 140 gives m = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q2/5

A plumber charges a $12 fee plus r dollars per hour. A 3-hour job costs $57.

Which equation fits?

A. 3r + 12 = 57

B. 3(r + 12) = 57

C. 12r + 3 = 57

D. r + 3 + 12 = 57

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q3/5

209

QUESTION ID

S5-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3r + 12 = 57

GRADING NOTE 3 hours at r dollars is 3r, plus the $12 fee.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q3/5

True or false?

x = 6 is the solution of 4x + 6 = 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q4/5

210

QUESTION ID

S5-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4 \cdot 6 + 6 = 30$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q4/5

True or false?

$x = 3$ is the solution of $2x - 5 = 11$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q5/5

211

QUESTION ID

S5-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2 \cdot 3 - 5 = 1$; the solution is $x = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q5/5

SET 5 • HSN-Q.A.2

FORWARD 1

Use geometric quantities to make design decisions • High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A rectangular garden needs an area of at least 200 square feet and can use at most 60 feet of fence.

Does a 10 ft by 20 ft garden meet both requirements? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q1/5

213

QUESTION ID

S5-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. Its area is 200 square feet and its perimeter is 60 feet.

GRADING NOTE Any equivalent wording. Both checks are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q1**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q1/5

A box with a square base 10 cm on each side must hold 1,000 cubic centimeters.

How tall must it be?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q2/5

214

QUESTION ID

S5-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10 cm

GRADING NOTE Any equivalent form. 10 × 10 × h = 1,000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q2/5

A rectangular pen must have an area of 100 square meters.

Which dimensions use the least fence?

A. 5 m by 20 m

B. 2 m by 50 m

C. 10 m by 10 m

D. 4 m by 25 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q3/5

215

QUESTION ID

S5-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 10 m by 10 m

GRADING NOTE The perimeters are 40, 50, 58, and 104 meters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q3/5

True or false?

A 6 by 6 square has more area than a 4 by 8 rectangle, even though their perimeters are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q4/5

216

QUESTION ID

S5-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 36 > 32, and both perimeters are 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q4/5

True or false?

Of all rectangles with a perimeter of 20, the 1 by 9 rectangle has the greatest area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q5/5

217

QUESTION ID

S5-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its area is 9; the 5 by 5 square has area 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSN-Q.A.2

FORWARD 2

Recognize surveys, experiments, and observational studies and the variables they measure • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Defining the quantities a model needs leads to deciding which variables a study measures and how they are collected.

A researcher randomly assigns 40 plants to two fertilizers and measures each plant's growth in centimeters.

What kind of study is this, and what is the response variable (with units)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 5 • FORWARD 2 • Q1/5 • NEAREST RELATED

219

QUESTION ID
S5-F2-Q1

TYPE
Short answer

CORRECT ANSWER
An experiment; the response is growth in centimeters.

GRADING NOTE Any equivalent wording. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 5 • FORWARD 2 • Q1/5 • NEAREST RELATED

A survey asks 200 randomly chosen students how many hours they sleep on a school night.

What kind of study is this, and what variable does it measure?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 5 • FORWARD 2 • Q2/5 • NEAREST RELATED

220

QUESTION ID
S5-F2-Q2

TYPE
Short answer

CORRECT ANSWER
A survey (sample survey); hours of sleep per school night.

GRADING NOTE Any equivalent wording. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 5 • FORWARD 2 • Q2/5 • NEAREST RELATED

Which kind of study can show cause and effect?

A. An observational study

B. A poll of volunteers

C. A randomized experiment

D. A survey

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 5 • FORWARD 2 • Q3/5 • NEAREST RELATED

221

QUESTION ID

S5-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A randomized experiment

GRADING NOTE Random assignment tends to balance other variables between the groups, so a difference can be credited to the treatment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 5 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

In an observational study, the researchers do not assign treatments.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 5 • FORWARD 2 • Q4/5 • NEAREST RELATED

222

QUESTION ID

S5-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They observe groups that already exist.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 5 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A survey of volunteers who answer an online poll is a random sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 5 • FORWARD 2 • Q5/5 • NEAREST RELATED

223

QUESTION ID

S5-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Volunteers choose themselves, so the sample is not random.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 5 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 6 • HSN-Q.A.3 • NUMBER & QUANTITY

Choose a level of accuracy that fits the limits of measurement

MAIN	Algebra I main question	**HSN-Q.A.3**
BACKWARD 1	Estimate to check whether an answer is reasonable	**7.EE.B.3**
BACKWARD 2	Round decimals to any place	**5.NBT.A.4**
FORWARD 1	Use volume formulas and report results to a sensible accuracy	**HSG-GMD.A.3**
FORWARD 2	Use the margin of error of a sample survey — nearest related	**HSS-IC.B.4**

SET 6 • HSN-Q.A.3

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3

A ruler measures to the nearest 0.1 cm. A rectangle measures 4.3 cm by 2.7 cm, and a calculator gives its area as 11.61 square cm.

Which report of the area best fits the accuracy of the measurements?

A. 11.610 square cm

B. 11.61 square cm

C. About 10 square cm

D. About 12 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q1/5

226

QUESTION ID

S6-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. About 12 square cm

GRADING NOTE Each length could be off by 0.05 cm, so the true area is between about 11.3 and 12.0 square cm; the hundredths digit is not known.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q1/5

A stopwatch reads to the nearest 0.1 second. A runner covers 100 m in 12.4 seconds, and a calculator gives 8.0645161 meters per second.

Report the speed to a sensible accuracy.

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q2/5

227

QUESTION ID

S6-M-Q2

TYPE

Short answer

CORRECT ANSWER

About 8.1 meters per second

GRADING NOTE Form required: a rounded value. 8.1 or 8.06 m/s is accepted; the true speed is between about 8.03 and 8.10 m/s, so more digits are not meaningful.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q2/5

A tape measure is marked in centimeters.

Which is the most sensible report of a person's height measured with it?

A. 167.3482 cm

B. 170.00 cm

C. 2 m

D. 167 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q3/5

228

QUESTION ID

S6-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 167 cm

GRADING NOTE The tape measures to the nearest centimeter, so report whole centimeters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q3/5

A bridge is 1,250 m long, measured to the nearest 10 m.

Between which two lengths is its true length?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q4/5

229

QUESTION ID

S6-M-Q4

TYPE

Short answer

CORRECT ANSWER

Between 1,245 m and 1,255 m

GRADING NOTE Any equivalent form. Measuring to the nearest 10 m allows an error of up to 5 m either way.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q4/5

True or false?

Giving more decimal places always makes a measured answer more accurate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q5/5

230

QUESTION ID

S6-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A result cannot be more accurate than the measurements it comes from.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q5/5

SET 6 • HSN-Q.A.3

BACKWARD 1

Estimate to check whether an answer is reasonable • Grade 7 • Expressions & Equations • 7.EE.B.3

Estimate.

4.9×21.2

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q1/5

232

QUESTION ID

S6-B1-Q1

TYPE

Short answer

CORRECT ANSWER

About 100

GRADING NOTE Any equivalent estimate near 100: $5 \times 20 = 100$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q1/5

Is 38.4 ÷ 4.8 = 80 reasonable? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q2/5

233

QUESTION ID

S6-B1-Q2

TYPE

Short answer

CORRECT ANSWER

No. 40 ÷ 5 = 8, so the answer should be about 8.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q2/5

Which is the best estimate of 19.8% of 503?

A. 100

B. 50

C. 1,000

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q3/5

234

QUESTION ID

S6-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 100

GRADING NOTE About 20% of 500 is 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q3/5

True or false?

$29.95 × 4 is about $120.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q4/5

235

QUESTION ID

S6-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $30 × 4 = $120.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q4/5

True or false?

0.48 × 61 is about 300.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q5/5

236

QUESTION ID

S6-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE About 0.5 × 60 = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q5/5

SET 6 • HSN-Q.A.3

BACKWARD 2

Round decimals to any place • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round 8.0645 to the nearest hundredth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q1/5

238

QUESTION ID

S6-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8.06

GRADING NOTE Form required: hundredths. The thousandths digit is 4, so round down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q1/5

Round 11.61 to the nearest whole number.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q2/5

239

QUESTION ID

S6-B2-Q2

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Form required: a whole number. The tenths digit is 6, so round up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q2/5

Round 0.0357 to the nearest thousandth.

A. 0.035

B. 0.0357

C. 0.036

D. 0.04

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q3/5

240

QUESTION ID

S6-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 0.036

GRADING NOTE The ten-thousandths digit is 7, so round up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q3/5

True or false?

3.149 rounded to the nearest tenth is 3.1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q4/5

241

QUESTION ID

S6-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The hundredths digit is 4, so round down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q4/5

True or false?

6.95 rounded to the nearest tenth is 6.9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q5/5

242

QUESTION ID

S6-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The hundredths digit is 5, so round up to 7.0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • HSN-Q.A.3

FORWARD 1

Use volume formulas and report results to a sensible accuracy • High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A cylinder has a radius of 3.0 cm and a height of 10.0 cm.

Find its volume to the nearest cubic centimeter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q1/5

244

QUESTION ID

S6-F1-Q1

TYPE

Short answer

CORRECT ANSWER

283 cubic centimeters

GRADING NOTE Form required: rounded to the nearest whole number. $V = \pi \cdot 3^2 \cdot 10 = 90\pi \approx 282.7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q1/5

A sphere has a radius of 5 m.

Find its volume exactly in terms of π, then to the nearest cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q2/5

245

QUESTION ID

S6-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$\frac{500}{3}\pi$ cubic meters, about 524 cubic meters

GRADING NOTE Form required: both the exact and the rounded value. V = (4/3)π · 125.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q2/5

A cone has a radius of 6 and a height of 4.

What is its volume?

A. 144π

B. 96π

C. 48π

D. 16π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q3/5

246

QUESTION ID

S6-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 48π

GRADING NOTE $V = (1/3)\pi \cdot 6^2 \cdot 4 = 48\pi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q3/5

True or false?

A cylinder with radius 2 cm and height 2 cm has a volume of 8π cm³, which is about 25 cm³ to the nearest cubic centimeter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q4/5

247

QUESTION ID
S6-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $V = \pi \cdot 2^2 \cdot 2 = 8\pi \approx 25.1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q4/5

True or false?

Doubling the radius of a sphere doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q5/5

248

QUESTION ID

S6-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Volume depends on r^3, so it is multiplied by 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q5/5

SET 6 • HSN-Q.A.3

FORWARD 2

Use the margin of error of a sample survey • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: The accuracy a measurement supports leads to the accuracy of an estimate from a sample, given by its margin of error.

A poll reports that 54% of voters support a plan, with a margin of error of 3 percentage points.

What interval of values for the true percent does this give?

Interval: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q1/5 • NEAREST RELATED

250

QUESTION ID

S6-F2-Q1

TYPE

Short answer

CORRECT ANSWER

51% to 57%

GRADING NOTE Any equivalent form. 54% − 3% to 54% + 3%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F2-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q1/5 • NEAREST RELATED

A poll reports 54% support with a margin of error of 3 percentage points.

Do the results suggest that a majority supports the plan? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q2/5 • NEAREST RELATED

251

QUESTION ID

S6-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. Every value in the interval, 51% to 57%, is above 50%.

GRADING NOTE Any equivalent explanation that uses the interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q2/5 • NEAREST RELATED

A poll reports 48% support with a margin of error of 4 percentage points.

Which statement fits?

A. The poll must be wrong.

B. Fewer than half support the plan, for certain.

C. The true percent could be above or below 50%.

D. The true percent is exactly 48%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q3/5 • NEAREST RELATED

252

QUESTION ID

S6-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. The true percent could be above or below 50%.

GRADING NOTE The interval 44% to 52% includes values on both sides of 50%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

Larger random samples tend to have smaller margins of error.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q4/5 • NEAREST RELATED

253

QUESTION ID

S6-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE More data gives less variation from sample to sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A margin of error of 3 percentage points means the poll is off by exactly 3 points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q5/5 • NEAREST RELATED

254

QUESTION ID
S6-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It gives a range of plausible values, not the exact error.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q5/5 • NEAREST RELATED

Interpret the terms, factors, and coefficients of an expression in context

MAIN	Algebra I main question	**HSA-SSE.A.1.a**
BACKWARD 1	Identify the parts of an expression	**6.EE.A.2.b**
BACKWARD 2	Rewrite an expression to show how quantities are related	**7.EE.A.2**
FORWARD 1	Rewrite a rational expression and interpret its parts	**HSA-APR.D.6**
FORWARD 2	Compose functions and interpret the composite in context	**HSF-BF.A.1.c**

SET 7 • HSA-SSE.A.1.a

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

A gym charges C = 35 + 20m dollars for m months of membership.

What do 35 and 20 represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q1/5

257

QUESTION ID

S7-M-Q1

TYPE

Short answer

CORRECT ANSWER

35 is a one-time joining fee in dollars; 20 is the cost in dollars for each month.

GRADING NOTE Any equivalent wording. Both meanings are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q1/5

The expression 1.08p gives the total cost, including sales tax, of an item with price p.

What does the factor 1.08 tell you?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q2/5

258

QUESTION ID

S7-M-Q2

TYPE

Short answer

CORRECT ANSWER

The total is 108% of the price: the price plus 8% tax (1.08p = p + 0.08p).

GRADING NOTE Any equivalent wording that names the 8% tax.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q2/5

The perimeter of a rectangle is 2(l + w).

What does the factor (l + w) represent?

A. The area

B. The length of one side

C. Half the perimeter: one length plus one width

D. Twice the perimeter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q3/5

259

QUESTION ID

S7-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Half the perimeter: one length plus one width

GRADING NOTE 2(l + w) is two copies of (l + w), so (l + w) is half of the perimeter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q3/5

For the expression $3x^2 - 5x + 7$, name its terms, the coefficient of x, and the constant term.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q4/5

260

QUESTION ID

S7-M-Q4

TYPE

Short answer

CORRECT ANSWER

Terms: $3x^2$, $-5x$, and 7; the coefficient of x is -5; the constant term is 7.

GRADING NOTE Any equivalent form. All three parts are required. The sign belongs to the term: the x-term is $-5x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q4/5

True or false?

A car's value is $500(0.9)^t$ dollars after t years. The factor 0.9 means the car loses 90% of its value each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q5/5

261

QUESTION ID
S7-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Multiplying by 0.9 keeps 90% of the value each year, so the car loses 10% each year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 1

Identify the parts of an expression • Grade 6 • Expressions & Equations • 6.EE.A.2.b

How many terms does 4a + 3b − 2 have?

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q1/5

263

QUESTION ID

S7-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. The terms are 4a, 3b, and −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q1/5

What is the coefficient of y in 7y − 2?

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q2/5

264

QUESTION ID

S7-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form. The coefficient is the number multiplying y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q2/5

Which description fits 5(x + 2)?

A. A product of two factors, 5 and (x + 2)

B. A sum of three terms

C. A single term with coefficient 2

D. A quotient of two numbers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q3/5

265

QUESTION ID

S7-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A product of two factors, 5 and (x + 2)

GRADING NOTE 5(x + 2) means 5 times (x + 2), a product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q3/5

True or false?

In 8 + 2n, the term 2n has a coefficient of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q4/5

266

QUESTION ID

S7-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $2n = 2 \cdot n$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q4/5

True or false?

The terms of $6(m - 1)$ are 6 and m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q5/5

267

QUESTION ID

S7-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $6(m - 1)$ is a product with the factors 6 and $(m - 1)$; expanded, its terms are $6m$ and -6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 2

Rewrite an expression to show how quantities are related • Grade 7 • Expressions & Equations • 7.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite a + 0.05a as a single term.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q1/5

269

QUESTION ID

S7-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1.05a

GRADING NOTE Form required: a single term. a + 0.05a = (1 + 0.05)a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A price p is lowered by 20%.

Write the new price as a product.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q2/5

270

QUESTION ID

S7-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.8p

GRADING NOTE Form required: a product. $p - 0.2p = 0.8p$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q2/5

Which expression shows a 15% increase of n?

A. 1.15n

B. 15n

C. n + 15

D. 0.15n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q3/5

271

QUESTION ID

S7-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1.15n

GRADING NOTE n + 0.15n = 1.15n.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q3/5

True or false?

$p - 0.25p = 0.75p$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q4/5

272

QUESTION ID

S7-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $p - 0.25p = (1 - 0.25)p$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q4/5

True or false?

$x + 0.3x = 0.3x^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q5/5

273

QUESTION ID

S7-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x + 0.3x = 1.3x$; adding like terms does not multiply them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • HSA-SSE.A.1.a

FORWARD 1

Rewrite a rational expression and interpret its parts • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

ANSWER KEY Reduced copy of the question page, with its answer

A company's average cost per item for n items is (500 + 3n) ÷ n dollars.

Rewrite it as a sum of two terms, and say what each term means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q1/5

275

QUESTION ID
S7-F1-Q1

TYPE
Short answer

CORRECT ANSWER
3 + 500/n: $3 is the cost of making each item, and 500/n is each item's share of a $500 fixed cost.

GRADING NOTE Form required: 3 + 500/n (in either order). Both meanings are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Assume x ≠ 0. Rewrite (2x + 5) ÷ x in the form a + b/x.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q2/5

276

QUESTION ID

S7-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2 + 5/x

GRADING NOTE Form required: a + b/x. (2x + 5)/x = 2x/x + 5/x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q2/5

Assume $x \neq 0$. Which expression is equal to $(x^2 + 3x) \div x$?

A. $x^2 + 3$

B. $x + 3x$

C. $x + 3$

D. $4x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q3/5

277

QUESTION ID
S7-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. x + 3

GRADING NOTE Divide each term by x: $x^2/x + 3x/x = x + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q3/5

True or false?

$(6x + 4) \div 2 = 3x + 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q4/5

278

QUESTION ID

S7-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Divide each term by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q4/5

True or false?

Assume $x \neq 0$. $(x + 6) \div x = 7$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q5/5

279

QUESTION ID
S7-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $(x + 6)/x = 1 + 6/x$, which depends on x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q5/5

SET 7 • HSA-SSE.A.1.a

FORWARD 2

Compose functions and interpret the composite in context • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c

T(h) = 20 + 5h is a temperature in °C after h hours, and h(t) = t ÷ 60 converts t minutes to hours.

Write T(h(t)) in simplest form, and say what it gives.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q1/5

281

QUESTION ID

S7-F2-Q1

TYPE

Short answer

CORRECT ANSWER

T(h(t)) = 20 + t/12: the temperature after t minutes.

GRADING NOTE Form required: simplest form, 20 + t/12 (or t/12 + 20); 20 + 5t/60 is equivalent but not simplified. The meaning is required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x + 3$ and $g(x) = 2x$.

Find $f(g(4))$.

$f(g(4)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q2/5

282

QUESTION ID

S7-F2-Q2

TYPE

Short answer

CORRECT ANSWER

11

GRADING NOTE Any equivalent form. g(4) = 8, and f(8) = 11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q2/5

$f(x) = x^2$ and $g(x) = x - 1$.

Which expression is f(g(x))?

A. $x - 1^2$

B. $x^2 - 1$

C. $x^2 - x$

D. $(x - 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q3/5

283

QUESTION ID

S7-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $(x - 1)^2$

GRADING NOTE Put g(x) into f: $f(x - 1) = (x - 1)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q3/5

True or false?

If f(x) = 3x and g(x) = x + 2, then g(f(1)) = 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q4/5

284

QUESTION ID

S7-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE f(1) = 3, and g(3) = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q4/5

True or false?

f(g(x)) is always equal to g(f(x)).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q5/5

285

QUESTION ID

S7-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE For $f(x) = x^2$ and $g(x) = x - 1$, $f(g(2)) = 1$ but $g(f(2)) = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q5/5

Interpret complicated expressions by viewing one or more of their parts as a single quantity

MAIN	Algebra I main question	**HSA-SSE.A.1.b**
BACKWARD 1	Apply the distributive property to write equivalent expressions	**6.EE.A.3**
BACKWARD 2	Evaluate expressions for given values	**6.EE.A.2.c**
FORWARD 1	Use the formula for the sum of a finite geometric series	**HSA-SSE.B.4**
FORWARD 2	Solve exponential growth models with logarithms	**HSF-LE.A.4**

SET 8 • HSA-SSE.A.1.b

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

An investment of P dollars is worth $P(1 + r)^n$ dollars after n years, where $r > 0$.

Treating $(1 + r)^n$ as one quantity, explain why the investment grows.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q1/5

288

QUESTION ID

S8-M-Q1

TYPE

Short answer

CORRECT ANSWER

Since $r > 0$, $1 + r > 1$, so $(1 + r)^n > 1$ for $n > 0$. P times a number greater than 1 is greater than P.

GRADING NOTE Any equivalent explanation that treats $(1 + r)^n$ as one factor greater than 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q1/5

For which value of x is $10 - (x - 3)^2$ greatest, and what is that greatest value? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q2/5

289

QUESTION ID

S8-M-Q2

TYPE

Short answer

CORRECT ANSWER

At x = 3 the value is 10. $(x - 3)^2$ is never negative and is 0 only at x = 3, so subtracting it gives at most 10.

GRADING NOTE Form required: the x-value, the greatest value, and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q2/5

Which expression is never greater than 4, for every real number x?

A. $(x + 1)^2 - 4$

B. $4 - (x + 1)^2$

C. $4(x + 1)^2$

D. $4 + (x + 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q3/5

290

QUESTION ID

S8-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $4 - (x + 1)^2$

GRADING NOTE $(x + 1)^2 \geq 0$, so $4 - (x + 1)^2 \leq 4$. The other expressions can be larger than 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A square picture with side s is surrounded by a frame 2 units wide on every side. The frame's area is $(s + 4)^2 - s^2$.

What do $(s + 4)^2$ and s^2 represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q4/5

291

QUESTION ID
S8-M-Q4

TYPE
Short answer

CORRECT ANSWER
$(s + 4)^2$ is the area of the whole framed square (side s + 2 + 2); s^2 is the area of the picture. Their difference is the frame.

GRADING NOTE Any equivalent wording. Both meanings are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q4/5

True or false?

For every real number x, $5 + 3(x - 2)^2 \geq 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q5/5

292

QUESTION ID
S8-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $3(x - 2)^2$ is never negative, so adding it to 5 gives at least 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 1

Apply the distributive property to write equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Expand.

3(2x + 5)

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q1/5

294

QUESTION ID

S8-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6x + 15

GRADING NOTE Form required: expanded. Multiply each term by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q1/5

Factor using the greatest common factor.

12y + 18

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q2/5

295

QUESTION ID

S8-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6(2y + 3)

GRADING NOTE Form required: factored with the greatest common factor 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which expression is equal to 4(a − 3) + 2a?

A. 6a + 12

B. 6a − 12

C. 6a − 3

D. 4a − 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q3/5

296

QUESTION ID

S8-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6a − 12

GRADING NOTE 4a − 12 + 2a = 6a − 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q3/5

True or false?

$2(x + y) = 2x + 2y$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q4/5

297

QUESTION ID

S8-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The 2 multiplies both terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q4/5

True or false?

$3(x + 4) = 3x + 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q5/5

298

QUESTION ID

S8-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3(x + 4) = 3x + 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 2

Evaluate expressions for given values • Grade 6 • Expressions & Equations • 6.EE.A.2.c

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $(x - 3)^2$ when $x = 7$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q1/5

300

QUESTION ID

S8-B2-Q1

TYPE

Short answer

CORRECT ANSWER

16

GRADING NOTE Any equivalent form. $(7 - 3)^2 = 4^2 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q1/5

Evaluate $100(1.05)^2$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q2/5

301

QUESTION ID

S8-B2-Q2

TYPE

Short answer

CORRECT ANSWER

110.25

GRADING NOTE Any equivalent form. $1.05^2 = 1.1025$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $10 - (x - 3)^2$ when $x = 1$.

A. 8

B. -6

C. 14

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q3/5

302

QUESTION ID

S8-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 6

GRADING NOTE $(1 - 3)^2 = 4$, and $10 - 4 = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q3/5

True or false?

When $x = 3$, $(x - 3)^2 = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q4/5

303

QUESTION ID

S8-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $3 - 3 = 0$, and $0^2 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q4/5

True or false?

When $x = -2$, $x^2 = -4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q5/5

304

QUESTION ID

S8-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(-2)^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q5/5

SET 8 • HSA-SSE.A.1.b

FORWARD 1

Use the formula for the sum of a finite geometric series • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Find the sum of the first 6 terms of 3 + 6 + 12 + ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q1/5

306

QUESTION ID

S8-F1-Q1

TYPE

Short answer

CORRECT ANSWER

189

GRADING NOTE Any equivalent form. $S = 3(2^6 - 1)/(2 - 1) = 3 \cdot 63 = 189$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q1/5

Use $S = a(1 - r^n) \div (1 - r)$ to find $2 + 1 + 0.5 + 0.25 + 0.125$.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q2/5

307

QUESTION ID

S8-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3.875

GRADING NOTE Any equivalent form. a = 2, r = 0.5, n = 5: 2(1 − 0.03125)/0.5 = 3.875.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q2/5

A geometric series has first term 5, ratio 3, and 4 terms.

What is its sum?

A. 405

B. 80

C. 200

D. 121

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q3/5

308

QUESTION ID

S8-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 200

GRADING NOTE $5(3^4 - 1)/(3 - 1) = 5 \cdot 80/2 = 200$ (5 + 15 + 45 + 135).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q3/5

True or false?

$1 + 2 + 4 + \ldots + 2^9 = 2^{10} - 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q4/5

309

QUESTION ID

S8-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The sum of 10 terms is $(2^{10} - 1)/(2 - 1) = 1{,}023$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q4/5

True or false?

The formula $a(1 - r^n) \div (1 - r)$ gives the sum of a geometric series with r = 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S8-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q5/5

310

QUESTION ID
S8-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE With r = 1 the formula divides by 0; the sum is simply na.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S8-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q5/5

SET 8 • HSA-SSE.A.1.b

FORWARD 2

Solve exponential growth models with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

An investment is worth $1{,}000(1.05)^n$ dollars after n years.

Write the number of years to reach $2,000 as an expression with logarithms, then round it to the nearest tenth.

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q1/5

312

QUESTION ID

S8-F2-Q1

TYPE

Short answer

CORRECT ANSWER

n = log 2 ÷ log 1.05

≈ 14.2 years

GRADING NOTE Form required: the logarithm expression and the rounded value. $1.05^n = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q1/5

A town of 5,000 people grows 3% per year:
$P = 5{,}000(1.03)^t$. When will it reach 8,000? Round to the nearest tenth of a year. (log means base 10.)

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q2/5

313

QUESTION ID
S8-F2-Q2

TYPE
Short answer

CORRECT ANSWER
15.9 years

GRADING NOTE Form required: rounded to the nearest tenth. $1.03^t = 1.6$, so $t = \log 1.6 \div \log 1.03 \approx 15.90$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q2/5

Solve $5 \cdot 2^t = 60$. (log means base 10.)

A. t = log 60 ÷ log 10 ≈ 1.78

B. t = 3

C. t = log 12 ÷ log 2 ≈ 3.58

D. t = 12 ÷ 2 = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q3/5

314

QUESTION ID

S8-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. t = log 12 ÷ log 2 ≈ 3.58

GRADING NOTE $2^t = 12$, so t = log 12 ÷ log 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q3/5

True or false?

$\log(1.05^n) = n \cdot \log 1.05$ (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q4/5

315

QUESTION ID

S8-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The power property of logarithms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q4/5

True or false?

The solution of $2^x = 10$ is $x = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S8-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q5/5

316

QUESTION ID

S8-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2^5 = 32$. The solution is $x = \log 10 \div \log 2 \approx 3.32$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S8-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q5/5

Add and subtract polynomials and explain why the result is a polynomial

MAIN	Algebra I main question	**HSA-APR.A.1**
BACKWARD 1	Add and subtract linear expressions	**7.EE.A.1**
BACKWARD 2	Combine like terms to write equivalent expressions	**6.EE.A.4**
FORWARD 1	Use the Remainder Theorem	**HSA-APR.B.2**
FORWARD 2	Rewrite a quotient of polynomials as q(x) + r(x)/b(x)	**HSA-APR.D.6**

SET 9 • HSA-APR.A.1

MAIN

Algebra I main question • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Add. Write the answer in standard form.

$(4x^2 - 3x + 5) + (2x^2 + 7x - 9)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q1/5

319

QUESTION ID

S9-M-Q1

TYPE

Short answer

CORRECT ANSWER

$6x^2 + 4x - 4$

GRADING NOTE Form required: simplified, in standard form (terms in descending powers of x). Add like terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-M-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q1/5

Subtract. Write the answer in standard form.

$(5y^3 + 2y - 1) - (y^3 - 4y^2 + 6y - 3)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q2/5

320

QUESTION ID

S9-M-Q2

TYPE

Short answer

CORRECT ANSWER

$4y^3 + 4y^2 - 4y + 2$

GRADING NOTE Form required: simplified, in standard form (descending powers of y). Subtracting changes the sign of every term of the second polynomial: $5y^3 - y^3 + 4y^2 + 2y - 6y - 1 + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q2/5

Which expression is equal to $(3a^2 - 2ab) - (a^2 + 5ab)$?

A. $2a^2 - 7ab$

B. $2a^2 - 3ab$

C. $2a^2 + 3ab$

D. $4a^2 + 3ab$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q3/5

321

QUESTION ID

S9-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $2a^2 - 7ab$

GRADING NOTE $3a^2 - a^2 = 2a^2$, and $-2ab - 5ab = -7ab$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q3/5

Explain why the sum of two polynomials is always a polynomial.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q4/5

322

QUESTION ID

S9-M-Q4

TYPE

Short answer

CORRECT ANSWER

Adding combines like terms by adding their coefficients. The coefficients stay real numbers and the exponents stay whole numbers, so the result is a polynomial.

GRADING NOTE Any equivalent explanation (closure: the result has the same form).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q4/5

True or false?

The difference of two polynomials is always a polynomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q5/5

323

QUESTION ID

S9-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Subtracting adds the opposite of each term, which is still a sum of terms with whole-number exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q5/5

SET 9 • HSA-APR.A.1

BACKWARD 1

Add and subtract linear expressions • Grade 7 • Expressions & Equations • 7.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Add.

$(3x + 4) + (5x - 9)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q1/5

325

QUESTION ID

S9-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8x − 5

GRADING NOTE Form required: simplified. Combine like terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q1/5

Subtract.

$(7a - 2) - (3a - 6)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q2/5

326

QUESTION ID

S9-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4a + 4

GRADING NOTE Form required: simplified. $-(3a - 6) = -3a + 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q2/5

Subtract.

$(2y + 1) - (y + 5)$

A. $y - 6$

B. $3y - 4$

C. $y + 6$

D. $y - 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q3/5

327

QUESTION ID

S9-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $y - 4$

GRADING NOTE $2y - y = y$, and $1 - 5 = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q3/5

True or false?

$-(x - 3) = -x + 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q4/5

328

QUESTION ID

S9-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The minus sign changes the sign of each term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q4/5

True or false?

$(4n + 2) - (n + 2) = 3n + 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q5/5

329

QUESTION ID

S9-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(4n + 2) - (n + 2) = 3n$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • HSA-APR.A.1

BACKWARD 2

Combine like terms to write equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.4

Simplify.

4m + 2 + m

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q1/5

331

QUESTION ID

S9-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5m + 2

GRADING NOTE Form required: simplified. 4m + m = 5m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q1/5

Are 2(3 + x) and 6 + 2x equivalent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q2/5

332

QUESTION ID

S9-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. By the distributive property, 2(3 + x) = 6 + 2x for every x.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q2/5

Which expression is equal to b + b + b + 4?

A. 7b

B. 3b + 4

C. 4b + 3

D. $b^3 + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q3/5

333

QUESTION ID

S9-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3b + 4

GRADING NOTE b + b + b = 3b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q3/5

True or false?

$x + x = 2x$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q4/5

334

QUESTION ID

S9-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Two copies of x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q4/5

True or false?

$5y - y = 4y$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q5/5

335

QUESTION ID

S9-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $5y - 1y = 4y$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • HSA-APR.A.1

FORWARD 1

Use the Remainder Theorem • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Find the remainder when $x^2 + 3x - 4$ is divided by $x - 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q1/5

337

QUESTION ID
S9-F1-Q1

TYPE
Short answer

CORRECT ANSWER
6

GRADING NOTE Any equivalent form. By the Remainder Theorem, the remainder is $p(2) = 4 + 6 - 4 = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q1/5

Is $x + 1$ a factor of $x^3 + 1$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q2/5

338

QUESTION ID
S9-F1-Q2

TYPE
Short answer

CORRECT ANSWER
Yes.
$p(-1) = -1 + 1 = 0$, so the remainder is 0.

GRADING NOTE Any equivalent explanation using $p(-1)$ or division.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q2/5

What is the remainder when $p(x) = x^3 - 2x + 5$ is divided by $x - 1$?

A. 8

B. 4

C. -4

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q3/5

339

QUESTION ID
S9-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. 4

GRADING NOTE $p(1) = 1 - 2 + 5 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q3/5

True or false?

If p(3) = 0, then x − 3 is a factor of p(x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q4/5

340

QUESTION ID

S9-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A remainder of 0 means x − 3 divides p(x) evenly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q4/5

True or false?

The remainder when p(x) is divided by x + 2 is p(2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q5/5

341

QUESTION ID
S9-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE x + 2 = x − (−2), so the remainder is p(−2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q5/5

SET 9 • HSA-APR.A.1

FORWARD 2

Rewrite a quotient of polynomials as q(x) + r(x)/b(x) • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

ANSWER KEY Reduced copy of the question page, with its answer

Assume $x \neq -2$. Write $(x^2 + 5x + 7) \div (x + 2)$ in the form $q(x) + r/(x + 2)$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q1/5

343

QUESTION ID
S9-F2-Q1

TYPE
Short answer

CORRECT ANSWER
x + 3 + 1/(x + 2)

GRADING NOTE Form required: $q(x) + r/(x + 2)$. $(x + 2)(x + 3) = x^2 + 5x + 6$, which leaves a remainder of 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Assume x ≠ 0. Write (3x + 7) ÷ x in the form a + b/x.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q2/5

344

QUESTION ID
S9-F2-Q2

TYPE
Short answer

CORRECT ANSWER
3 + 7/x

GRADING NOTE Form required: a + b/x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q2/5

Assume $x \neq 1$. Which expression is equal to $(x^2 - 1) \div (x - 1)$?

A. $x + 1$

B. x

C. $x^2 + 1$

D. $x - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q3/5

345

QUESTION ID
S9-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. x + 1

GRADING NOTE $x^2 - 1 = (x - 1)(x + 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q3/5

True or false?

Assume $x \neq 0$. $(2x + 1) \div x = 2 + 1/x$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q4/5

346

QUESTION ID
S9-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $2x/x + 1/x = 2 + 1/x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q4/5

True or false?

Assume $x \neq -2$. $(x^2 + 4) \div (x + 2) = x + 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q5/5

347

QUESTION ID
S9-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $(x + 2)^2 = x^2 + 4x + 4$, not $x^2 + 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q5/5

Multiply polynomials

MAIN	Algebra I main question	**HSA-APR.A.1**
BACKWARD 1	Apply the distributive property to linear expressions	**6.EE.A.3**
BACKWARD 2	Multiply powers with the same base	**8.EE.A.1**
FORWARD 1	Prove polynomial identities	**HSA-APR.C.4**
FORWARD 2	Expand powers of binomials with the binomial theorem	**HSA-APR.C.5**

SET 10 • HSA-APR.A.1

MAIN

Algebra I main question • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply. Write the answer in standard form.

$(2x + 3)(x - 5)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q1/5

350

QUESTION ID

S10-M-Q1

TYPE

Short answer

CORRECT ANSWER

$2x^2 - 7x - 15$

GRADING NOTE Form required: expanded, simplified, in standard form (descending powers). $2x^2 - 10x + 3x - 15$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-M-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q1/5

Multiply. Write the answer in standard form.

$(x - 4)^2$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q2/5

351

QUESTION ID

S10-M-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 - 8x + 16$

GRADING NOTE Form required: expanded, simplified, in standard form (descending powers). $(x - 4)(x - 4) = x^2 - 4x - 4x + 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-M-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q2/5

Multiply. Write the answer in standard form.

$(y + 2)(y^2 - 3y + 1)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q3/5

352

QUESTION ID

S10-M-Q3

TYPE

Short answer

CORRECT ANSWER

$y^3 - y^2 - 5y + 2$

GRADING NOTE Form required: expanded, simplified, in standard form (descending powers). $y^3 - 3y^2 + y + 2y^2 - 6y + 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-M-Q3**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q3/5

Multiply.

$(3a - 1)(3a + 1)$

A. $9a^2 - 1$

B. $9a^2 - 6a - 1$

C. $6a - 1$

D. $9a^2 + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q4/5

353

QUESTION ID

S10-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $9a^2 - 1$

GRADING NOTE The middle terms cancel: $9a^2 + 3a - 3a - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q4/5

True or false?

$(x + 3)^2 = x^2 + 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q5/5

354

QUESTION ID

S10-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 3)^2 = x^2 + 6x + 9$; the middle term 6x comes from 3x + 3x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q5/5

SET 10 • HSA-APR.A.1

BACKWARD 1

Apply the distributive property to linear expressions • Grade 6 • Expressions & Equations • 6.EE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Expand.

$4(3x - 2)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q1/5

356

QUESTION ID

S10-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$12x - 8$

GRADING NOTE Form required: expanded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand.

$-2(5 - 3a)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q2/5

357

QUESTION ID

S10-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−10 + 6a

GRADING NOTE Form required: expanded (6a − 10 is the same). −2 times −3a is +6a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q2/5

Expand.

$7(2b + 1)$

A. $14b + 8$

B. $14b + 7$

C. $9b + 8$

D. $14b + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q3/5

358

QUESTION ID

S10-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 14b + 7

GRADING NOTE $7 \cdot 2b = 14b$ and $7 \cdot 1 = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q3/5

True or false?

$5(m - 2) = 5m - 10$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q4/5

359

QUESTION ID

S10-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 5 multiplies both terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q4/5

True or false?

$3(2x + 4) = 6x + 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q5/5

360

QUESTION ID

S10-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3(2x + 4) = 6x + 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • HSA-APR.A.1

BACKWARD 2

Multiply powers with the same base • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write as a single power.

$7^2 \cdot 7^3$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q1/5

362

QUESTION ID

S10-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7^5

GRADING NOTE Form required: a single power. 2 + 3 = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write as a single power.

$10^4 \cdot 10$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q2/5

363

QUESTION ID

S10-B2-Q2

TYPE

Short answer

CORRECT ANSWER

10^5

GRADING NOTE Form required: a single power. $10 = 10^1$, and $4 + 1 = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Write as a single power.

$6^3 \cdot 6^3$

A. 6^6

B. 6^9

C. 12^6

D. 36^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q3/5

364

QUESTION ID

S10-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6^6

GRADING NOTE Add the exponents: 3 + 3 = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q3/5

True or false?

$9 \cdot 9^4 = 9^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q4/5

365

QUESTION ID

S10-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $9^1 \cdot 9^4 = 9^5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q4/5

True or false?

$2^3 \cdot 2^3 = 2^9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q5/5

366

QUESTION ID

S10-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2^3 \cdot 2^3 = 2^6$; exponents are added, not multiplied.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • HSA-APR.A.1

FORWARD 1

Prove polynomial identities • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Show that $(a + b)^2 - (a - b)^2 = 4ab$ for all numbers a and b.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q1/5

368

QUESTION ID
S10-F1-Q1

TYPE
Answer with required work

CORRECT ANSWER
$(a + b)^2 - (a - b)^2 = 4ab$

GRADING NOTE REQUIRED WORK: Expand both squares: $(a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = 4ab$. The steps must be shown.

Any correct algebraic proof is accepted; checking numbers alone is not a proof.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q1/5

Use the identity $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$ to factor $x^3 - 8$.

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q2/5

369

QUESTION ID
S10-F1-Q2

TYPE
Short answer

CORRECT ANSWER
$(x - 2)(x^2 + 2x + 4)$

GRADING NOTE Form required: factored. Here y = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q2/5

Which identity is true for all numbers a and b?

A. $(a - b)^2 = a^2 - b^2$

B. $(a + b)(a - b) = a^2 - b^2$

C. $(a + b)^2 = a^2 + b^2$

D. $a^2 + b^2 = (a + b)(a + b)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q3/5

370

QUESTION ID
S10-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. $(a + b)(a - b) = a^2 - b^2$

GRADING NOTE $(a + b)(a - b) = a^2 - ab + ab - b^2 = a^2 - b^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q3/5

True or false?

$(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ for all x and y.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q4/5

371

QUESTION ID
S10-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Both sides expand to $x^4 + 2x^2y^2 + y^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q4/5

True or false?

$(x + 1)^3 = x^3 + 1$ for all x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q5/5

372

QUESTION ID
S10-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $(x + 1)^3 = x^3 + 3x^2 + 3x + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q5/5

SET 10 • HSA-APR.A.1

FORWARD 2

Expand powers of binomials with the binomial theorem • High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Expand. Write the answer in standard form.

$(2x - 1)^3$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q1/5

374

QUESTION ID
S10-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$8x^3 - 12x^2 + 6x - 1$

GRADING NOTE Form required: expanded, in standard form (descending powers). Coefficients 1, 3, 3, 1 with $(2x)^k$ and $(-1)^{3-k}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q1/5

What is the coefficient of the x^3y^2 term in the expansion of $(x + y)^5$?

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q2/5

375

QUESTION ID
S10-F2-Q2

TYPE
Short answer

CORRECT ANSWER
10

GRADING NOTE Any equivalent form. It is 5 choose 2 = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q2/5

How many terms does the expansion of $(a + b)^7$ have?

A. 128

B. 8

C. 14

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q3/5

376

QUESTION ID
S10-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. 8

GRADING NOTE $(a + b)^n$ has n + 1 terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q3/5

True or false?

The number 5 choose 2 equals 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q4/5

377

QUESTION ID
S10-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $5 \cdot 4 \div 2 = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q4/5

True or false?

Every coefficient in the expansion of $(a - b)^4$ is positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q5/5

QUESTION ID
S10-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $(a - b)^4 = a^4 - 4a^3b + 6a^2b^2 - 4ab^3 + b^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q5/5

Use structure to factor with a greatest common factor and the difference of squares

MAIN	Algebra I main question	**HSA-SSE.A.2**
BACKWARD 1	Use the greatest common factor and the distributive property	**6.NS.B.4**
BACKWARD 2	Multiply a sum and a difference	**HSA-APR.A.1**
FORWARD 1	Find the zeros of a polynomial from its factors	**HSA-APR.B.3**
FORWARD 2	Factor sums of squares with complex numbers	**HSN-CN.C.8**

SET 11 • HSA-SSE.A.2

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor completely.

$6x^3 - 15x^2$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q1/5

381

QUESTION ID

S11-M-Q1

TYPE

Short answer

CORRECT ANSWER

$3x^2(2x - 5)$

GRADING NOTE Form required: factored completely. The greatest common factor is $3x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor completely.

$49y^2 - 25$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q2/5

382

QUESTION ID

S11-M-Q2

TYPE

Short answer

CORRECT ANSWER

(7y − 5)(7y + 5)

GRADING NOTE Form required: factored completely (factors in either order). It is a difference of squares: $(7y)^2 - 5^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q2/5

Factor completely.

$x^4 - 16$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q3/5

383

QUESTION ID

S11-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(x^2 + 4)(x + 2)(x - 2)$

GRADING NOTE Form required: factored completely over the integers. $x^4 - 16 = (x^2)^2 - 4^2 = (x^2 + 4)(x^2 - 4)$, and $x^2 - 4$ factors again; $x^2 + 4$ does not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q3/5

Which is the complete factorization of $2x^2 - 18$?

A. $(2x - 6)(x + 3)$

B. $2(x - 3)(x + 3)$

C. $(x - 3)(x + 3)$

D. $2(x^2 - 9)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q4/5

384

QUESTION ID

S11-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $2(x - 3)(x + 3)$

GRADING NOTE Take out the common factor 2, then factor $x^2 - 9$. $(2x - 6)(x + 3)$ and $2(x^2 - 9)$ are equal to $2x^2 - 18$ but not completely factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q4/5

True or false?

$x^2 + 9$ can be factored as $(x + 3)(x - 3)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q5/5

385

QUESTION ID

S11-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 3)(x - 3) = x^2 - 9$. A sum of squares like $x^2 + 9$ does not factor over the real numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q5/5

SET 11 • HSA-SSE.A.2

BACKWARD 1

Use the greatest common factor and the distributive property • Grade 6 • The Number System • 6.NS.B.4

Find the greatest common factor of 24 and 36.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q1/5

387

QUESTION ID

S11-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the greatest common factor to write 42 + 28 as a product.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q2/5

388

QUESTION ID

S11-B1-Q2

TYPE

Short answer

CORRECT ANSWER

14(3 + 2)

GRADING NOTE Form required: the GCF times a sum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q2/5

What is the greatest common factor of 15 and 40?

A. 5

B. 8

C. 120

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q3/5

389

QUESTION ID

S11-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

GRADING NOTE $15 = 3 \cdot 5$ and $40 = 8 \cdot 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q3/5

True or false?

$18 + 12 = 6(3 + 2)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q4/5

390

QUESTION ID

S11-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $6 \cdot 3 = 18$ and $6 \cdot 2 = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q4/5

True or false?

The greatest common factor of 9 and 16 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q5/5

391

QUESTION ID

S11-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 9 and 16 share no factor greater than 1, so their GCF is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • HSA-SSE.A.2

BACKWARD 2

Multiply a sum and a difference • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$(x - 7)(x + 7)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q1/5

393

QUESTION ID

S11-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 - 49$

GRADING NOTE Form required: expanded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-B2-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$(2y + 3)(2y - 3)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q2/5

394

QUESTION ID

S11-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$4y^2 - 9$

GRADING NOTE Form required: expanded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-B2-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q2/5

Multiply.

$(x + 4)(x - 4)$

A. $x^2 - 8$

B. $x^2 + 16$

C. $x^2 - 8x - 16$

D. $x^2 - 16$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q3/5

395

QUESTION ID

S11-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x^2 - 16$

GRADING NOTE The middle terms 4x and −4x cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-B2-Q3**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q3/5

True or false?

$(a - 5)(a + 5) = a^2 - 25$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q4/5

396

QUESTION ID

S11-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $a^2 + 5a - 5a - 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q4/5

True or false?

$(x + 3)(x - 3) = x^2 - 6x - 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q5/5

397

QUESTION ID

S11-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 3)(x - 3) = x^2 - 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • HSA-SSE.A.2

FORWARD 1

Find the zeros of a polynomial from its factors • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

What are the zeros of p(x) = x(x − 2)(x + 5)?

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q1/5

399

QUESTION ID
S11-F1-Q1

TYPE
Short answer

CORRECT ANSWER
0, 2, and −5

GRADING NOTE Any order. All three are required. Each factor equals 0 at one zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F1-Q1**
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q1/5

Factor $x^3 - 9x$ completely, then give all of its zeros.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q2/5

400

QUESTION ID
S11-F1-Q2

TYPE
Short answer

CORRECT ANSWER
$x(x - 3)(x + 3)$; zeros 0, 3, and −3

GRADING NOTE Form required: factored completely, and all three zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q2/5

Which polynomial has the zeros −1, 0, and 4?

A. (x + 1)(x − 4)

B. x(x + 1)(x + 4)

C. x(x + 1)(x − 4)

D. x(x − 1)(x + 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q3/5

401

QUESTION ID
S11-F1-Q3

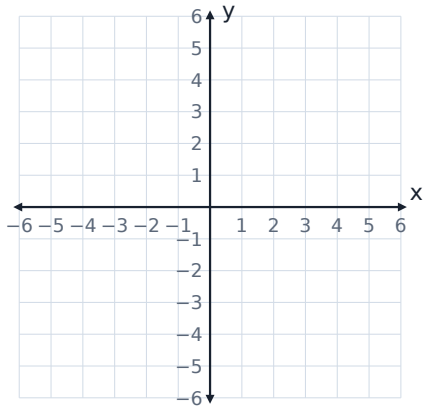
TYPE
Multiple choice

CORRECT ANSWER
C. x(x + 1)(x − 4)

GRADING NOTE A zero r comes from the factor (x − r): x, (x + 1), and (x − 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q3/5

Sketch a rough graph of $p(x) = x(x - 2)(x + 5)$. Mark the zeros and show the end behavior. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q4/5

402

QUESTION ID

S11-F1-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

A curve that crosses the x-axis at −5, 0, and 2, falls to the left and rises to the right, with one turn between −5 and 0 and one between 0 and 2

GRADING NOTE Any correct sketch: crossings at −5, 0, and 2, and end behavior of a cubic with positive leading coefficient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q4/5

True or false?

$p(x) = (x - 3)^2(x + 1)$ has three different zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q5/5

403

QUESTION ID
S11-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its zeros are 3 (twice) and −1, only two different values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q5/5

SET 11 • HSA-SSE.A.2

FORWARD 2

Factor sums of squares with complex numbers • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^2 + 25$ using complex numbers.

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q1/5

405

QUESTION ID

S11-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 5i)(x - 5i)$

GRADING NOTE Form required: two linear factors. $(x + 5i)(x - 5i) = x^2 - 25i^2 = x^2 + 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 16 = 0$. Give all solutions.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q2/5

406

QUESTION ID

S11-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x = 4i and x = −4i

GRADING NOTE Any equivalent form (±4i). Both solutions are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q2/5

Which is the complete factorization of $x^4 - 16$ into linear factors with complex numbers?

A. $(x - 4)(x + 4)$

B. $(x^2 - 4)(x^2 + 4)$

C. $(x - 2)(x + 2)(x - 2i)(x + 2i)$

D. $(x - 2)^2(x + 2)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q3/5

407

QUESTION ID

S11-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $(x - 2)(x + 2)(x - 2i)(x + 2i)$

GRADING NOTE $x^2 + 4 = (x - 2i)(x + 2i)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q3/5

True or false?

$(x + i)(x - i) = x^2 + 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S11-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q4/5

408

QUESTION ID

S11-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x^2 - i^2 = x^2 + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S11-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q4/5

True or false?

$x^2 + 4$ has no linear factors, even with complex numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q5/5

409

QUESTION ID

S11-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 + 4 = (x + 2i)(x - 2i)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q5/5

Factor quadratic trinomials

MAIN	Algebra I main question	**HSA-SSE.A.2**
BACKWARD 1	Find factor pairs of a whole number	**4.OA.B.4**
BACKWARD 2	Multiply two binomials	**HSA-APR.A.1**
FORWARD 1	Use factored form to describe the graph of a polynomial function	**HSF-IF.C.7.c**
FORWARD 2	Write a polynomial from its zeros and find zeros by factoring	**HSA-APR.B.3**

SET 12 • HSA-SSE.A.2

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor.

$x^2 + 7x + 12$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q1/5

412

QUESTION ID

S12-M-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 3)(x + 4)

GRADING NOTE Form required: factored (factors in either order). 3 · 4 = 12 and 3 + 4 = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-M-Q1**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q1/5

Factor.

$x^2 - 2x - 15$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q2/5

413

QUESTION ID

S12-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 5)(x + 3)$

GRADING NOTE Form required: factored. $-5 \cdot 3 = -15$ and $-5 + 3 = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-M-Q2**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q2/5

Factor completely.

$2x^2 + 7x + 3$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q3/5

414

QUESTION ID

S12-M-Q3

TYPE

Short answer

CORRECT ANSWER

(2x + 1)(x + 3)

GRADING NOTE Form required: factored. Check: $2x^2 + 6x + x + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q3/5

Which is a factor of $3x^2 - 10x - 8$?

A. $3x - 2$

B. $x + 4$

C. $x - 4$

D. $x - 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q4/5

415

QUESTION ID

S12-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $x - 4$

GRADING NOTE $3x^2 - 10x - 8 = (3x + 2)(x - 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q4/5

True or false?

$x^2 + 5x + 6 = (x + 1)(x + 6)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q5/5

416

QUESTION ID

S12-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 1)(x + 6) = x^2 + 7x + 6$. The correct factoring is $(x + 2)(x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q5/5

SET 12 • HSA-SSE.A.2

BACKWARD 1

Find factor pairs of a whole number • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all the factor pairs of 12.

Factor pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q1/5

418

QUESTION ID

S12-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1 and 12, 2 and 6, 3 and 4

GRADING NOTE Any order. All three pairs are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q1/5

Two whole numbers multiply to 15 and add to 8.

What are they?

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q2/5

419

QUESTION ID

S12-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3 and 5

GRADING NOTE Any order. 3 · 5 = 15 and 3 + 5 = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q2/5

Which factor pair of 24 has a sum of 10?

A. 4 and 6

B. 3 and 8

C. 1 and 24

D. 2 and 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q3/5

420

QUESTION ID

S12-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4 and 6

GRADING NOTE $4 + 6 = 10$, and $4 \cdot 6 = 24$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q3/5

True or false?

3 and 7 are a factor pair of 21.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q4/5

421

QUESTION ID

S12-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $3 \cdot 7 = 21$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q4/5

True or false?

4 and 5 are a factor pair of 25.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q5/5

422

QUESTION ID

S12-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4 \cdot 5 = 20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • HSA-SSE.A.2

BACKWARD 2

Multiply two binomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$(x + 2)(x + 9)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q1/5

424

QUESTION ID

S12-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + 11x + 18$

GRADING NOTE Form required: expanded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-B2-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q1/5

Multiply.

$(x - 6)(x + 1)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q2/5

425

QUESTION ID

S12-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 - 5x - 6$

GRADING NOTE Form required: expanded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-B2-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q2/5

Multiply.

$(3x + 1)(x + 2)$

A. $3x^2 + 6x + 2$

B. $3x^2 + 7x + 2$

C. $4x^2 + 7x + 2$

D. $3x^2 + 3x + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q3/5

426

QUESTION ID

S12-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $3x^2 + 7x + 2$

GRADING NOTE $3x^2 + 6x + x + 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-B2-Q3**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q3/5

True or false?

$(x - 2)(x - 6) = x^2 - 8x + 12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q4/5

427

QUESTION ID

S12-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x^2 - 6x - 2x + 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q4/5

True or false?

$(x + 1)(x - 4) = x^2 - 3x + 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q5/5

428

QUESTION ID

S12-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The constant is $1 \cdot (-4) = -4$: $x^2 - 3x - 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • HSA-SSE.A.2

FORWARD 1

Use factored form to describe the graph of a polynomial function • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

$p(x) = (x + 2)(x - 1)(x - 3)$.

What are its zeros and its y-intercept?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q1/5

430

QUESTION ID

S12-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Zeros −2, 1, and 3; y-intercept (0, 6)

GRADING NOTE Any equivalent form. All parts are required. $p(0) = 2 \cdot (-1) \cdot (-3) = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q1/5

Describe the end behavior of $y = x^3 - 4x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q2/5

431

QUESTION ID

S12-F1-Q2

TYPE

Short answer

CORRECT ANSWER

As x increases without bound, y increases without bound; as x decreases without bound, y decreases without bound.

GRADING NOTE Any equivalent wording (up to the right, down to the left). The leading term x^3 decides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q2/5

Which describes the end behavior of $y = -x^3$?

A. It rises to the left and falls to the right.

B. It rises on both ends.

C. It falls to the left and rises to the right.

D. It falls on both ends.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q3/5

QUESTION ID

S12-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

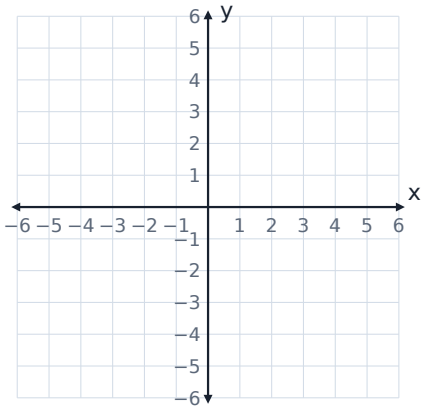
A. It rises to the left and falls to the right.

GRADING NOTE For large positive x, $-x^3$ is very negative; for large negative x, it is very positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q3/5

Graph
$p(x) = (x + 2)(x - 1)(x - 3)$.
Mark the zeros and the y-intercept, and show the end behavior. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q4/5

433

QUESTION ID
S12-F1-Q4

TYPE
Drawing / plotting

CORRECT ANSWER

A curve crossing the x-axis at −2, 1, and 3, through (0, 6), falling to the left and rising to the right

GRADING NOTE Any correct graph. $p(0) = (2)(-1)(-3) = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q4/5

True or false?

The graph of $y = (x - 1)^2$ crosses the x-axis at $x = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q5/5

434

QUESTION ID

S12-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It touches the x-axis at $x = 1$ and turns back; $(x - 1)^2$ is never negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q5/5

SET 12 • HSA-SSE.A.2

FORWARD 2

Write a polynomial from its zeros and find zeros by factoring • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Write a polynomial function of degree 3 whose only zeros are 2, −3, and 5.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q1/5

436

QUESTION ID
S12-F2-Q1

TYPE
Short answer

CORRECT ANSWER
(x − 2)(x + 3)(x − 5)

GRADING NOTE Any equivalent form, or any nonzero constant multiple, such as 2(x − 2)(x + 3)(x − 5). A degree-3 polynomial with these three zeros has exactly these factors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q1/5

Factor $x^3 - x^2 - 6x$ completely, then give its zeros.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q2/5

437

QUESTION ID
S12-F2-Q2

TYPE
Short answer

CORRECT ANSWER
x(x − 3)(x + 2);
zeros 0, 3, and −2

GRADING NOTE Form required: factored completely, and all three zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q2/5

What are the zeros of p(x) = (2x − 1)(x + 4)?

A. 2 and −4

B. 1 and 4

C. $\frac{1}{2}$ and −4

D. $-\frac{1}{2}$ and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q3/5

438

QUESTION ID
S12-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. 1/2 and −4

GRADING NOTE 2x − 1 = 0 gives x = 1/2, and x + 4 = 0 gives x = −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q3/5

True or false?

If $(x - 7)$ is a factor of $p(x)$, then 7 is a zero of p.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q4/5

439

QUESTION ID
S12-F2-Q4

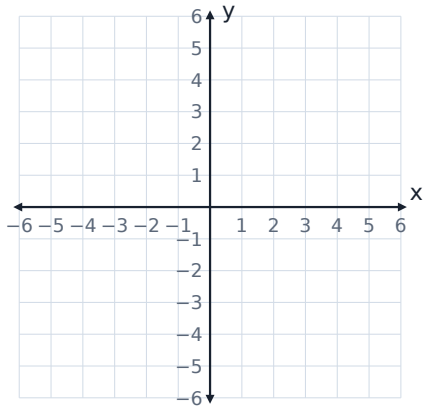
TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $p(7)$ has the factor $7 - 7 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q4/5

Sketch a rough graph of $p(x) = (x - 1)^2(x + 2)(x - 3)$. Show what the graph does at each zero and its end behavior. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q5/5

440

QUESTION ID
S12-F2-Q5

TYPE
Drawing / plotting

CORRECT ANSWER
A curve rising on both ends, crossing the x-axis at −2 and 3 and touching it at 1 (double zero)

GRADING NOTE Any correct sketch: degree 4 with positive leading coefficient, so it rises on both ends; even multiplicity at 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q5/5

Use structure to factor perfect squares and expressions in quadratic form

MAIN	Algebra I main question	**HSA-SSE.A.2**
BACKWARD 1	Find squares of whole numbers and decimals	**6.EE.A.1**
BACKWARD 2	Square a binomial	**HSA-APR.A.1**
FORWARD 1	Find the zeros of polynomial functions in quadratic form	**HSF-IF.C.7.c**
FORWARD 2	Factor quadratic expressions with complex numbers	**HSN-CN.C.8**

SET 13 • HSA-SSE.A.2

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor.

$x^2 + 10x + 25$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q1/5

443

QUESTION ID

S13-M-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 5)^2$

GRADING NOTE Form required: factored; $(x + 5)(x + 5)$ is also correct. It is a perfect square: $x^2 + 2 \cdot 5 \cdot x + 5^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q1/5

Factor completely. (Think of x^2 as one quantity.)

$x^4 - 5x^2 + 4$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q2/5

444

QUESTION ID

S13-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 1)(x + 1)(x - 2)(x + 2)$

GRADING NOTE Form required: factored completely. With $u = x^2$: $u^2 - 5u + 4 = (u - 1)(u - 4) = (x^2 - 1)(x^2 - 4)$, and each factors again.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q2/5

Factor, and simplify each factor. (Think of x + 1 as one quantity.)

$(x + 1)^2 - 9$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q3/5

445

QUESTION ID

S13-M-Q3

TYPE

Short answer

CORRECT ANSWER

(x − 2)(x + 4)

GRADING NOTE Form required: factored and simplified. It is a difference of squares: ((x + 1) − 3)((x + 1) + 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q3/5

Which expression is a perfect square trinomial?

A. $x^2 + 9$

B. $4x^2 - 12x + 9$

C. $4x^2 + 6x + 9$

D. $x^2 - 6x - 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q4/5

446

QUESTION ID

S13-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $4x^2 - 12x + 9$

GRADING NOTE $4x^2 - 12x + 9 = (2x - 3)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q4/5

True or false?

$9x^2 + 30x + 25 = (3x + 5)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q5/5

447

QUESTION ID

S13-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(3x + 5)^2 = 9x^2 + 2 \cdot 3x \cdot 5 + 25 = 9x^2 + 30x + 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q5/5

SET 13 • HSA-SSE.A.2

BACKWARD 1

Find squares of whole numbers and decimals • Grade 6 • Expressions & Equations • 6.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

15^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q1/5

449

QUESTION ID

S13-B1-Q1

TYPE

Short answer

CORRECT ANSWER

225

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What whole number squared is 169?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q2/5

450

QUESTION ID

S13-B1-Q2

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form. 13 · 13 = 169.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

2.5^2

A. 5

B. 4.25

C. 62.5

D. 6.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q3/5

451

QUESTION ID

S13-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 6.25

GRADING NOTE $2.5 \cdot 2.5 = 6.25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q3/5

True or false?

$11^2 = 121$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q4/5

452

QUESTION ID

S13-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $11 \cdot 11 = 121$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q4/5

True or false?

$6^2 = 12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q5/5

453

QUESTION ID

S13-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $6^2 = 6 \cdot 6 = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • HSA-SSE.A.2

BACKWARD 2

Square a binomial • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$(x + 5)^2$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q1/5

455

QUESTION ID

S13-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + 10x + 25$

GRADING NOTE Form required: expanded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-B2-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q1/5

Multiply.

$(3y - 2)^2$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q2/5

456

QUESTION ID

S13-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$9y^2 - 12y + 4$

GRADING NOTE Form required: expanded. $(3y)^2 - 2 \cdot 3y \cdot 2 + 2^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-B2-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q2/5

Multiply.

$(x - 6)^2$

A. $x^2 - 6x + 36$

B. $x^2 - 36$

C. $x^2 + 36$

D. $x^2 - 12x + 36$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q3/5

457

QUESTION ID

S13-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x^2 - 12x + 36$

GRADING NOTE $x^2 - 6x - 6x + 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-B2-Q3**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q3/5

True or false?

$(a + 1)^2 = a^2 + 2a + 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q4/5

458

QUESTION ID

S13-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(a + 1)(a + 1) = a^2 + a + a + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q4/5

True or false?

$(x - 3)^2 = x^2 - 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q5/5

459

QUESTION ID

S13-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x - 3)^2 = x^2 - 6x + 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • HSA-SSE.A.2

FORWARD 1

Find the zeros of polynomial functions in quadratic form • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

Find all real zeros of $f(x) = x^4 - 5x^2 + 4$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q1/5

461

QUESTION ID

S13-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−2, −1, 1, and 2

GRADING NOTE Any order. All four are required. $f(x) = (x^2 - 1)(x^2 - 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q1/5

Find all real zeros of $y = x^4 - 9x^2$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q2/5

462

QUESTION ID

S13-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0, 3, and −3

GRADING NOTE Any order. All three are required. $x^4 - 9x^2 = x^2(x - 3)(x + 3)$; 0 is a double zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q2/5

How many real zeros does $y = x^4 - 1$ have?

A. 2

B. 1

C. 0

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q3/5

463

QUESTION ID

S13-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

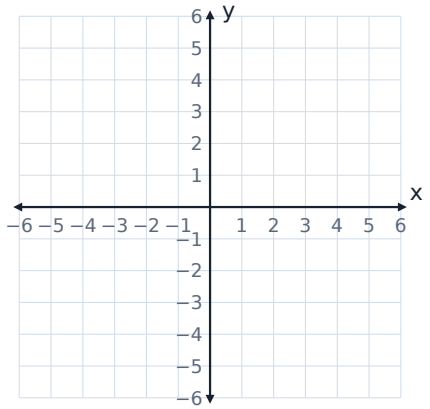
A. 2

GRADING NOTE $x^4 - 1 = (x^2 + 1)(x - 1)(x + 1)$: only 1 and −1 are real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q3/5

Graph $f(x) = x^4 - 5x^2 + 4 = (x^2 - 1)(x^2 - 4)$. Mark the zeros and the y-intercept, and show the end behavior.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q4/5

464

QUESTION ID

S13-F1-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

A W-shaped curve crossing the x-axis at −2, −1, 1, and 2, with a high point at (0, 4), low points near (±1.6, −2.25), rising on both ends

GRADING NOTE Any correct graph showing all four zeros, the y-intercept 4, and both ends rising.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q4/5

True or false?

$y = x^4 + 1$ has a real zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q5/5

465

QUESTION ID

S13-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^4 + 1 \geq 1$ for every real x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q5/5

SET 13 • HSA-SSE.A.2

FORWARD 2

Factor quadratic expressions with complex numbers • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 + 49$ using complex numbers.

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q1/5

467

QUESTION ID

S13-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 7i)(x − 7i)

GRADING NOTE Form required: two linear factors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q1/5

The solutions of $x^2 + 2x + 5 = 0$ are $-1 + 2i$ and $-1 - 2i$.

Write $x^2 + 2x + 5$ as a product of two linear factors.

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q2/5

468

QUESTION ID

S13-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$(x + 1 - 2i)(x + 1 + 2i)$

GRADING NOTE Form required: two linear factors. A zero r gives the factor $(x - r)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q2/5

Multiply.

$(x - 3i)(x + 3i)$

A. $x^2 - 9$

B. $x^2 + 6ix - 9$

C. $x^2 + 9$

D. $x^2 - 9i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q3/5

469

QUESTION ID

S13-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $x^2 + 9$

GRADING NOTE $x^2 - 9i^2 = x^2 + 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q3/5

True or false?

$x^4 - 1 = (x - 1)(x + 1)(x - i)(x + i)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q4/5

470

QUESTION ID

S13-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x^4 - 1 = (x^2 - 1)(x^2 + 1)$, and $x^2 + 1 = (x - i)(x + i)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q4/5

True or false?

$(x + 2i)^2 = x^2 - 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q5/5

471

QUESTION ID

S13-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 2i)^2 = x^2 + 4ix + 4i^2 = x^2 + 4ix - 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q5/5

Factor a quadratic expression to reveal the zeros of the function it defines

MAIN	Algebra I main question	**HSA-SSE.B.3.a**
BACKWARD 1	Factor trinomials with a leading coefficient of 1	**HSA-SSE.A.2**
BACKWARD 2	Use substitution to decide which values make an equation true	**6.EE.B.5**
FORWARD 1	Find zeros of cubic functions by factoring	**HSA-APR.B.3**
FORWARD 2	Sketch polynomial graphs from their zeros	**HSF-IF.C.7.c**

SET 14 • HSA-SSE.B.3.a

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a

$f(x) = x^2 - x - 12$.

Factor f(x), then give the zeros of f.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q1/5

474

QUESTION ID

S14-M-Q1

TYPE

Short answer

CORRECT ANSWER

(x − 4)(x + 3); zeros 4 and −3

GRADING NOTE Form required: the factored form and both zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q1/5

A ball's height is $h(t) = -16t^2 + 48t$ feet after t seconds.

Factor h(t), and use the factors to say when the ball is at ground level.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q2/5

475

QUESTION ID

S14-M-Q2

TYPE

Short answer

CORRECT ANSWER

h(t) = −16t(t − 3); the height is 0 at t = 0 seconds (when it is thrown) and t = 3 seconds (when it lands).

GRADING NOTE Form required: the factored form and both times.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q2/5

What are the zeros of $g(x) = 2x^2 + 5x - 3$?

A. $\frac{1}{2}$ and -3

B. $-\frac{1}{2}$ and 3

C. 2 and -3

D. 1 and -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q3/5

476

QUESTION ID

S14-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1/2 and −3

GRADING NOTE $2x^2 + 5x - 3 = (2x - 1)(x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q3/5

Write a quadratic function, in factored form with a leading coefficient of 1, whose zeros are 2 and −5.

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q4/5

477

QUESTION ID

S14-M-Q4

TYPE

Short answer

CORRECT ANSWER

(x − 2)(x + 5)

GRADING NOTE Form required: factored form. Factors in either order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q4/5

True or false?

The zeros of f(x) = (x − 6)(x + 1) are 6 and −1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q5/5

478

QUESTION ID

S14-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each factor is 0 at one zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q5/5

SET 14 • HSA-SSE.B.3.a

BACKWARD 1

Factor trinomials with a leading coefficient of 1 • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor.

$x^2 + 9x + 20$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q1/5

480

QUESTION ID

S14-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 4)(x + 5)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q1/5

Factor.

$x^2 - 9x + 14$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q2/5

481

QUESTION ID

S14-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(x − 2)(x − 7)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-B1-Q2**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q2/5

Factor.

$x^2 + x - 6$

A. $(x - 6)(x + 1)$

B. $(x - 3)(x + 2)$

C. $(x + 3)(x - 2)$

D. $(x + 6)(x - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q3/5

482

QUESTION ID

S14-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (x + 3)(x − 2)

GRADING NOTE $3 \cdot (-2) = -6$ and $3 + (-2) = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q3/5

True or false?

$x^2 - 4x - 21 = (x - 7)(x + 3)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q4/5

483

QUESTION ID

S14-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $-7 \cdot 3 = -21$ and $-7 + 3 = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q4/5

True or false?

$x^2 + 8x + 15 = (x + 3)(x + 6)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q5/5

484

QUESTION ID
S14-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $(x + 3)(x + 6) = x^2 + 9x + 18$; the correct factors are $(x + 3)(x + 5)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • HSA-SSE.B.3.a

BACKWARD 2

Use substitution to decide which values make an equation true • Grade 6 • Expressions & Equations • 6.EE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Is $x = 4$ a solution of $x^2 = 16$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q1/5

486

QUESTION ID

S14-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. $4^2 = 16$.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q1/5

Which of the values 1, 2, and 3 make $x^2 - 5x + 6 = 0$ true?

Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q2/5

487

QUESTION ID

S14-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2 and 3

GRADING NOTE Any equivalent form. Both are required. $4 - 10 + 6 = 0$ and $9 - 15 + 6 = 0$; $x = 1$ gives 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q2/5

Which value makes $3x - 6 = 0$ true?

A. 3

B. -2

C. 2

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q3/5

488

QUESTION ID

S14-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2

GRADING NOTE $3 \cdot 2 - 6 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q3/5

True or false?

$x = 5$ makes $x(x - 5) = 0$ true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q4/5

489

QUESTION ID

S14-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $5 \cdot 0 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q4/5

True or false?

$x = -2$ makes $x + 2 = 4$ true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q5/5

490

QUESTION ID

S14-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-2 + 2 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • HSA-SSE.B.3.a

FORWARD 1

Find zeros of cubic functions by factoring • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Find the zeros of $f(x) = x^3 + 2x^2 - 15x$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q1/5

492

QUESTION ID
S14-F1-Q1

TYPE
Short answer

CORRECT ANSWER
0, 3, and −5

GRADING NOTE Any order. All three are required. $x(x^2 + 2x - 15) = x(x + 5)(x - 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q1/5

$g(x) = (x - 1)(x^2 - 4)$.

List all the zeros of g.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q2/5

493

QUESTION ID
S14-F1-Q2

TYPE
Short answer

CORRECT ANSWER
1, 2, and −2

GRADING NOTE Any order. All three are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F1-Q2**
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q2/5

Which function has a double zero at x = 2?

A. $f(x) = (x - 2)^2(x + 1)$

B. $f(x) = 2x(x - 1)$

C. $f(x) = (x + 2)^2(x - 1)$

D. $f(x) = (x - 2)(x + 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q3/5

494

QUESTION ID
S14-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
A.
$f(x) = (x - 2)^2(x + 1)$

GRADING NOTE The factor $(x - 2)$ appears twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q3/5

True or false?

$f(x) = x^3 - x$ has three real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q4/5

495

QUESTION ID
S14-F1-Q4

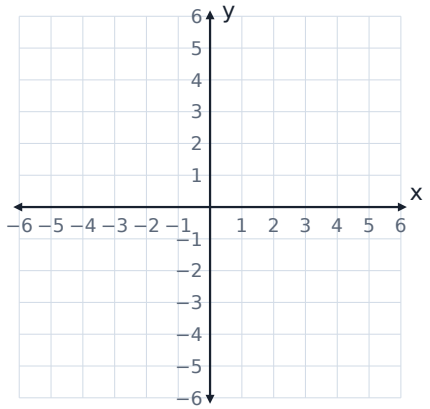
TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $x(x - 1)(x + 1)$ has zeros -1, 0, and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q4/5

Sketch a rough graph of $f(x) = (x - 2)^2(x + 1)$. Mark the zeros and the y-intercept, and show what the graph does at each zero. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S14-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q5/5

496

QUESTION ID

S14-F1-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

A curve that crosses the x-axis at −1, touches it at 2 and turns back up (double zero), passes through (0, 4), falls to the left and rises to the right

GRADING NOTE Any correct sketch: a crossing at −1, a touch at 2 (even multiplicity), y-intercept 4, and cubic end behavior.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S14-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q5/5

SET 14 • HSA-SSE.B.3.a

FORWARD 2

Sketch polynomial graphs from their zeros • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

ANSWER KEY Reduced copy of the question page, with its answer

Describe where f(x) = (x + 1)(x − 2)(x − 4) is above the x-axis.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q1/5

498

QUESTION ID

S14-F2-Q1

TYPE

Short answer

CORRECT ANSWER

For −1 < x < 2 and for x > 4.

GRADING NOTE Any equivalent form. Test a value in each interval: f(0) = 8 > 0, f(3) = −4 < 0, f(5) = 12 > 0, f(−2) = −24 < 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q1/5

A cubic function has zeros −3, 0, and 3 and a positive leading coefficient.

Describe its end behavior.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q2/5

499

QUESTION ID
S14-F2-Q2

TYPE
Short answer

CORRECT ANSWER
It falls to the left and rises to the right.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

How many turning points can the graph of a cubic function have, at most?

A. 2

B. 1

C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q3/5

500

QUESTION ID

S14-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

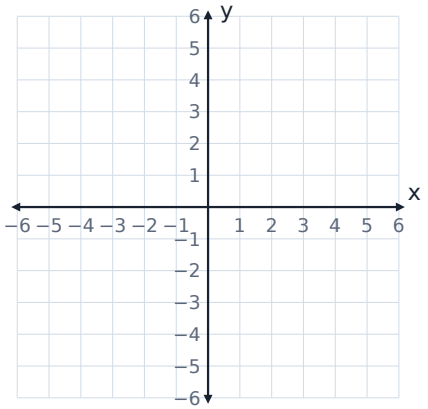
A. 2

GRADING NOTE A polynomial of degree n has at most n − 1 turning points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q3/5

Graph
$f(x) = (x + 1)(x - 2)(x - 4)$.
Mark the zeros and the y-intercept, and show the end behavior. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q4/5

501

QUESTION ID
S14-F2-Q4

TYPE
Drawing / plotting

CORRECT ANSWER

A curve crossing the x-axis at −1, 2, and 4, through (0, 8), falling to the left and rising to the right; above the axis for $-1 < x < 2$ and $x > 4$

GRADING NOTE Any correct graph. $f(0) = (1)(-2)(-4) = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q4/5

True or false?

A polynomial function with a negative leading coefficient always falls on both ends.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q5/5

502

QUESTION ID
S14-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $y = -x^3$ rises on the left and falls on the right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q5/5

Complete the square to reveal the maximum or minimum value of a quadratic function

MAIN	Algebra I main question	**HSA-SSE.B.3.b**
BACKWARD 1	Recognize and complete perfect square trinomials	**HSA-SSE.A.2**
BACKWARD 2	Describe where a graph rises, falls, and reaches its highest point	**8.F.B.5**
FORWARD 1	Complete the square to find the center and radius of a circle	**HSG-GPE.A.1**
FORWARD 2	Derive the equations of ellipses and hyperbolas from their foci	**HSG-GPE.A.3**

SET 15 • HSA-SSE.B.3.b

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b

Write f(x) = x² + 6x + 5 in vertex form by completing the square. Show the steps.

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q1/5

505

QUESTION ID

S15-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

(x + 3)² − 4

GRADING NOTE

REQUIRED WORK: The completed square shown: x² + 6x + 9 − 9 + 5 (add and subtract (6 ÷ 2)² = 9), then (x + 3)² − 4.

Form required: vertex form a(x − h)² + k, with the completing-the-square steps. x² + 6x + 9 − 9 + 5 = (x + 3)² − 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q1/5

$g(x) = x^2 - 10x + 30$.

Complete the square to write g in vertex form, then give the minimum value of g and where it happens. Show the steps.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q2/5

506

QUESTION ID

S15-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$g(x) = (x - 5)^2 + 5$; the minimum is 5, at $x = 5$.

GRADING NOTE REQUIRED WORK: The completed square shown: $x^2 - 10x + 25 - 25 + 30 = (x - 5)^2 + 5$, and the minimum read from the vertex form.

Form required: the vertex form with its steps, the minimum value, and the x-value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q2/5

A ball's height is $h(t) = -t^2 + 8t + 2$ meters after t seconds.

Complete the square (write h in vertex form) to find the greatest height and when it happens. Show the steps.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q3/5

507

QUESTION ID

S15-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

$h(t) = -(t - 4)^2 + 18$; the greatest height is 18 meters, at t = 4 seconds.

GRADING NOTE

REQUIRED WORK: The completed square shown: $-(t^2 - 8t) + 2 = -(t^2 - 8t + 16) + 16 + 2 = -(t - 4)^2 + 18$, and the maximum read from it.

Form required: the vertex form with its steps, the height, and the time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q3/5

Which is $x^2 - 4x + 7$ in vertex form?

A. $(x - 4)^2 + 7$

B. $(x - 2)^2 + 7$

C. $(x - 2)^2 + 3$

D. $(x + 2)^2 + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q4/5

508

QUESTION ID

S15-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $(x - 2)^2 + 3$

GRADING NOTE $x^2 - 4x + 4 + 3 = (x - 2)^2 + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q4/5

True or false?

The minimum value of $(x + 1)^2 - 6$ is -6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q5/5

509

QUESTION ID

S15-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x + 1)^2 \geq 0$, so the value is at least -6, reached at $x = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q5/5

SET 15 • HSA-SSE.B.3.b

BACKWARD 1

Recognize and complete perfect square trinomials • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor.

$x^2 + 14x + 49$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q1/5

511

QUESTION ID

S15-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 7)^2$

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q1/5

What number c makes $x^2 - 8x + c$ a perfect square trinomial?

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q2/5

512

QUESTION ID

S15-B1-Q2

TYPE

Short answer

CORRECT ANSWER

16

GRADING NOTE Any equivalent form. $(-8/2)^2 = 16$, giving $(x - 4)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q2/5

Which number completes $x^2 + 12x + ___$ to a perfect square trinomial?

A. 36

B. 6

C. 24

D. 144

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q3/5

513

QUESTION ID

S15-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 36

GRADING NOTE $(12/2)^2 = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q3/5

True or false?

$x^2 - 2x + 1 = (x - 1)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q4/5

514

QUESTION ID

S15-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x - 1)^2 = x^2 - 2x + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q4/5

True or false?

$x^2 + 6x + 6$ is a perfect square trinomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q5/5

515

QUESTION ID

S15-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 3)^2 = x^2 + 6x + 9$, not + 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q5/5

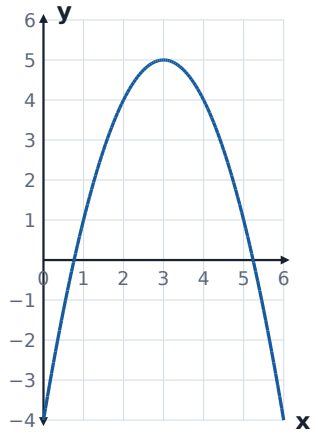
SET 15 • HSA-SSE.B.3.b

BACKWARD 2

Describe where a graph rises, falls, and reaches its highest point • Grade 8 • Functions • 8.F.B.5

Where is the highest point of the graph?

Point: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q1/5

517

QUESTION ID

S15-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 5)

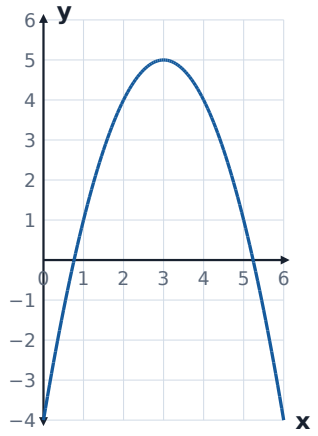
GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q1/5

For which x-values shown is the graph rising?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q2

Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q2/5

518

QUESTION ID

S15-B2-Q2

TYPE

Short answer

CORRECT ANSWER

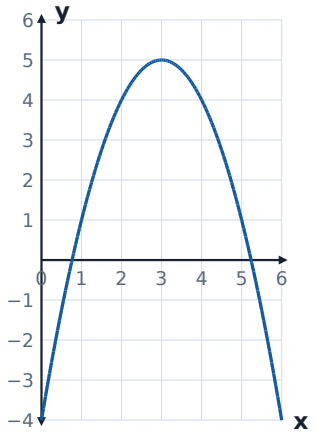
From x = 0 to x = 3

GRADING NOTE Any equivalent form (0 < x < 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q2

Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q2/5

For which x-values shown is the graph falling?



A. From x = 3 to x = 6

B. It never falls.

C. From x = 0 to x = 3

D. From x = 0 to x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q3/5

519

QUESTION ID

S15-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. From x = 3 to x = 6

GRADING NOTE After its highest point at x = 3, the graph goes down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

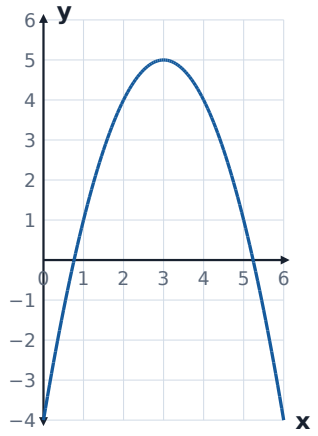
Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q3/5

True or false?

The graph has the same height at x = 2 and at x = 4.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q4/5

520

QUESTION ID

S15-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The graph is symmetric about x = 3, and 2 and 4 are each 1 unit from 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

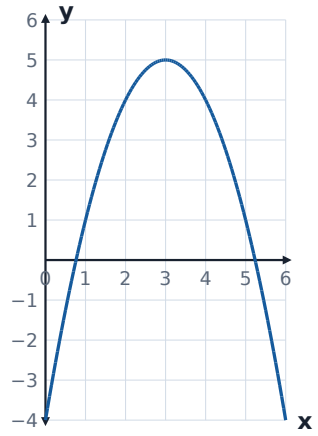
Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q4/5

True or false?

The graph is a straight line.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q5/5

521

QUESTION ID

S15-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It curves: it rises and then falls.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q5/5

SET 15 • HSA-SSE.B.3.b

FORWARD 1

Complete the square to find the center and radius of a circle • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Find the center and radius of the circle
$x^2 + y^2 - 6x + 4y - 3 = 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q1/5

523

QUESTION ID
S15-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Center (3, −2), radius 4

GRADING NOTE Any equivalent form. Both are required. $(x - 3)^2 + (y + 2)^2 = 3 + 9 + 4 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an equation of the circle with center (−1, 4) and radius 3 in standard form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q2/5

524

QUESTION ID
S15-F1-Q2

TYPE
Short answer

CORRECT ANSWER
$(x + 1)^2 + (y - 4)^2 = 9$

GRADING NOTE Form required: standard form $(x - h)^2 + (y - k)^2 = r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q2/5

Find the center and radius of $x^2 + y^2 + 8x = 9$.

A. Center (4, 0), radius 5

B. Center (0, 0), radius 9

C. Center (−4, 0), radius 3

D. Center (−4, 0), radius 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q3/5

525

QUESTION ID
S15-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. Center (−4, 0), radius 5

GRADING NOTE $(x + 4)^2 + y^2 = 9 + 16 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F1-Q3**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q3/5

True or false?

The circle $(x - 2)^2 + (y + 1)^2 = 4$ has a radius of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q4/5

526

QUESTION ID
S15-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $r^2 = 4$, so $r = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q4/5

True or false?

The circle $x^2 + y^2 = 25$ has its center at (5, 5).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q5/5

527

QUESTION ID
S15-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its center is (0, 0), and its radius is 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q5/5

SET 15 • HSA-SSE.B.3.b

FORWARD 2

Derive the equations of ellipses and hyperbolas from their foci • High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

An ellipse has foci (4, 0) and (−4, 0), and the sum of the distances from any point on it to the foci is 10.

Write its equation in standard form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q1/5

529

QUESTION ID
S15-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$x^2/25 + y^2/9 = 1$

GRADING NOTE Form required: standard form. $a = 5$, $c = 4$, and $b^2 = a^2 - c^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q1/5

A hyperbola has foci (5, 0) and (−5, 0), and the absolute value of the difference of the distances from any point on it to the foci is 8.

Write its equation in standard form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q2/5

530

QUESTION ID
S15-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$x^2/16 - y^2/9 = 1$

GRADING NOTE Form required: standard form. $a = 4$, $c = 5$, and $b^2 = c^2 - a^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q2/5

The ellipse $x^2 \div 25 + y^2 \div 9 = 1$ has foci (4, 0) and (−4, 0).

Show that (3, 2.4) is on the ellipse and that its distances to the two foci add to 10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q3/5

531

QUESTION ID
S15-F2-Q3

TYPE
Answer with required work

CORRECT ANSWER
It satisfies the equation, and 2.6 + 7.4 = 10.

GRADING NOTE REQUIRED WORK: $9 \div 25 + 5.76 \div 9 = 0.36 + 0.64 = 1$ shown; the distances $\sqrt{(1 + 5.76)} = 2.6$ and $\sqrt{(49 + 5.76)} = 7.4$ computed and added.

Form required: the substitution and both distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q3/5

True or false?

$x^2/25 + y^2/9 = 1$ is the equation of an ellipse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q4/5

532

QUESTION ID
S15-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE A sum of two squared terms with different denominators, equal to 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q4/5

True or false?

For every point on an ellipse, the difference of the distances to the two foci is constant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q5/5

533

QUESTION ID
S15-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE For an ellipse the sum is constant; for a hyperbola the difference is.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q5/5

Use the properties of exponents to transform exponential expressions

MAIN	Algebra I main question	**HSA-SSE.B.3.c**
BACKWARD 1	Use the meaning of rational exponents and the power-of-a-power…	**HSN-RN.A.1**
BACKWARD 2	Solve percent increase and decrease problems	**7.RP.A.3**
FORWARD 1	Solve exponential equations with logarithms	**HSF-LE.A.4**
FORWARD 2	Use the sum of a geometric series in a growth model	**HSA-SSE.B.4**

SET 16 • HSA-SSE.B.3.c

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c

An account grows by 6% each year: $A = 500(1.06)^t$ after t years.

Rewrite the expression to show the monthly growth factor, rounded to four decimal places.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q1/5

536

QUESTION ID

S16-M-Q1

TYPE

Short answer

CORRECT ANSWER

$500(1.0049)^{12t}$

GRADING NOTE Form required: 500 times (monthly factor)^{12t}. The factor is $1.06^{1/12} \approx 1.0049$, so $A = 500(1.06^{1/12})^{12t}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q1/5

Use the properties of exponents to explain why $1.21^t = 1.1^{2t}$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q2/5

537

QUESTION ID

S16-M-Q2

TYPE

Short answer

CORRECT ANSWER

$1.21 = 1.1^2$, so $1.21^t = (1.1^2)^t = 1.1^{2t}$ by the power-of-a-power property.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q2/5

A population doubles every 5 years: $P = 300 \cdot 2^{t/5}$ after t years.

Find the yearly growth factor to three decimal places, and the yearly percent growth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q3/5

538

QUESTION ID

S16-M-Q3

TYPE

Short answer

CORRECT ANSWER

$2^{1/5} \approx 1.149$, so about 14.9% growth per year.

GRADING NOTE Form required: the factor to three decimal places and the percent. $2^{t/5} = (2^{1/5})^t$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which expression is equal to 8^t?

A. 2^{t+3}

B. 3^{2t}

C. 4^{2t}

D. 2^{3t}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q4/5

539

QUESTION ID

S16-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 2^{3t}

GRADING NOTE $8 = 2^3$, so $8^t = (2^3)^t = 2^{3t}$. ($4^{2t} = 16^t$.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q4/5

True or false?

In $y = 100(0.5)^{t/2}$, the amount is cut in half every 2 time units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q5/5

540

QUESTION ID

S16-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE When t increases by 2, the exponent t/2 increases by 1, so y is multiplied by 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q5/5

SET 16 • HSA-SSE.B.3.c

BACKWARD 1

Use the meaning of rational exponents and the power-of-a-power property • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1

Find the value.

$4^{1/2}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q1/5

542

QUESTION ID

S16-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. $4^{1/2} = \sqrt{4}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write $(1.1^2)^t$ as a single power of 1.1.

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q2/5

543

QUESTION ID

S16-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1.1^{2t}

GRADING NOTE Form required: a single power. Multiply the exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B1-Q2**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q2/5

Find the value.

$9^{1/2}$

A. $\frac{1}{9}$

B. 4.5

C. 3

D. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q3/5

544

QUESTION ID

S16-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3

GRADING NOTE $9^{1/2} = \sqrt{9} = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q3/5

True or false?

$(2^3)^t = 2^{3t}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q4/5

545

QUESTION ID

S16-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Power of a power: multiply the exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q4/5

True or false?

$5^{1/2} = 2.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q5/5

546

QUESTION ID

S16-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $5^{1/2} = \sqrt{5} \approx 2.24$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • HSA-SSE.B.3.c

BACKWARD 2

Solve percent increase and decrease problems • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Increase $500 by 6%.

Amount: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q1/5

548

QUESTION ID

S16-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$530

GRADING NOTE Any equivalent form. 500 · 1.06.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q1/5

Decrease 80 by 25%.

Amount: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q2/5

549

QUESTION ID

S16-B2-Q2

TYPE

Short answer

CORRECT ANSWER

60

GRADING NOTE Any equivalent form. 80 · 0.75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What do you multiply by to increase an amount by 4%?

A. 0.04

B. 1.4

C. 1.04

D. 0.96

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q3/5

550

QUESTION ID

S16-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1.04

GRADING NOTE 100% + 4% = 104% = 1.04.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q3/5

True or false?

Decreasing an amount by 10% is the same as multiplying it by 0.9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q4/5

551

QUESTION ID

S16-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 100% − 10% = 90%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q4/5

True or false?

An increase of 5% followed by a decrease of 5% gives back the original amount.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q5/5

552

QUESTION ID

S16-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $1.05 \cdot 0.95 = 0.9975$, slightly less than the original.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • HSA-SSE.B.3.c

FORWARD 1

Solve exponential equations with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $500(1.06)^t = 800$. Round t to the nearest tenth.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q1/5

554

QUESTION ID

S16-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8.1

GRADING NOTE Form required: rounded to the nearest tenth. $1.06^t = 1.6$, so $t = \log 1.6 \div \log 1.06 \approx 8.07$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q1/5

A population is $P = 300 \cdot 2^{t/5}$ after t years. When is $P = 1{,}000$? Round to the nearest tenth. (log means base 10.)

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q2/5

555

QUESTION ID
S16-F1-Q2

TYPE
Short answer

CORRECT ANSWER
8.7 years

GRADING NOTE Form required: rounded to the nearest tenth. $2^{t/5} = 10/3$, so $t = 5 \cdot \log(10/3) \div \log 2 \approx 8.68$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q2/5

Solve $10^{2x} = 50$. (log means base 10.)

A. x = 2.5

B. x = (log 50) ÷ 2 ≈ 0.85

C. x = 25

D. x = log 25 ≈ 1.40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q3/5

556

QUESTION ID

S16-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = (log 50) ÷ 2 ≈ 0.85

GRADING NOTE 2x = log 50 ≈ 1.699.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q3/5

True or false?

If $3^x = 3^5$, then x = 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q4/5

557

QUESTION ID

S16-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equal powers of the same base (other than 1) have equal exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q4/5

True or false?

The solution of $1.06^t = 2$ is $t = 2 \div 1.06$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S16-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q5/5

558

QUESTION ID

S16-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $t = \log 2 \div \log 1.06 \approx 11.9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S16-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q5/5

SET 16 • HSA-SSE.B.3.c

FORWARD 2

Use the sum of a geometric series in a growth model • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Deposits of \$100, \$100(1.005), \$100(1.005)^2, ..., \$100(1.005)^{11} are added. Use $S = a(1 - r^n) \div (1 - r)$ to find the total, to the nearest dollar.

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q1/5

560

QUESTION ID
S16-F2-Q1

TYPE
Short answer

CORRECT ANSWER
\$1,234

GRADING NOTE Form required: to the nearest dollar. a = 100, r = 1.005, n = 12: $100(1 - 1.005^{12}) \div (1 - 1.005) \approx 1{,}233.56$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the sum 5 + 10 + 20 + ... + 640.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q2/5

561

QUESTION ID

S16-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1,275

GRADING NOTE Any equivalent form. There are 8 terms ($640 = 5 \cdot 2^7$): $5(2^8 - 1) = 1{,}275$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q2/5

What is 1 + 3 + 9 + 27 + 81?

A. 81

B. 121

C. 243

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q3/5

562

QUESTION ID

S16-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 121

GRADING NOTE $(3^5 - 1) \div (3 - 1) = 242 \div 2 = 121$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q3/5

True or false?

The series 2 + 6 + 18 + 54 has a common ratio of 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q4/5

563

QUESTION ID

S16-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each term is 3 times the one before.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q4/5

True or false?

The sum of the first n terms of a geometric series with first term a and ratio $r \neq 1$ is $a \cdot r^n$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q5/5

QUESTION ID

S16-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $a \cdot r^n$ is not the sum; the sum is $a(1 - r^n) \div (1 - r)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q5/5

Explain each step in solving an equation and justify the solution method

MAIN	Algebra I main question	**HSA-REI.A.1**
BACKWARD 1	Use substitution to decide whether a number makes an equation…	**6.EE.B.5**
BACKWARD 2	Solve linear equations by expanding and collecting like terms	**8.EE.C.7.b**
FORWARD 1	Justify the steps of a proof about lines and angles	**HSG-CO.C.9**
FORWARD 2	Solve radical and rational equations and identify extraneous solut..	**HSA-REI.A.2**

SET 17 • HSA-REI.A.1

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

Solve 3(x − 4) + 5 = 2x + 7. Give a reason for each step.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q1/5

567

QUESTION ID

S17-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

x = 14

GRADING NOTE REQUIRED WORK: Each step written as an equation with its reason: distributive property (3x − 12 + 5), combining like terms (3x − 7 = 2x + 7), subtraction property of equality (x − 7 = 7), addition property of equality (x = 14).

Form required: the solution with a reason for every step. Check: 3(10) + 5 = 35 and 2(14) + 7 = 35.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q1/5

A student solves 5x − 8 = 12.

Step 1: 5x = 20

Step 2: x = 4

Which property justifies Step 1?

A. Distributive property

B. Addition property of equality: add 8 to both sides

C. Subtraction property of equality: subtract 8 from both sides

D. Multiplication property of equality: multiply both sides by 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q2/5

568

QUESTION ID

S17-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Addition property of equality: add 8 to both sides

GRADING NOTE Adding 8 to both sides of 5x − 8 = 12 gives 5x = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q2/5

A student wrote:

4(x + 2) = 20

4x + 2 = 20

4x = 18

x = 4.5

Which step has the error? What is the correct solution?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q3/5

569

QUESTION ID

S17-M-Q3

TYPE

Short answer

CORRECT ANSWER

The step from 4(x + 2) = 20 to 4x + 2 = 20: the 4 must multiply both terms, giving 4x + 8 = 20. The solution is x = 3.

GRADING NOTE Form required: the step with the error and the correct solution. 4x + 8 = 20, so 4x = 12 and x = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q3/5

True or false?

If a = b, then a ÷ c = b ÷ c for every number c.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q4/5

570

QUESTION ID

S17-M-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Division by 0 is undefined, so the property holds only when c ≠ 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q4/5

True or false?

Subtracting 2x from both sides of $2(x + 1) = 2x + 2$ gives $2 = 2$, so every real number is a solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q5/5

571

QUESTION ID

S17-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $2(x + 1) = 2x + 2$ is true for every x. The step leaves a true statement with no variable, so the solution set is all real numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q5/5

SET 17 • HSA-REI.A.1

BACKWARD 1

Use substitution to decide whether a number makes an equation true • Grade 6 • Expressions & Equations • 6.EE.B.5

Is x = 3 a solution of 4x − 5 = 7? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q1/5

573

QUESTION ID

S17-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes.

4(3) − 5 = 12 − 5 = 7.

GRADING NOTE Any equivalent explanation that substitutes 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q1/5

Which number from 1, 2, 3, 4 makes 2x + 1 = 9 true?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q2/5

574

QUESTION ID

S17-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. 2(4) + 1 = 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q2/5

Which value of y makes 15 − y = 9 true?

A. 6

B. −6

C. 9

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q3/5

575

QUESTION ID

S17-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6

GRADING NOTE 15 − 6 = 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q3/5

True or false?

$x = 5$ is a solution of $3x = 15$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q4/5

576

QUESTION ID

S17-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $3 \cdot 5 = 15$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q4/5

True or false?

n = 2 is a solution of n + 7 = 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q5/5

577

QUESTION ID

S17-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2 + 7 = 9, not 5. The solution is n = −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • HSA-REI.A.1

BACKWARD 2

Solve linear equations by expanding and collecting like terms • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve 2(x + 3) = 14.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q1/5

579

QUESTION ID

S17-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. 2x + 6 = 14, so 2x = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 5y − 2y + 4 = 19.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q2/5

580

QUESTION ID

S17-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. 3y + 4 = 19, so 3y = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q2/5

Solve 3(n − 2) = n + 8.

A. n = 7

B. n = 5

C. n = 3

D. n = 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q3/5

581

QUESTION ID

S17-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. n = 7

GRADING NOTE 3n − 6 = n + 8, so 2n = 14 and n = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q3/5

True or false?

The solution of $0.5x + 1 = 4$ is $x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q4/5

582

QUESTION ID

S17-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $0.5x = 3$, so $x = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q4/5

True or false?

The solution of $4(x - 1) = 12$ is $x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q5/5

583

QUESTION ID

S17-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4x - 4 = 12$, so $4x = 16$ and $x = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q5/5

SET 17 • HSA-REI.A.1

FORWARD 1

Justify the steps of a proof about lines and angles • High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9

Angles 1 and 2 are vertical angles. Angle 1 and angle 3 form a linear pair, and so do angle 2 and angle 3.

Explain why m∠1 = m∠2.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q1/5

585

QUESTION ID

S17-F1-Q1

TYPE

Short answer

CORRECT ANSWER

m∠1 + m∠3 = 180° and m∠2 + m∠3 = 180°, so m∠1 + m∠3 = m∠2 + m∠3. Subtracting m∠3 from both sides gives m∠1 = m∠2.

GRADING NOTE Any equivalent explanation that uses both linear pairs and the subtraction property of equality.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q1/5

In a proof, the line m∠A + 40° = 90° is followed by m∠A = 50°.

Which reason justifies that step?

A. Definition of a linear pair

B. Vertical angles are congruent

C. Addition property of equality

D. Subtraction property of equality

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q2/5

586

QUESTION ID

S17-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Subtraction property of equality

GRADING NOTE 40° is subtracted from both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q2/5

Two angles form a linear pair. One measures (3x + 10)° and the other measures (2x + 20)°.

Find x and both angle measures.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q3/5

587

QUESTION ID

S17-F1-Q3

TYPE

Short answer

CORRECT ANSWER

x = 30; the angles are 100° and 80°.

GRADING NOTE Form required: x and both angle measures. 5x + 30 = 180, so x = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q3/5

True or false?

Vertical angles have equal measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q4/5

588

QUESTION ID

S17-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the vertical angles theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q4/5

True or false?

If two angles have equal measures, they must be vertical angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q5/5

589

QUESTION ID

S17-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The converse is false: for example, two angles in different parts of a figure can both measure 40°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q5/5

SET 17 • HSA-REI.A.1

FORWARD 2

Solve radical and rational equations and identify extraneous solutions • High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve √(x + 3) = 5.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q1/5

591

QUESTION ID

S17-F2-Q1

TYPE

Short answer

CORRECT ANSWER

22

GRADING NOTE Any equivalent form. Square both sides: x + 3 = 25. Check: √25 = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q1**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q1/5

Solve $\sqrt{(2x + 3)} = x$. Name any extraneous solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q2/5

592

QUESTION ID

S17-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x = 3; x = −1 is extraneous.

GRADING NOTE Form required: the solution and the extraneous value. Squaring gives $x^2 - 2x - 3 = 0$, so $x = 3$ or $x = -1$. For $x = -1$, $\sqrt{1} = 1 \neq -1$, so -1 is not a solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q2/5

Assume $x \neq 0$. Solve $6 \div x = 3$.

A. $x = \frac{1}{2}$

B. $x = 3$

C. $x = 18$

D. $x = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q3/5

593

QUESTION ID

S17-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 2

GRADING NOTE Multiply both sides by x: 6 = 3x, so x = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q3/5

True or false?

$x = 4$ is a solution of $\sqrt{x} = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q4/5

594

QUESTION ID

S17-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{4} = 2$, and a principal square root is never negative, so the equation has no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q4**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q4/5

True or false?

Squaring both sides of an equation can produce a value that does not satisfy the original equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q5/5

595

QUESTION ID

S17-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example, $x = 2$ has one solution, but $x^2 = 4$ also has $x = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q5/5

Solve linear equations in one variable, including equations with letter coefficients

MAIN	Algebra I main question	**HSA-REI.B.3**
BACKWARD 1	Name the property that justifies a step	**HSA-REI.A.1**
BACKWARD 2	Solve two-step equations from word problems	**7.EE.B.4.a**
FORWARD 1	Prove and apply theorems about triangles	**HSG-CO.C.10**
FORWARD 2	Solve rational equations that lead to linear equations	**HSA-REI.A.2**

SET 18 • HSA-REI.B.3

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\frac{2}{3}x - 5 = \frac{1}{4}x + 5$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q1/5

598

QUESTION ID

S18-M-Q1

TYPE

Short answer

CORRECT ANSWER

24

GRADING NOTE Any equivalent form. Multiply both sides by 12: 8x − 60 = 3x + 60, so 5x = 120. Check: 16 − 5 = 11 and 6 + 5 = 11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q1/5

Solve 0.4(3x − 5) = 2.2x + 1.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q2/5

599

QUESTION ID

S18-M-Q2

TYPE

Short answer

CORRECT ANSWER

−3

GRADING NOTE Any equivalent form. 1.2x − 2 = 2.2x + 1, so −x = 3. Check: 0.4(−14) = −5.6 and 2.2(−3) + 1 = −5.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

How many solutions does $5x - 2(x + 4) = 3x + 1$ have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q3/5

600

QUESTION ID

S18-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. None

GRADING NOTE The left side is $3x - 8$, so $3x - 8 = 3x + 1$ gives $-8 = 1$, which is false for every x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve ax + b = c for x, where a ≠ 0.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q4/5

601

QUESTION ID

S18-M-Q4

TYPE

Short answer

CORRECT ANSWER

x = (c − b) ÷ a

GRADING NOTE Any equivalent form, such as c/a − b/a. Subtract b, then divide by a (allowed because a ≠ 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q4/5

True or false?

The equation $2(x - 3) = 2x - 6$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q5/5

602

QUESTION ID
S18-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Both sides equal $2x - 6$ for every x, so every real number is a solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q5/5

SET 18 • HSA-REI.B.3

BACKWARD 1

Name the property that justifies a step • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

Name the property used to go from 3x + 4 = 19 to 3x = 15.

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q1/5

604

QUESTION ID

S18-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Subtraction property of equality (subtract 4 from both sides)

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q1/5

Name the property used to go from $\frac{x}{5} = 3$ to $x = 15$.

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q2/5

605

QUESTION ID

S18-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Multiplication property of equality (multiply both sides by 5)

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q2/5

Which step keeps the same solutions as 2x = 10?

A. Divide both sides by 2

B. Square the right side only

C. Add 2 to the left side only

D. Multiply both sides by 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q3/5

606

QUESTION ID

S18-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Divide both sides by 2

GRADING NOTE Doing the same nonzero operation to both sides keeps the solutions. The other steps change the equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q3/5

True or false?

Adding the same number to both sides of an equation keeps the same solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q4/5

607

QUESTION ID

S18-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Addition property of equality.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q4/5

True or false?

Multiplying both sides of an equation by 0 keeps the same solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q5/5

608

QUESTION ID

S18-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It turns every equation into 0 = 0, which every number satisfies.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • HSA-REI.B.3

BACKWARD 2

Solve two-step equations from word problems • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A taxi charges $3 plus $2 per mile. A ride cost $19.

How many miles was the ride?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q1/5

610

QUESTION ID

S18-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form. 3 + 2m = 19, so 2m = 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q1/5

Solve −4x + 9 = 1.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q2/5

611

QUESTION ID

S18-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. −4x = −8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\frac{1}{2}(x + 6) = 10$.

A. x = 14

B. x = 17

C. x = 2

D. x = 26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q3/5

612

QUESTION ID

S18-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = 14

GRADING NOTE x + 6 = 20, so x = 14.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q3/5

True or false?

The solution of $6x - 4 = 20$ is $x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q4/5

613

GRADING NOTE $6x = 24$.

QUESTION ID

S18-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q4/5

True or false?

The solution of $2x + 5 = -1$ is $x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q5/5

614

QUESTION ID

S18-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2x = -6$, so $x = -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q5/5

SET 18 • HSA-REI.B.3

FORWARD 1

Prove and apply theorems about triangles • High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10

Prove that the angles of any triangle ABC add to 180°.
Start by drawing the line through B parallel to AC.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q1/5

616

QUESTION ID
S18-F1-Q1

TYPE
Answer with required work

CORRECT ANSWER
m∠A + m∠B + m∠C = 180°

GRADING NOTE REQUIRED WORK: The parallel line through B makes two angles with BA and BC; each equals ∠A or ∠C because they are alternate interior angles; the three angles at B form a straight angle (180°).

Form required: the proof with reasons (parallel postulate, alternate interior angles, straight angle).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q1/5

Explain why an exterior angle of a triangle equals the sum of the two remote interior angles.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q2/5

617

QUESTION ID

S18-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The exterior angle and the adjacent interior angle form a linear pair, so they add to 180°. The three interior angles also add to 180°. Subtracting the adjacent angle from both shows the exterior angle equals the other two interior angles together.

GRADING NOTE Any equivalent explanation that uses the linear pair and the triangle angle sum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q2/5

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C. (Draw the bisector of ∠A, meeting BC at D.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q3/5

618

QUESTION ID

S18-F1-Q3

TYPE

Answer with required work

CORRECT ANSWER

∠B ≅ ∠C

GRADING NOTE REQUIRED WORK: Triangles ABD and ACD are congruent by SAS (AB = AC, ∠BAD = ∠CAD, AD = AD), so the corresponding angles ∠B and ∠C are congruent.

Form required: the congruence with its reasons and the corresponding-parts conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q3/5

D and E are the midpoints of AB and AC in triangle ABC, and BC = 18.

What does the midsegment theorem let you conclude?

A. DE ⊥ BC and DE = 9

B. DE ∥ BC and DE = 18

C. DE = 9, but DE need not be parallel to BC

D. DE ∥ BC and DE = 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q4/5

619

QUESTION ID

S18-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. DE ∥ BC and DE = 9

GRADING NOTE The segment joining the midpoints of two sides is parallel to the third side and half as long.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q4/5

True or false?

The three medians of a triangle meet at one point, and that point is $\frac{2}{3}$ of the way from each vertex to the midpoint of the opposite side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q5/5

620

QUESTION ID

S18-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the centroid theorem, one of the triangle theorems of HSG-CO.C.10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q5/5

SET 18 • HSA-REI.B.3

FORWARD 2

Solve rational equations that lead to linear equations • High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Assume $x \neq 0$. Solve $3 \div x + 2 = 5 \div x$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q1/5

622

QUESTION ID

S18-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form. Multiply by x: 3 + 2x = 5, so x = 1. Check: 3 + 2 = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Assume $x \neq 2$. Solve $(x + 1) \div (x - 2) = 4$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q2/5

623

QUESTION ID

S18-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. $x + 1 = 4x - 8$, so $x = 3$. Check: $4 \div 1 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F2-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q2/5

Assume $x \neq 3$. Solve $x \div (x - 3) = 3 \div (x - 3)$.

A. $x = 0$

B. No solution: $x = 3$ is extraneous

C. $x = 3$

D. Every real number except 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q3/5

624

QUESTION ID

S18-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. No solution: x = 3 is extraneous

GRADING NOTE Multiplying by $x - 3$ gives $x = 3$, but 3 makes the denominator 0, so it is not a solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q3/5

True or false?

$x = 1$ is a solution of $2x \div (x + 1) = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q4/5

625

QUESTION ID

S18-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $2 \div 2 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q4/5

True or false?

The solution of $6 \div (x + 1) = 2$ is $x = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S18-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q5/5

626

QUESTION ID

S18-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x + 1 = 3$, so $x = 2$. For $x = 3$, $6 \div 4 = 1.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S18-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q5/5

Solve linear inequalities and compound inequalities in one variable

MAIN	Algebra I main question	**HSA-REI.B.3**
BACKWARD 1	Solve two-step inequalities from word problems	**7.EE.B.4.b**
BACKWARD 2	Write and graph simple inequalities	**6.EE.B.8**
FORWARD 1	Use inequalities to meet constraints in a design problem	**HSG-MG.A.3**
FORWARD 2	Write a margin-of-error interval as a compound inequality — neare…	**HSS-IC.B.4**

SET 19 • HSA-REI.B.3

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $5 - 3x \leq 17$.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q1/5

629

QUESTION ID

S19-M-Q1

TYPE

Short answer

CORRECT ANSWER

$x \geq -4$

GRADING NOTE Form required: an inequality in x ($x \geq -4$ or $-4 \leq x$). $-3x \leq 12$; dividing by -3 reverses the sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $-2 < 3x + 4 \leq 13$.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q2/5

630

QUESTION ID

S19-M-Q2

TYPE

Short answer

CORRECT ANSWER

$-2 < x \leq 3$

GRADING NOTE Form required: a compound inequality. Subtract 4 from all three parts ($-6 < 3x \leq 9$), then divide by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q2/5

The number line shows the solution set of which inequality?



A. $x + 5 > 7$

B. $4x + 3 < 2x + 7$

C. $3x \leq 6$

D. $-2x < -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q3/5

631

QUESTION ID

S19-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $4x + 3 < 2x + 7$

GRADING NOTE The graph is $x < 2$. $4x + 3 < 2x + 7$ gives $2x < 4$, so $x < 2$. $x + 5 > 7$ and $-2x < -4$ give $x > 2$, and $3x \leq 6$ gives $x \leq 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\frac{x}{4} - 1 > \frac{x}{2} + 2$.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q4/5

632

QUESTION ID

S19-M-Q4

TYPE

Short answer

CORRECT ANSWER

x < −12

GRADING NOTE Form required: an inequality in x. Multiply by 4: x − 4 > 2x + 8, so −x > 12 and x < −12. Check x = −16: −5 > −6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q4/5

True or false?

The solution of $-4x > 8$ is $x > -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q5/5

633

QUESTION ID
S19-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Dividing by -4 reverses the sign: $x < -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q5/5

SET 19 • HSA-REI.B.3

BACKWARD 1

Solve two-step inequalities from word problems • Grade 7 • Expressions & Equations • 7.EE.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve 3x + 2 > 14.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q1/5

635

QUESTION ID

S19-B1-Q1

TYPE

Short answer

CORRECT ANSWER

x > 4

GRADING NOTE Form required: an inequality in x. 3x > 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q1/5

A phone plan costs $20 plus $5 per gigabyte g. Lena can spend at most $45.

Write and solve an inequality for g.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q2/5

636

QUESTION ID

S19-B1-Q2

TYPE

Short answer

CORRECT ANSWER

20 + 5g ≤ 45; g ≤ 5, so at most 5 gigabytes.

GRADING NOTE Form required: the inequality and its solution. 5g ≤ 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q2/5

Solve $-2x + 1 \geq 9$.

A. $x \leq -4$

B. $x \geq 5$

C. $x \leq 4$

D. $x \geq -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q3/5

637

QUESTION ID

S19-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $x \leq -4$

GRADING NOTE $-2x \geq 8$; dividing by -2 reverses the sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q3/5

True or false?

$x = 6$ is a solution of $2x - 3 > 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q4/5

638

QUESTION ID

S19-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $2(6) - 3 = 9$, and $9 > 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q4/5

True or false?

Dividing both sides of an inequality by a positive number reverses the inequality sign.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q5/5

639

QUESTION ID

S19-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only multiplying or dividing by a negative number reverses the sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q5/5

SET 19 • HSA-REI.B.3

BACKWARD 2

Write and graph simple inequalities • Grade 6 • Expressions & Equations • 6.EE.B.8

A rider must be at least 48 inches tall.

Write an inequality for the rider's height h.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q1/5

641

QUESTION ID

S19-B2-Q1

TYPE

Short answer

CORRECT ANSWER

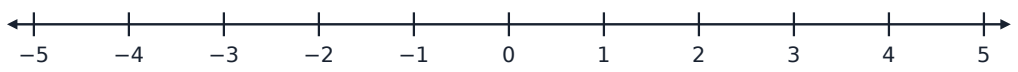
h ≥ 48

GRADING NOTE Form required: an inequality. "At least" includes 48.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q1/5

Graph x < 3 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q2/5

642

QUESTION ID

S19-B2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

An open circle at 3 and a ray shaded to the left

GRADING NOTE Any correct graph. The open circle shows 3 is not included.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q2/5

Which number is a solution of $x > -2$?

A. -2

B. -3

C. -5

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q3/5

643

QUESTION ID

S19-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0

GRADING NOTE 0 is greater than −2; −2 itself is not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q3/5

True or false?

The inequality x > 5 has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q4/5

644

QUESTION ID

S19-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every number greater than 5 is a solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q4/5

True or false?

On a number line, $x \leq 2$ is shown with an open circle at 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q5/5

645

QUESTION ID

S19-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE ≤ includes 2, so the circle is closed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • HSA-REI.B.3

FORWARD 1

Use inequalities to meet constraints in a design problem • High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A rectangular garden is 12 ft long. Its perimeter can be at most 50 ft.

Write and solve an inequality for the width w.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q1/5

647

QUESTION ID

S19-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$24 + 2w \leq 50$;

$0 < w \leq 13$ feet

GRADING NOTE Form required: the inequality and its solution. A width must be positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q1**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q1/5

A box with a 10 in by 10 in base must hold at least 600 in^3.

Write and solve an inequality for the height h.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q2/5

QUESTION ID

S19-F1-Q2

TYPE

Short answer

CORRECT ANSWER

100h ≥ 600; h ≥ 6 inches

GRADING NOTE Form required: the inequality and its solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q2/5

A poster 15 in wide must have an area of at least 300 in².

Which lengths L work?

A. L ≥ 285 inches

B. L ≥ 20 inches

C. L ≥ 4.5 inches

D. L ≤ 20 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q3/5

649

QUESTION ID

S19-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. L ≥ 20 inches

GRADING NOTE 15L ≥ 300, so L ≥ 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q3/5

True or false?

A rectangle 8 cm long and w cm wide has an area of at most 40 cm² exactly when $0 < w \leq 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q4/5

650

QUESTION ID
S19-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $8w \leq 40$ gives $w \leq 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q4/5

True or false?

A square with side s cm has a perimeter less than 20 cm exactly when $s < 20$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q5/5

651

QUESTION ID
S19-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $4s < 20$ gives $s < 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q5/5

SET 19 • HSA-REI.B.3

FORWARD 2

Write a margin-of-error interval as a compound inequality • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: A margin-of-error interval is written as a compound inequality like those solved in the MAIN set.

A survey estimates that 54% of voters support a plan, with a margin of error of 3 percentage points.

Write the interval for the true proportion p as a compound inequality.

Interval: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q1/5 • NEAREST RELATED

653

QUESTION ID

S19-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$0.51 \leq p \leq 0.57$

GRADING NOTE Any equivalent form, such as 51% ≤ p ≤ 57%. 54 − 3 = 51 and 54 + 3 = 57.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F2-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

A poll reports 62% with a margin of error of 4 percentage points.

Is 60% inside the interval? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q2/5 • NEAREST RELATED

654

QUESTION ID

S19-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. The interval is 58% ≤ p ≤ 66%, and 60% is in it.

GRADING NOTE Any equivalent explanation that states the interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q2/5 • NEAREST RELATED

The interval $0.40 \leq p \leq 0.48$ is an estimate plus or minus a margin of error.

What are the estimate and the margin of error?

A. 0.44 and 0.08

B. 0.40 and 0.08

C. 0.44 and 0.04

D. 0.48 and 0.04

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q3/5 • NEAREST RELATED

655

QUESTION ID
S19-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. 0.44 and 0.04

GRADING NOTE The estimate is the midpoint, 0.44; the margin is half the width, 0.04.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

An estimate of 70% with a margin of error of 5 percentage points gives $65\% \le p \le 75\%$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q4/5 • NEAREST RELATED

656

QUESTION ID
S19-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $70 - 5 = 65$ and $70 + 5 = 75$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

If the interval is $0.30 \leq p \leq 0.36$, the margin of error is 0.06.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q5/5 • NEAREST RELATED

657

QUESTION ID

S19-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The margin is half the width: $0.06 \div 2 = 0.03$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q5/5 • NEAREST RELATED

Create linear equations and inequalities in one variable and use them to solve problems

MAIN	Algebra I main question	**HSA-CED.A.1**
BACKWARD 1	Solve linear equations and inequalities	**HSA-REI.B.3**
BACKWARD 2	Write and solve one-step equations	**6.EE.B.7**
FORWARD 1	Write and solve equations from similar triangles	**HSG-SRT.B.5**
FORWARD 2	Write and solve an equation for an expected value — nearest relat.	**HSS-MD.A.2**

SET 20 • HSA-CED.A.1

MAIN

Algebra I main question • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A plumber charges a $65 call fee plus $48 per hour. A bill is $257.

Write an equation for the number of hours h, and solve it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q1/5

660

QUESTION ID

S20-M-Q1

TYPE

Short answer

CORRECT ANSWER

65 + 48h = 257;
h = 4 hours

GRADING NOTE Form required: an equation and its solution. 48h = 192.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q1/5

The sum of three consecutive integers is 141.

Write an equation and find the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q2/5

661

QUESTION ID

S20-M-Q2

TYPE

Short answer

CORRECT ANSWER

n + (n + 1) + (n + 2) = 141; the integers are 46, 47, and 48.

GRADING NOTE Form required: an equation and all three integers. 3n + 3 = 141, so n = 46.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q2/5

Maya has $180 and saves $15 each week. She wants at least $400.

Write and solve an inequality for the number of weeks w. How many whole weeks must she save?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q3/5

662

QUESTION ID

S20-M-Q3

TYPE

Short answer

CORRECT ANSWER

180 + 15w ≥ 400; w ≥ $14\frac{2}{3}$, so at least 15 weeks.

GRADING NOTE Form required: the inequality, its solution, and the whole number of weeks. 15w ≥ 220. After 14 weeks she has only $390.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q3/5

A rental costs $40 per day plus a one-time $25 fee.

Which inequality gives the number of days d you can rent for at most $265?

A. 40(d + 25) ≤ 265

B. 40d + 25 ≥ 265

C. 40d + 25 ≤ 265

D. 25d + 40 ≤ 265

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q4/5

663

QUESTION ID

S20-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 40d + 25 ≤ 265

GRADING NOTE The cost 40d + 25 must be at most (≤) 265.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q4/5

True or false?

Plan A costs 30 + 0.10m dollars and Plan B costs 0.25m dollars for m minutes. Plan A costs less when m < 200.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q5/5

664

QUESTION ID

S20-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 30 + 0.10m < 0.25m gives 30 < 0.15m, so m > 200. For fewer than 200 minutes, Plan B costs less.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q5/5

SET 20 • HSA-CED.A.1

BACKWARD 1

Solve linear equations and inequalities • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve 7x − 4 = 3x + 20.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q1/5

666

QUESTION ID

S20-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. 4x = 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-B1-Q1**

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2(5 − x) ≥ 16.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q2/5

667

QUESTION ID

S20-B1-Q2

TYPE

Short answer

CORRECT ANSWER

x ≤ −3

GRADING NOTE Form required: an inequality in x. 10 − 2x ≥ 16, so −2x ≥ 6 and x ≤ −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 0.25t + 3 = 7.

A. t = 1

B. t = 40

C. t = 2.5

D. t = 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q3/5

668

QUESTION ID

S20-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. t = 16

GRADING NOTE 0.25t = 4, so t = 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q3/5

True or false?

$x = -3$ is a solution of $4 - x > 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q4/5

669

QUESTION ID

S20-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4 - (-3) = 7$, and $7 > 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q4/5

True or false?

The solution of $12 - 3x < 0$ is $x < 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q5/5

670

QUESTION ID
S20-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $-3x < -12$; dividing by -3 reverses the sign, so $x > 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • HSA-CED.A.1

BACKWARD 2

Write and solve one-step equations • Grade 6 • Expressions & Equations • 6.EE.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve x + 17 = 42.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q1/5

672

QUESTION ID

S20-B2-Q1

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Any equivalent form. Subtract 17.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Tickets cost $9 each. Sam spent $63.

Write an equation and find the number of tickets t.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q2/5

673

QUESTION ID

S20-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9t = 63; t = 7

GRADING NOTE Form required: an equation and its solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q2/5

Solve 4.5 = 1.5y.

A. y = 6.75

B. y = 6

C. y = 3

D. y = 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q3/5

674

QUESTION ID

S20-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. y = 3

GRADING NOTE 4.5 ÷ 1.5 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q3/5

True or false?

If n − 8 = 15, then n = 23.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q4/5

675

QUESTION ID

S20-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Add 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q4/5

True or false?

If 6p = 3, then p = 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q5/5

676

QUESTION ID

S20-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $p = 3 \div 6 = \frac{1}{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • HSA-CED.A.1

FORWARD 1

Write and solve equations from similar triangles • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

A 6-ft person casts a 4-ft shadow at the same time a tree casts a 22-ft shadow.

Write a proportion and find the tree's height.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q1/5

678

QUESTION ID

S20-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6/4 = h/22; h = 33 feet

GRADING NOTE Form required: a proportion and the height. The triangles formed by the sun's rays are similar.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F1-Q1**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q1/5

Triangle ABC ~ triangle DEF with AB = 8, BC = 12, and DE = 10.

Find EF.

EF = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q2/5

679

QUESTION ID

S20-F1-Q2

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. 8/10 = 12/EF, so EF = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q2/5

Triangle PQR ~ triangle STU, with PQ = 5, ST = 15, QR = x + 2, and TU = 2x + 9.

What is x?

A. x = 1

B. x = 4.5

C. x = 3

D. x = 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q3/5

680

QUESTION ID

S20-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 3

GRADING NOTE The scale factor is 3, so 3(x + 2) = 2x + 9 and x = 3 (QR = 5, TU = 15).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q3/5

True or false?

If two triangles are similar, their corresponding side lengths are proportional.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q4/5

681

QUESTION ID

S20-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is what similarity means for triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q4/5

True or false?

In similar triangles with a ratio of 2 : 3, a side of 10 corresponds to a side of 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q5/5

682

QUESTION ID

S20-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $10 \cdot \frac{3}{2} = 15$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q5/5

SET 20 • HSA-CED.A.1

FORWARD 2

Write and solve an equation for an expected value • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Writing and solving a linear equation from a situation is applied to an expected-value condition.

ANSWER KEY Reduced copy of the question page, with its answer

A game pays $10 with probability 0.2 and $0 otherwise.

What is the expected payout?

Expected payout: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q1/5 • NEAREST RELATED

684

QUESTION ID

S20-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$2

GRADING NOTE Any equivalent form. 10(0.2) + 0(0.8) = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F2-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q1/5 • NEAREST RELATED

A game pays x dollars with probability 0.25 and $0 otherwise.

Write and solve an equation for x so that the expected payout is $3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q2/5 • NEAREST RELATED

685

QUESTION ID

S20-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.25x = 3; x = 12

GRADING NOTE Form required: an equation and its solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q2/5 • NEAREST RELATED

A random variable is 1 with probability 0.5, 2 with probability 0.3, and 4 with probability 0.2.

What is its expected value?

A. 7

B. 1.9

C. 2

D. 2.33

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q3/5 • NEAREST RELATED

686

QUESTION ID
S20-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. 1.9

GRADING NOTE $1(0.5) + 2(0.3) + 4(0.2) = 0.5 + 0.6 + 0.8 = 1.9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

The expected value of one roll of a fair number cube numbered 1 to 6 is 3.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q4/5 • NEAREST RELATED

687

QUESTION ID
S20-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $(1 + 2 + 3 + 4 + 5 + 6) \div 6 = 3.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A game costs $5 to play. It pays $20 with probability 0.2 and $0 otherwise. The expected net gain is $4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q5/5 • NEAREST RELATED

688

QUESTION ID
S20-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The expected payout is 20(0.2) + 0(0.8) = $4, so the expected net gain is 4 − 5 = −$1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q5/5 • NEAREST RELATED

Rearrange formulas to highlight a quantity of interest

MAIN	Algebra I main question	**HSA-CED.A.4**
BACKWARD 1	Solve linear equations with letter coefficients	**HSA-REI.B.3**
BACKWARD 2	Evaluate formulas for given values	**6.EE.A.2.c**
FORWARD 1	Rearrange the arc-length formula $s = r\theta$	**HSG-C.B.5**
FORWARD 2	Solve exponential formulas for the exponent with logarithms	**HSF-LE.A.4**

SET 21 • HSA-CED.A.4

MAIN

Algebra I main question • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4

Solve $V = \pi r^2 h$ for h. ($r > 0$)

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q1/5

691

QUESTION ID

S21-M-Q1

TYPE

Short answer

CORRECT ANSWER

$h = V \div (\pi r^2)$

GRADING NOTE Any equivalent form. Divide both sides by πr^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $C = \frac{5}{9}(F - 32)$ for F.

F = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q2/5

692

QUESTION ID

S21-M-Q2

TYPE

Short answer

CORRECT ANSWER

$F = \frac{9}{5}C + 32$

GRADING NOTE Any equivalent form, such as F = 1.8C + 32. Multiply by $\frac{9}{5}$, then add 32.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve A = P(1 + rt) for t. (P ≠ 0, r ≠ 0)

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q3/5

693

QUESTION ID

S21-M-Q3

TYPE

Short answer

CORRECT ANSWER

t = (A − P) ÷ (Pr)

GRADING NOTE Any equivalent form, such as (A/P − 1)/r. Divide by P: A/P = 1 + rt; subtract 1, then divide by r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q3/5

Ohm's law is V = IR. Solve for R. (I ≠ 0)

A. $R = \frac{I}{V}$

B. $R = V - I$

C. $R = VI$

D. $R = \frac{V}{I}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q4/5

694

QUESTION ID

S21-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. R = V/I

GRADING NOTE Divide both sides by I.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q4/5

True or false?

Solving $y = mx + b$ for x ($m \neq 0$) gives $x = y \div m - b$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q5/5

695

QUESTION ID

S21-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Subtract b first, then divide: $x = (y - b) \div m$. The given form divides only y by m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q5/5

SET 21 • HSA-CED.A.4

BACKWARD 1

Solve linear equations with letter coefficients • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve 3x + k = 10 for x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q1/5

697

QUESTION ID

S21-B1-Q1

TYPE

Short answer

CORRECT ANSWER

x = (10 − k) ÷ 3

GRADING NOTE Any equivalent form. Subtract k, then divide by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve ax = bx + c for x, where a ≠ b.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q2/5

698

QUESTION ID

S21-B1-Q2

TYPE

Short answer

CORRECT ANSWER

x = c ÷ (a − b)

GRADING NOTE Any equivalent form. (a − b)x = c, and a − b ≠ 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x \div m = n$ for x. ($m \neq 0$)

A. $x = m + n$

B. $x = mn$

C. $x = n \div m$

D. $x = m \div n$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q3/5

699

QUESTION ID
S21-B1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. x = mn

GRADING NOTE Multiply both sides by m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If 2x − p = q, then x = (p + q) ÷ 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q4/5

700

QUESTION ID

S21-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 2x = p + q.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q4/5

True or false?

If $kx = 5$ and $k \neq 0$, then $x = 5k$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q5/5

701

QUESTION ID

S21-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x = 5 \div k$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • HSA-CED.A.4

BACKWARD 2

Evaluate formulas for given values • Grade 6 • Expressions & Equations • 6.EE.A.2.c

ANSWER KEY Reduced copy of the question page, with its answer

Use A = lw to find A when l = 9 and w = 4.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q1/5

703

QUESTION ID

S21-B2-Q1

TYPE

Short answer

CORRECT ANSWER

36

GRADING NOTE Any equivalent form. 9 · 4 = 36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q1/5

Use d = rt to find d when r = 55 and t = 3.

d = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q2/5

704

QUESTION ID

S21-B2-Q2

TYPE

Short answer

CORRECT ANSWER

165

GRADING NOTE Any equivalent form. 55 · 3 = 165.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q2/5

Use P = 2l + 2w to find P when l = 7 and w = 3.

A. 10

B. 21

C. 20

D. 42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q3/5

705

QUESTION ID

S21-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 20

GRADING NOTE 14 + 6 = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q3/5

True or false?

$V = s^3$ gives $V = 27$ when $s = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q4/5

706

QUESTION ID

S21-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $3^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q4/5

True or false?

$A = \frac{1}{2}bh$ gives $A = 40$ when $b = 10$ and $h = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q5/5

707

QUESTION ID

S21-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\frac{1}{2} \cdot 10 \cdot 4 = 20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q5/5

SET 21 • HSA-CED.A.4

FORWARD 1

Rearrange the arc-length formula s = rθ • High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has radius 6 cm. An arc has length 9 cm.

Use $s = r\theta$ to find the central angle θ in radians.

$\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q1/5

709

QUESTION ID

S21-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1.5 radians

GRADING NOTE Any equivalent form. $\theta = s \div r = 9 \div 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q1/5

Solve s = rθ for r. Then find r when s = 10π and θ = π ÷ 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q2/5

710

QUESTION ID

S21-F1-Q2

TYPE

Short answer

CORRECT ANSWER

r = s ÷ θ; r = 20

GRADING NOTE Form required: the formula for r and the value. 10π ÷ (π ÷ 2) = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q2/5

The area of a sector is $A = \frac{1}{2}r^2\theta$. Solve for θ. ($r > 0$)

A. $\theta = A - \frac{1}{2}r^2$

B. $\theta = 2Ar^2$

C. $\theta = A \div (2r^2)$

D. $\theta = 2A \div r^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q3/5

711

QUESTION ID

S21-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $\theta = 2A \div r^2$

GRADING NOTE Multiply by 2, then divide by r^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q3/5

True or false?

An arc of length 2πr has a central angle of 2π radians.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q4/5

712

QUESTION ID

S21-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE θ = 2πr ÷ r = 2π, the whole circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q4/5

True or false?

If $s = r\theta$, then $\theta = r \div s$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q5/5

713

QUESTION ID

S21-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\theta = s \div r$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q5/5

SET 21 • HSA-CED.A.4

FORWARD 2

Solve exponential formulas for the exponent with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $A = Pe^{rt}$ for t. (A, P, r > 0)

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q1/5

715

QUESTION ID

S21-F2-Q1

TYPE

Short answer

CORRECT ANSWER

t = ln(A ÷ P) ÷ r

GRADING NOTE Any equivalent form, such as (ln A − ln P) ÷ r. Divide by P, take ln of both sides, then divide by r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q1/5

$500 grows to $1,000 with continuous interest at r = 0.05.

Find t to the nearest tenth of a year.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q2/5

716

QUESTION ID

S21-F2-Q2

TYPE

Short answer

CORRECT ANSWER

13.9 years

GRADING NOTE Form required: rounded to the nearest tenth. t = ln 2 ÷ 0.05 ≈ 13.86.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q2/5

Solve $2^x = y$ for x. ($y > 0$)

A. $x = y^2$

B. $x = \sqrt{y}$

C. $x = y \div 2$

D. $x = \log_2 y$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q3/5

717

QUESTION ID

S21-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x = \log_2 y$

GRADING NOTE By definition, x is the exponent on 2 that gives y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q3/5

True or false?

If $10^k = 1{,}000$, then k = 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S21-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q4/5

718

QUESTION ID

S21-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $10^3 = 1{,}000$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S21-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q4/5

True or false?

Solving $N = 5 \cdot 3^t$ for t gives $t = \log_3 N - 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q5/5

719

QUESTION ID

S21-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Divide by 5 first: $3^t = N \div 5$, so $t = \log_3(N \div 5)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q5/5

Complete the square to solve quadratic equations and derive the quadratic formula

MAIN	Algebra I main question	**HSA-REI.B.4.a**
BACKWARD 1	Complete the square to write a quadratic in vertex form	**HSA-SSE.B.3.b**
BACKWARD 2	Solve $x^2 = p$ and $x^3 = p$ with square and cube roots	**8.EE.A.2**
FORWARD 1	Complete the square to find the center and radius of a circle	**HSG-GPE.A.1**
FORWARD 2	Complete the square to write the equation of an ellipse	**HSG-GPE.A.3**

SET 22 • HSA-REI.B.4.a

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 6x - 7 = 0$ by completing the square.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q1/5

722

QUESTION ID

S22-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

x = 1 or x = −7

GRADING NOTE REQUIRED WORK: The equation rewritten as $(x + 3)^2 = 16$, then the square root of both sides ($x + 3 = 4$ or $x + 3 = -4$).

Form required: both solutions with the completed square shown. $x^2 + 6x = 7$; add 9 to both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $x^2 - 10x + 18 = 0$ in the form $(x - p)^2 = q$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q2/5

723

QUESTION ID

S22-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 5)^2 = 7$

GRADING NOTE Form required: $(x - p)^2 = q$. $x^2 - 10x = -18$; add 25 to both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q2/5

Rewrite $2x^2 + 12x + 5 = 0$ in the form $(x - p)^2 = q$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q3/5

724

QUESTION ID

S22-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(x + 3)^2 = 6.5$

GRADING NOTE Form required: $(x - p)^2 = q$, with q as any equivalent number (13/2 or 6.5). Divide by 2: $x^2 + 6x = -2.5$; add 9 to both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-M-Q3**

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q3/5

Which number completes the square?

$x^2 + 8x + ___$

A. 8

B. 16

C. 4

D. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q4/5

725

QUESTION ID

S22-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 16

GRADING NOTE $(8 \div 2)^2 = 16$, and $x^2 + 8x + 16 = (x + 4)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q4/5

True or false?

Completing the square on $ax^2 + bx + c = 0$ ($a \neq 0$) gives $(x + b \div (2a))^2 = (b^2 - 4ac) \div (4a^2)$, which leads to the quadratic formula.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q5/5

726

QUESTION ID

S22-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Divide by a, move c ÷ a, and add $(b \div (2a))^2$ to both sides. Taking square roots gives $x = (-b \pm \sqrt{(b^2 - 4ac)}) \div (2a)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q5/5

SET 22 • HSA-REI.B.4.a

BACKWARD 1

Complete the square to write a quadratic in vertex form • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b

ANSWER KEY Reduced copy of the question page, with its answer

Write $x^2 + 4x + 1$ in the form $(x - h)^2 + k$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q1/5

728

QUESTION ID

S22-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 2)^2 - 3$

GRADING NOTE Form required: $(x - h)^2 + k$. $x^2 + 4x + 4 - 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q1/5

Write $x^2 - 6x + 11$ in the form $(x - h)^2 + k$, and give its minimum value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q2/5

729

QUESTION ID
S22-B1-Q2

TYPE
Short answer

CORRECT ANSWER
$(x - 3)^2 + 2$; the minimum value is 2.

GRADING NOTE Form required: the completed square and the minimum. $(x - 3)^2$ is never negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q2/5

Which expression is equal to $x^2 + 2x + 5$?

A. $(x + 2)^2 + 1$

B. $(x + 1)^2 + 5$

C. $(x + 1)^2 + 4$

D. $(x - 1)^2 + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q3/5

730

QUESTION ID

S22-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $(x + 1)^2 + 4$

GRADING NOTE $(x + 1)^2 = x^2 + 2x + 1$, and $1 + 4 = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q3/5

True or false?

$(x - 4)^2 = x^2 - 8x + 16$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q4/5

731

QUESTION ID

S22-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x - 4)(x - 4) = x^2 - 4x - 4x + 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q4/5

True or false?

$x^2 + 10x = (x + 5)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q5/5

732

QUESTION ID

S22-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 5)^2 = x^2 + 10x + 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q5/5

SET 22 • HSA-REI.B.4.a

BACKWARD 2

Solve $x^2 = p$ and $x^3 = p$ with square and cube roots • Grade 8 • Expressions & Equations • 8.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 = 49$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q1/5

734

QUESTION ID

S22-B2-Q1

TYPE

Short answer

CORRECT ANSWER

x = 7 or x = −7

GRADING NOTE Form required: both solutions. $7^2 = 49$ and $(-7)^2 = 49$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^3 = -64$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q2/5

735

QUESTION ID

S22-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−4

GRADING NOTE Any equivalent form. $(-4)^3 = -64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q2/5

What are the solutions of $x^2 = 81$?

A. 9 and −9

B. 9 only

C. −9 only

D. 40.5 and −40.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q3/5

736

QUESTION ID

S22-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 9 and −9

GRADING NOTE Both 9^2 and $(-9)^2$ equal 81.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q3/5

True or false?

$\sqrt{64} = 8$ and $\sqrt[3]{64} = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q4/5

737

QUESTION ID

S22-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $8^2 = 64$ and $4^3 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The only solution of $x^2 = 25$ is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q5/5

738

QUESTION ID

S22-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE −5 is also a solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • HSA-REI.B.4.a

FORWARD 1

Complete the square to find the center and radius of a circle • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Find the center and radius of the circle
$x^2 + y^2 - 6x + 4y - 12 = 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q1/5

740

QUESTION ID
S22-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Center (3, −2), radius 5

GRADING NOTE Form required: the center and the radius. $(x - 3)^2 + (y + 2)^2 = 12 + 9 + 4 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q1/5

Write an equation of the circle with center (2, −5) and radius 6 in standard form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q2/5

741

QUESTION ID
S22-F1-Q2

TYPE
Short answer

CORRECT ANSWER
$(x - 2)^2 + (y + 5)^2 = 36$

GRADING NOTE Form required: standard form $(x - h)^2 + (y - k)^2 = r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F1-Q2**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q2/5

What is the center of the circle $x^2 + 8x + y^2 = 9$?

A. $(-8, 0)$

B. $(-4, 0)$

C. $(4, 0)$

D. $(0, -4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q3/5

742

QUESTION ID
S22-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. (−4, 0)

GRADING NOTE $(x + 4)^2 + y^2 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F1-Q3**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q3/5

True or false?

The circle $(x - 2)^2 + y^2 = 16$ has radius 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q4/5

743

QUESTION ID
S22-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $r^2 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q4/5

True or false?

The circle $x^2 + (y + 5)^2 = 9$ has center (0, 5).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q5/5

744

QUESTION ID
S22-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $y + 5 = y - (-5)$, so the center is $(0, -5)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q5/5

SET 22 • HSA-REI.B.4.a

FORWARD 2

Complete the square to write the equation of an ellipse • High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

Write $x^2 + 4y^2 - 6x - 7 = 0$ in standard form, and give the center of the ellipse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q1/5

746

QUESTION ID
S22-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$(x - 3)^2 \div 16 + y^2 \div 4 = 1$; center $(3, 0)$

GRADING NOTE Form required: standard form and the center. $(x - 3)^2 + 4y^2 = 16$; divide by 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q1/5

Derive the equation of the set of points whose distances to the foci (3, 0) and (−3, 0) add to 10.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q2/5

747

QUESTION ID
S22-F2-Q2

TYPE
Answer with required work

CORRECT ANSWER
$x^2 \div 25 + y^2 \div 16 = 1$

GRADING NOTE REQUIRED WORK: $\sqrt{((x-3)^2+y^2)} + \sqrt{((x+3)^2+y^2)} = 10$ written; one radical isolated and both sides squared twice; simplified to $16x^2 + 25y^2 = 400$.

Form required: the derivation and the standard form. $a = 5$, $c = 3$, and $b^2 = 25 - 9 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q2/5

The ellipse $(x - 1)^2 \div 25 + (y + 3)^2 \div 9 = 1$ has a horizontal major axis.

How long is the major axis?

A. 6

B. 5

C. 25

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q3/5

748

QUESTION ID
S22-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. 10

GRADING NOTE The semi-major axis is $\sqrt{25} = 5$, so the major axis is 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q3/5

True or false?

Completing the square on $x^2 - 8x$ gives $(x - 4)^2 - 16$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S22-F2-Q4

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q4/5

749

QUESTION ID

S22-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x - 4)^2 = x^2 - 8x + 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S22-F2-Q4

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q4/5

True or false?

The ellipse $x^2 \div 4 + y^2 \div 9 = 1$ is wider than it is tall.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q5/5

750

QUESTION ID
S22-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It reaches 2 units left and right but 3 units up and down, so it is taller.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q5/5

Derive the quadratic formula by completing the square

MAIN	Algebra I main question	**HSA-REI.B.4.a**
BACKWARD 1	Complete the square in expressions	**HSA-SSE.B.3.b**
BACKWARD 2	Solve literal equations for a variable	**HSA-CED.A.4**
FORWARD 1	Use the quadratic formula when the discriminant is negative	**HSN-CN.C.7**
FORWARD 2	Count the complex solutions of quadratic equations	**HSN-CN.C.9**

SET 71 • HSA-REI.B.4.a

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

Derive the quadratic formula. Start from $ax^2 + bx + c = 0$ with $a \neq 0$, and complete the square to solve for x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q1/5

753

QUESTION ID

S71-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$x = (-b \pm \sqrt{(b^2 - 4ac)}) \div (2a)$

GRADING NOTE REQUIRED WORK: Every step shown: divide by a; subtract c ÷ a; add (b ÷ (2a))² to both sides; write $(x + b \div (2a))^2 = (b^2 - 4ac) \div (4a^2)$; take square roots, using ±; subtract b ÷ (2a).

Form required: the complete derivation with every step, ending in the quadratic formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q1/5

The equation $x^2 + (b \div a)x = -c \div a$ comes from $ax^2 + bx + c = 0$. What must be added to both sides to complete the square, and what does the left side become?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q2/5

754

QUESTION ID

S71-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Add $b^2 \div (4a^2)$; the left side becomes $(x + b \div (2a))^2$.

GRADING NOTE REQUIRED WORK: Half of the x-coefficient, $b \div (2a)$, squared: $(b \div (2a))^2 = b^2 \div (4a^2)$; then the left side factored as a square.

Form required: the number added and the squared binomial.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q2/5

After completing the square, $(x + b \div (2a))^2 = (b^2 - 4ac) \div (4a^2)$. Explain how taking square roots of both sides gives the ± and the denominator 2a.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q3/5

755

QUESTION ID

S71-M-Q3

TYPE

Short answer

CORRECT ANSWER

Both the positive and the negative square roots of the right side work, so $x + b \div (2a) = \pm\sqrt{(b^2 - 4ac)} \div \sqrt{(4a^2)}$. $\sqrt{(4a^2)} = 2|a|$, and the ± already covers both signs, so the denominator can be written 2a.

GRADING NOTE Any equivalent explanation that mentions both square roots and $\sqrt{(4a^2)} = 2|a|$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q3/5

In the derivation of the quadratic formula, why must a ≠ 0?

A. Because the discriminant must be positive.

B. Because b must be positive.

C. Because c must be 0.

D. The first step divides by a, and if a = 0 the equation is not quadratic.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q4/5

756

QUESTION ID
S71-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. The first step divides by a, and if a = 0 the equation is not quadratic.

GRADING NOTE Dividing by 0 is not allowed; with a = 0 the equation is linear.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q4/5

True or false?

The derivation shows that when $b^2 - 4ac < 0$, the equation has no real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q5/5

757

QUESTION ID

S71-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The right side $(b^2 - 4ac) \div (4a^2)$ is negative, but a square of a real number is never negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q5/5

SET 71 • HSA-REI.B.4.a

BACKWARD 1

Complete the square in expressions • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b

Complete the square.

$x^2 + 12x + 7$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q1/5

759

QUESTION ID

S71-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 6)^2 - 29$

GRADING NOTE Form required: $(x - h)^2 + k$. $x^2 + 12x + 36 - 36 + 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Complete the square.

$x^2 - 5x$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q2/5

760

QUESTION ID

S71-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 2.5)^2 - 6.25$

GRADING NOTE Form required: $(x - h)^2 + k$, with fractions or decimals: $(x - \frac{5}{2})^2 - \frac{25}{4}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What number completes the square for $x^2 + \frac{2}{3}x$?

A. $\frac{1}{9}$

B. $\frac{2}{3}$

C. $\frac{4}{9}$

D. $\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q3/5

761

QUESTION ID

S71-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1/9

GRADING NOTE Half of $\frac{2}{3}$ is $\frac{1}{3}$, and $(\frac{1}{3})^2 = \frac{1}{9}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q3/5

True or false?

$x^2 + bx + (b \div 2)^2 = (x + b \div 2)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q4/5

762

QUESTION ID

S71-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x + b \div 2)^2 = x^2 + bx + b^2 \div 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q4/5

True or false?

To complete the square on $2x^2 + 8x$, add 16 to the expression.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q5/5

763

QUESTION ID

S71-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Factor 2 first: $2(x^2 + 4x + 4) - 8 = 2(x + 2)^2 - 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q5/5

SET 71 • HSA-REI.B.4.a

BACKWARD 2

Solve literal equations for a variable • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $ax + b = 0$ for x. ($a \neq 0$)

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q1/5

765

QUESTION ID

S71-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$-b \div a$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q1/5

Solve $(x + p)^2 = q$ for x. ($q \geq 0$)

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q2/5

766

QUESTION ID

S71-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$-p + \sqrt{q}$ or $-p - \sqrt{q}$

GRADING NOTE Form required: both solutions ($x = -p \pm \sqrt{q}$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q2/5

Solve $y = kx^2$ for x, where $k > 0$ and $y \geq 0$.

A. $x = \pm\sqrt{(ky)}$

B. $x = \sqrt{y} \div k$

C. $x = \pm\sqrt{(y \div k)}$

D. $x = y \div k$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q3/5

767

QUESTION ID

S71-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $x = \pm\sqrt{(y \div k)}$

GRADING NOTE $x^2 = y \div k$, so $x = \pm\sqrt{(y \div k)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q3/5

True or false?

Solving $A = \pi r^2$ for r (with $r > 0$) gives $r = \sqrt{(A \div \pi)}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q4/5

768

QUESTION ID

S71-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $r^2 = A \div \pi$, and r is positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q4/5

True or false?

Solving $2x - c = d$ for x gives $x = (d - c) \div 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q5/5

769

QUESTION ID

S71-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2x = c + d$, so $x = (c + d) \div 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q5/5

SET 71 • HSA-REI.B.4.a

FORWARD 1

Use the quadratic formula when the discriminant is negative • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7

Use the quadratic formula to solve $x^2 + 6x + 13 = 0$.
Write the solutions as a + bi.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q1/5

771

QUESTION ID
S71-F1-Q1

TYPE
Short answer

CORRECT ANSWER
x = −3 + 2i or
x = −3 − 2i

GRADING NOTE Form required: a + bi (both). $x = (-6 \pm \sqrt{(36 - 52)}) \div 2 = (-6 \pm 4i) \div 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q1/5

Solve $2x^2 - 2x + 5 = 0$. Write the solutions as a + bi.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q2/5

772

QUESTION ID

S71-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 0.5 + 1.5i or
x = 0.5 − 1.5i

GRADING NOTE Form required: a + bi (both), with fractions or decimals. x = (2 ± √(4 − 40)) ÷ 4 = (2 ± 6i) ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-F1-Q2**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q2/5

What is the discriminant of $3x^2 + x + 2$?

A. 25

B. −23

C. −25

D. 23

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q3/5

773

QUESTION ID

S71-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −23

GRADING NOTE $1 - 4(3)(2) = 1 - 24 = -23$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q3/5

True or false?

When $b^2 - 4ac < 0$, the quadratic formula gives two complex solutions that are conjugates.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q4/5

774

QUESTION ID
S71-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The ± gives p + qi and p − qi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q4/5

True or false?

The solutions of $x^2 + 9 = 0$ are 3 and -3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q5/5

775

QUESTION ID

S71-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 = -9$, so $x = 3i$ or $x = -3i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-F1-Q5**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q5/5

SET 71 • HSA-REI.B.4.a

FORWARD 2

Count the complex solutions of quadratic equations • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9

ANSWER KEY Reduced copy of the question page, with its answer

How many complex solutions, counted with multiplicity, does every quadratic equation have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q1/5

777

QUESTION ID

S71-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. A degree-2 polynomial has 2 complex zeros counted with multiplicity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q1/5

$x^2 - 6x + 9 = 0$ has one real solution. How is it counted by the Fundamental Theorem of Algebra?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q2/5

778

QUESTION ID

S71-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x = 3 is a double solution (multiplicity 2): $(x - 3)^2 = 0$.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q2/5

A quadratic equation with real coefficients has the solution 2 + i. What is its other solution?

A. −2 − i

B. 2 − i

C. −2 + i

D. 2 + i again

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q3/5

779

QUESTION ID

S71-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2 − i

GRADING NOTE Nonreal solutions of real quadratics come in conjugate pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q3/5

True or false?

Every quadratic equation with real coefficients has two solutions in the complex numbers, counted with multiplicity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q4/5

780

QUESTION ID
S71-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE This is the Fundamental Theorem of Algebra for degree 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q4/5

True or false?

A quadratic equation can have three different complex solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q5/5

781

QUESTION ID
S71-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A degree-2 equation has at most 2 solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-F2-Q5**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q5/5

Solve quadratic equations by inspection, square roots, and factoring

MAIN	Algebra I main question	**HSA-REI.B.4.b**
BACKWARD 1	Factor quadratic expressions	**HSA-SSE.A.2**
BACKWARD 2	Solve the linear equations that come from each factor	**HSA-REI.B.3**
FORWARD 1	Find the zeros of a polynomial from its factors	**HSA-APR.B.3**
FORWARD 2	Use the Fundamental Theorem of Algebra to count complex zeros	**HSN-CN.C.9**

SET 23 • HSA-REI.B.4.b

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve by factoring.

$x^2 - 5x - 24 = 0$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q1/5

784

QUESTION ID

S23-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

x = 8 or x = −3

GRADING NOTE REQUIRED WORK: The factored form (x − 8)(x + 3) = 0, and each factor set equal to 0.

Form required: both solutions, with the factoring shown. A product is 0 only when a factor is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $3(x - 2)^2 = 75$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q2/5

785

QUESTION ID

S23-M-Q2

TYPE

Short answer

CORRECT ANSWER

$x = 7$ or $x = -3$

GRADING NOTE Form required: both solutions. $(x - 2)^2 = 25$, so $x - 2 = 5$ or $x - 2 = -5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2x^2 = 7x$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q3/5

786

QUESTION ID

S23-M-Q3

TYPE

Short answer

CORRECT ANSWER

x = 0 or x = 3.5

GRADING NOTE Form required: both solutions (3.5 or 7/2). $2x^2 - 7x = 0$, so $x(2x - 7) = 0$. Dividing both sides by x would lose x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q3/5

Solve $6x^2 + x - 2 = 0$.

A. $x = 2$ or $x = -1$

B. $x = -\frac{2}{3}$ or $x = \frac{1}{2}$

C. $x = \frac{2}{3}$ or $x = -\frac{1}{2}$

D. $x = -2$ or $x = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q4/5

787

QUESTION ID

S23-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. x = −2/3 or x = 1/2

GRADING NOTE $6x^2 + x - 2 = (3x + 2)(2x - 1)$, so $3x + 2 = 0$ or $2x - 1 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q4/5

True or false?

A student divides both sides of $x^2 = 4x$ by x and gets $x = 4$. This finds every solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q5/5

788

QUESTION ID

S23-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 - 4x = x(x - 4) = 0$, so $x = 0$ is also a solution. Dividing by x assumed $x \neq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q5/5

SET 23 • HSA-REI.B.4.b

BACKWARD 1

Factor quadratic expressions • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor completely.

$x^2 + 2x - 15$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q1/5

790

QUESTION ID

S23-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 5)(x − 3)

GRADING NOTE Form required: factored completely. $5 \cdot (-3) = -15$ and $5 + (-3) = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q1/5

Factor completely.

$4x^2 - 9$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q2/5

791

QUESTION ID

S23-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(2x + 3)(2x − 3)

GRADING NOTE Form required: factored completely. A difference of squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q2/5

Factor $x^2 - 9x + 20$.

A. $(x - 4)(x + 5)$

B. $(x - 2)(x - 10)$

C. $(x - 4)(x - 5)$

D. $(x + 4)(x + 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q3/5

792

QUESTION ID

S23-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $(x - 4)(x - 5)$

GRADING NOTE $-4 \cdot -5 = 20$ and $-4 + -5 = -9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q3/5

True or false?

$3x^2 - 12x = 3x(x - 4)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q4/5

793

QUESTION ID

S23-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $3x \cdot x = 3x^2$ and $3x \cdot (-4) = -12x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q4/5

True or false?

$x^2 + 9 = (x + 3)(x - 3)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q5/5

794

QUESTION ID

S23-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 3)(x - 3) = x^2 - 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • HSA-REI.B.4.b

BACKWARD 2

Solve the linear equations that come from each factor • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2x + 7 = 0.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q1/5

796

QUESTION ID

S23-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−3.5

GRADING NOTE Any equivalent form, such as −7/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q1/5

Solve $3x - 4 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q2/5

797

QUESTION ID

S23-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4/3

GRADING NOTE Any equivalent form, such as $1\frac{1}{3}$. A rounded decimal is not exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $5 - 2x = 0$.

A. $x = 3$

B. $x = -2.5$

C. $x = 2.5$

D. $x = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q3/5

798

QUESTION ID

S23-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 2.5

GRADING NOTE $-2x = -5$, so $x = 2.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q3/5

True or false?

If $x - 6 = 0$, then $x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q4/5

799

QUESTION ID

S23-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Add 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If 4x = 0, then x = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q5/5

800

QUESTION ID
S23-B2-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Divide both sides by 4: 0 ÷ 4 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q5/5